

FANUC Robot **series**

**R-30*i*A/R-30*i*A Mate/R-30*i*B/R-30*i*B Mate/
R-30*i*B Plus/R-30*i*B Mate Plus
CONTROLLER**

Initial Setup Function

OPERATOR'S MANUAL

MAROCINSE06091E REV. D

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FANUC manuals present descriptions, specifications, drawings, schematics, bills of material, parts, connections and/or procedures for installing, disassembling, connecting, operating and programming FANUC products and/or systems. Such systems consist of robots, extended axes, robot controllers, application software, the KAREL® programming language, INSIGHT® vision equipment, and special tools.

FANUC recommends that only persons who have been trained in one or more approved FANUC Training Course(s) be permitted to install, operate, use, perform procedures on, repair, and/or maintain FANUC products and/or systems and their respective components. Approved training necessitates that the courses selected be relevant to the type of system installed and application performed at the customer site.



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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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For customer assistance, including Technical Support, Service, Parts & Part Repair, and Marketing Requests, contact the Customer Resource Center, 24 hours a day, at 1-800-47-ROBOT (1-800-477-6268). International customers should call 011-1-248-377-7159.

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Patents

One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

- **Original Instructions**

Before using the Robot, be sure to read the "FANUC Robot Safety Manual (B-80687EN)" and understand the content.

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Should you wish to export or re-export these products, please contact FANUC for advice.

In this manual we have tried as much as possible to describe all the various matters.

However, we cannot describe all the matters which must not be done, or which cannot be done, because there are so many possibilities.

Therefore, matters which are not especially described as possible in this manual should be regarded as "impossible".

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

SAFETY PRECAUTIONS

This chapter describes the precautions which must be followed to ensure the safe use of the robot. Before using the robot, be sure to read this chapter thoroughly.

For detailed functions of the robot operation, read the relevant operator's manual to understand fully its specification.

For the safety of the operator and the system, follow all safety precautions when operating a robot and its peripheral equipment installed in a work cell.

In addition, refer to the "FANUC Robot SAFETY HANDBOOK (B-80687EN)".

1 DEFINITION OF USER

The user can be defined as follows.

Operator:

- Turns ON/OFF power to the robot
- Starts the robot program from the operator's panel

Programmer:

- Operates the robot
- Teaches the robot inside the safety fence

Maintenance engineer:

- Operates the robot
- Teaches the robot inside the safety fence
- Performs maintenance (repair, adjustment, replacement)

- Operator is not allowed to work in the safety fence.
- Programmers and maintenance engineers are allowed to work in the safety fence. The work inside the safety fence includes lifting, setting, teaching, adjustment, maintenance, etc.
- To work inside the safety fence, the person must receive a professional training for the robot.

During the operation, programming, and maintenance of your robotic system, the programmer, operator, and maintenance engineer should take additional care of their safety by wearing the following safety items.

- Adequate clothes for the operation
- Safety shoes
- A helmet

2 DEFINITION OF SAFETY NOTATIONS

To ensure the safety of users and prevent damage to the machine, this manual indicates each precaution on safety with "WARNING" or "CAUTION" according to its severity. Supplementary information is indicated by "NOTE". Read the contents of each "WARNING", "CAUTION" and "NOTE" before using the robot.

Symbol	Definitions
 WARNING	Used if hazard resulting in the death or serious injury of the user will be expected to occur if he or she fails to follow the approved procedure.
 CAUTION	Used if a hazard resulting in the minor or moderate injury of the user, or equipment damage may be expected to occur if he or she fails to follow the approved procedure.
NOTE	Used if a supplementary explanation not related to any of WARNING and CAUTION is to be indicated.

- Check this manual thoroughly, and keep it handy for the future reference.

3 SAFETY OF THE USER

User safety is the primary safety consideration. Because it is very dangerous to enter the operating space of the robot during automatic operation, adequate safety precautions must be observed. The following lists the general safety precautions. Careful consideration must be made to ensure user safety.

- (1) Have the robot system users attend the training courses held by FANUC.

FANUC provides various training courses. Contact our sales office for details.

- (2) Even when the robot is stationary, it is possible that the robot is still in a ready to move state, and is waiting for a signal. In this state, the robot is regarded as still in motion. To ensure user safety, provide the system with an alarm to indicate visually or aurally that the robot is in motion.
- (3) Install a safety fence with a gate so that no user can enter the work area without passing through the gate. Install an interlocking device, a safety plug, and so forth in the safety gate so that the robot is stopped as the safety gate is opened.

The controller is designed to receive this interlocking signal of the door switch. When the gate is opened and this signal received, the controller stops the robot (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type). For connection, see Fig. 3 (b).

- (4) Provide the peripheral equipment with appropriate earth (Class A, Class B, Class C, and Class D).
- (5) Try to install the peripheral equipment outside the robot operating space.
- (6) Draw an outline on the floor, clearly indicating the range of the robot operating space, including the tools such as a hand.
- (7) Install a mat switch or photoelectric switch on the floor with an interlock to a visual or aural alarm that stops the robot when a user enters the work area.
- (8) If necessary, install a safety lock so that no one except the user in charge can turn on the power of the robot.

The circuit breaker installed in the controller is designed to disable anyone from turning it on when it is locked with a padlock.

- (9) When adjusting each peripheral equipment independently, be sure to turn off the power of the robot.
- (10) Operators should be ungloved while manipulating the operator panel or teach pendant. Operation with gloved fingers could cause an operation error.
- (11) Programs, system variables, and other information can be saved on memory card or USB memories. Be sure to save the data periodically in case the data is lost in an accident. (refer to Controller OPERATOR'S MANUAL.)
- (12) The robot should be transported and installed by accurately following the procedures recommended by FANUC. Wrong transportation or installation may cause the robot to fall, resulting in severe injury to workers.
- (13) In the first operation of the robot after installation, the operation should be restricted to low speeds. Then, the speed should be gradually increased to check the operation of the robot.
- (14) Before the robot is started, it should be checked that no one is inside the safety fence. At the same time, a check must be made to ensure that there is no risk of hazardous situations. If detected, such a situation should be eliminated before the operation.
- (15) When the robot is used, the following precautions should be taken. Otherwise, the robot and peripheral equipment can be adversely affected, or workers can be severely injured.
 - Avoid using the robot in a flammable environment.
 - Avoid using the robot in an explosive environment.
 - Avoid using the robot in an environment full of radiation.
 - Avoid using the robot under water or at high humidity.
 - Avoid using the robot to carry a person or animal.
 - Avoid using the robot as a stepladder. (Never climb up on or hang from the robot.)
 - Outdoor
- (16) When connecting the peripheral equipment related to stop (safety fence etc.) and each signal (external emergency, fence etc.) of robot, be sure to confirm the stop movement and do not take the wrong connection.
- (17) When preparing footstep, please consider security for installation and maintenance work in high place according to Fig. 3 (c). Please consider footstep and safety belt mounting position.

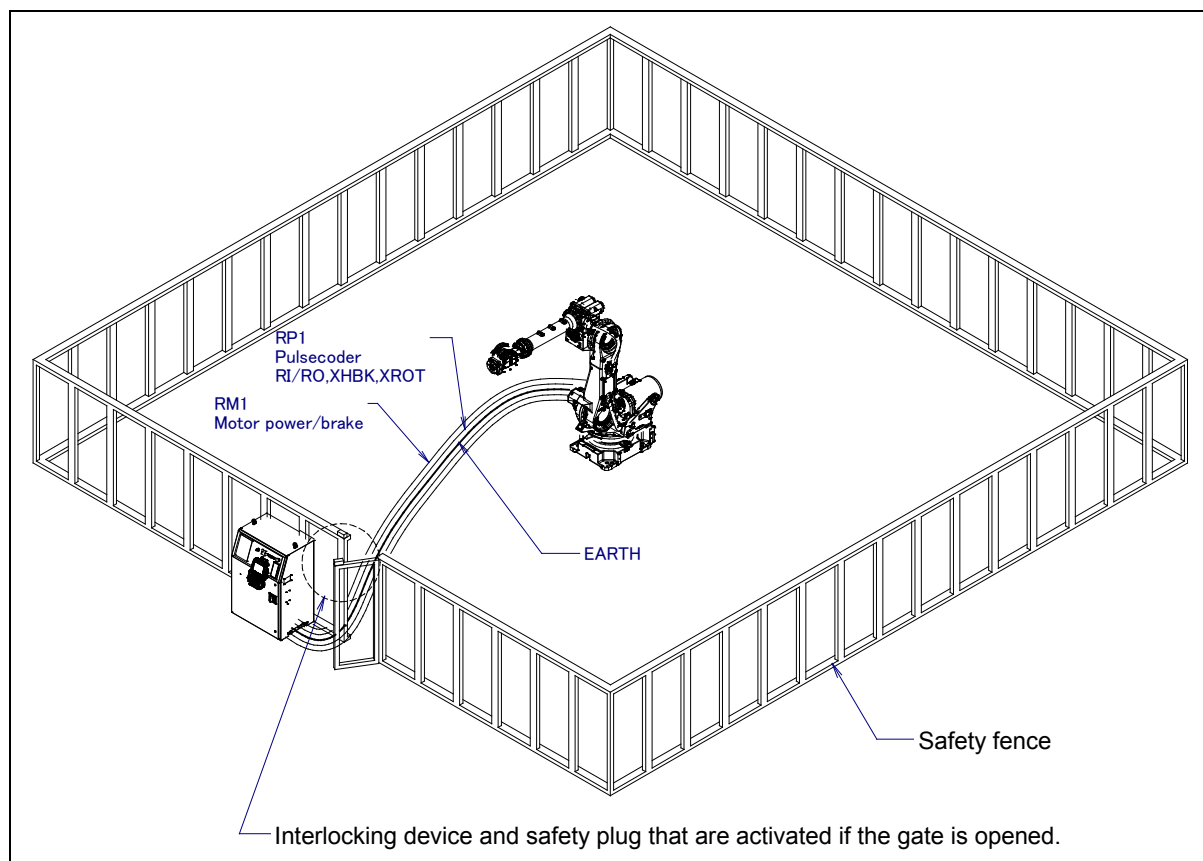


Fig. 3 (a) Safety fence and safety gate

**WARNING**

When you close a fence, please confirm that there is not a person from all directions of the robot.

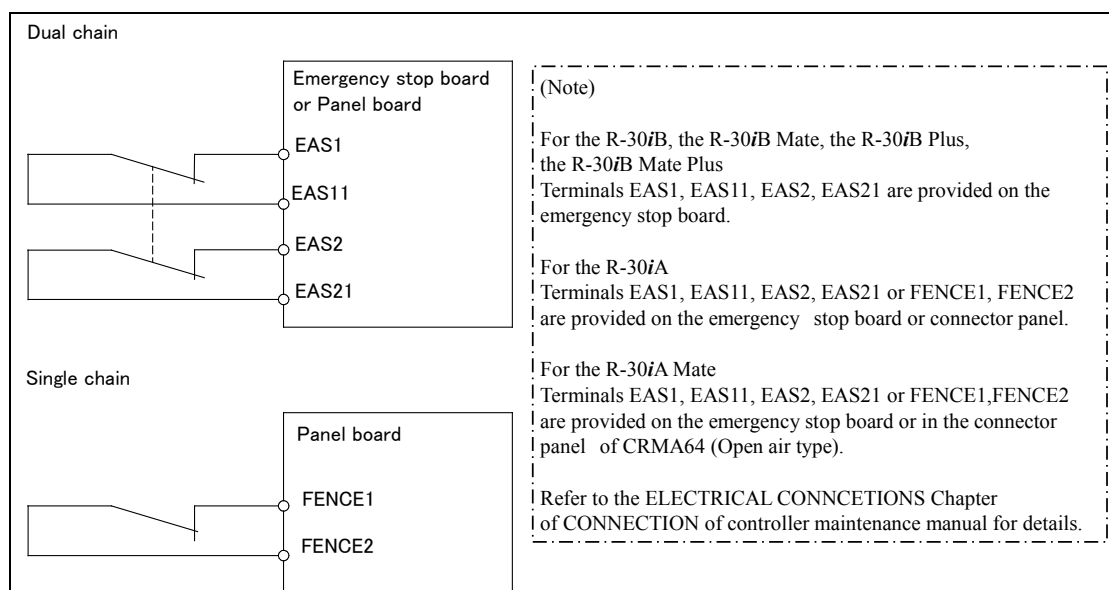


Fig. 3 (b) Connection diagram for the signal of safety fence

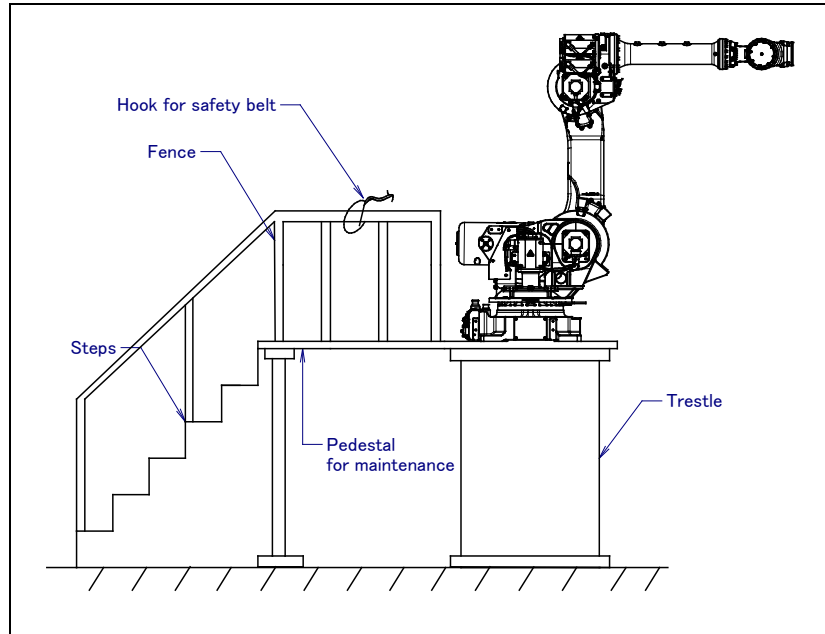


Fig. 3 (c) Pedestal for maintenance

3.1 SAFETY OF THE OPERATOR

An operator refers to a person who turns on and off the robot system and starts a robot program from, for example, the operator panel during daily operation.

Operators cannot work inside of the safety fence.

- (1) If the robot does not need to be operated, turn off the robot controller power or press the EMERGENCY STOP button during working.
- (2) Operate the robot system outside the operating space of the robot.
- (3) Install a safety fence or safety door to avoid the accidental entry of a person other than an operator in charge or keep operator out from the hazardous place.
- (4) Install the EMERGENCY STOP button within the operator's reach.

The robot controller is designed to be connected to an external EMERGENCY STOP button. With this connection, the controller stops the robot operation (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type) when the external EMERGENCY STOP button is pressed. See the diagram below for connection.

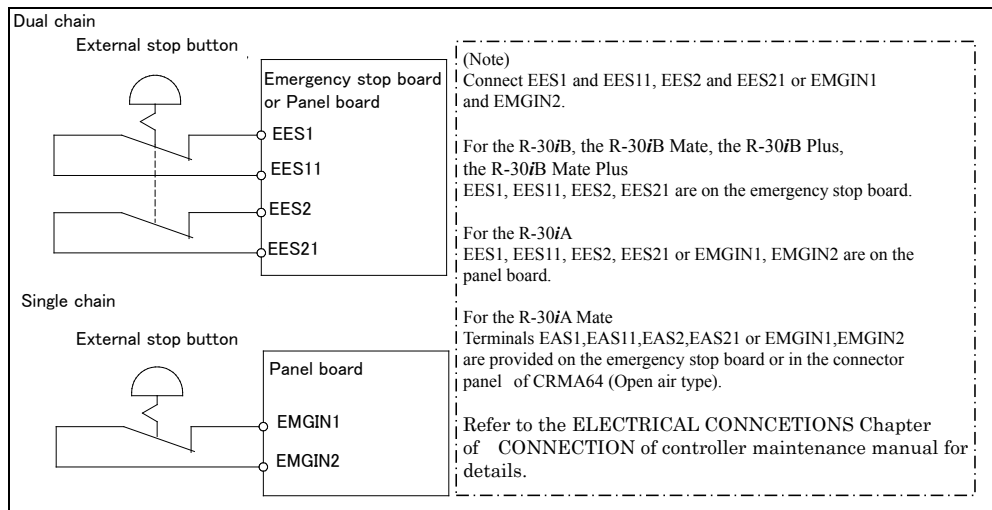


Fig. 3.1 Connection diagram for external emergency stop button

3.2 SAFETY OF THE PROGRAMMER

While teaching the robot, the operator may need to enter the robot operation area. The programmer must ensure the safety especially.

- (1) Unless it is specifically necessary to enter the robot operating space, carry out all tasks outside the operating space.
- (2) Before teaching the robot, check that the robot and its peripheral equipment are all in the normal operating condition.
- (3) If it is inevitable to enter the robot operating space to teach the robot, check the locations, settings, and other conditions of the safety devices (such as the EMERGENCY STOP button, the DEADMAN switch on the teach pendant) before entering the area.
- (4) The programmer must be extremely careful not to let anyone else enter the robot operating space.
- (5) Programming should be done outside the area of the safety fence as far as possible. If programming needs to be done inside the safety fence, the programmer should take the following precautions:
 - Before entering the area of the safety fence, ensure that there is no risk of dangerous situations in the area.
 - Be prepared to press the emergency stop button whenever necessary.
 - Robot motions should be made at low speeds.
 - Before starting programming, check the whole robot system status to ensure that no remote instruction to the peripheral equipment or motion would be dangerous to the user.

Our operator panel is provided with an emergency stop button and a key switch (mode switch) for selecting the automatic operation mode (AUTO) and the teach modes (T1 and T2). Before entering the inside of the safety fence for the purpose of teaching, set the switch to a teach mode, remove the key from the mode switch to prevent other people from changing the operation mode carelessly, then open the safety gate. If the safety gate is opened with the automatic operation mode set, the robot stops (Please refer to "**STOP TYPE OF ROBOT**" in SAFETY PRECAUTIONS for detail of stop type). After the switch is set to a teach mode, the safety gate is disabled. The programmer should understand that the safety gate is disabled and is responsible for keeping other people from entering the inside of the safety fence.

Teach pendant is provided with a switch to enable/disable robot operation from teach pendant and DEADMAN switch as well as emergency stop button. These button and switch function as follows:

- (1) Emergency stop button: Causes the stop of the robot (Please refer to "STOP TYPE OF ROBOT" in SAFETY PRECAUTIONS for detail of stop type) when pressed.
- (2) DEADMAN switch: Functions are different depending on the teach pendant enable/disable switch setting status.
 - (a) **Enable:** Servo power is turned off and robot stops when the operator releases the DEADMAN switch or when the operator presses the switch strongly.
 - (b) **Disable:** The DEADMAN switch is disabled.

(Note) The DEADMAN switch is provided to stop the robot when the operator releases the teach pendant or presses the pendant strongly in case of emergency. The R-30iB Plus/R-30iB Mate Plus /R-30iB/R-30iB Mate/R-30iA/R-30iA Mate employs a 3-position DEADMAN switch, which allows the robot to operate when the 3-position DEADMAN switch is pressed to its intermediate point. When the operator releases the DEADMAN switch or presses the switch strongly, the robot stops immediately.

The operator's intention of starting teaching is determined by the controller through the dual operation of setting the teach pendant enable/disable switch to the enable position and pressing the DEADMAN switch. The operator should make sure that the robot could operate in such conditions and be responsible in carrying out tasks safely.

Based on the risk assessment by FANUC, number of operation of DEADMAN switch should not exceed about 10000 times per year.

The teach pendant, operator panel, and peripheral device interface send each robot start signal. However the validity of each signal changes as follows depending on the mode switch and the DEADMAN switch of the operator panel, the teach pendant enable switch and the remote condition on the software.

**For the R-30iB Plus/R-30iB Mate Plus/R-30iB/R-30iB Mate/R-30iA Controller
or CE or RIA specification of the R-30iA Mate Controller**

Mode	Teach pendant enable switch	Software remote condition	Teach pendant	Operator panel	Peripheral device
AUTO mode	On	Local	Not allowed	Not allowed	Not allowed
		Remote	Not allowed	Not allowed	Not allowed
	Off	Local	Not allowed	Allowed to start	Not allowed
		Remote	Not allowed	Not allowed	Allowed to start
T1, T2 mode	On	Local	Allowed to start	Not allowed	Not allowed
		Remote	Allowed to start	Not allowed	Not allowed
	Off	Local	Not allowed	Not allowed	Not allowed
		Remote	Not allowed	Not allowed	Not allowed

T1,T2 mode: DEADMAN switch is effective.

For the standard specification of R-30iA Mate Controller

Teach pendant enable switch	Software remote condition	Teach pendant	Peripheral device
On	Ignored	Allowed to start	Not allowed
Off	Local	Not allowed	Not allowed
	Remote	Not allowed	Allowed to start

- (6) (Only when R-30iB Plus/R-30iB Mate Plus/R-30iB/R-30iB Mate /R-30iA Controller or CE or RIA specification of R-30iA Mate controller is selected.) To start the system using the operator panel, make certain that nobody is in the robot operating space and that there are no abnormal conditions in the robot operating space.
- (7) When a program is completed, be sure to carry out the test operation according to the following procedure.
 - (a) Run the program for at least one operation cycle in the single step mode at low speed.
 - (b) Run the program for at least one operation cycle in the continuous operation mode at low speed.
 - (c) Run the program for one operation cycle in the continuous operation mode at the intermediate speed and check that no abnormalities occur due to a delay in timing.
 - (d) Run the program for one operation cycle in the continuous operation mode at the normal operating speed, and check that the system operates automatically without trouble.
 - (e) After checking the completeness of the program through the test operation above, execute it in the automatic operation mode.
- (8) While operating the system in the automatic operation mode, the teach pendant operator must leave the safety fence.

3.3 SAFETY OF THE MAINTENANCE ENGINEER

For the safety of maintenance engineer personnel, pay utmost attention to the following.

- (1) During operation, never enter the robot operating space.
- (2) A hazardous situation may arise when the robot or the system, are kept with their power-on during maintenance operations. Therefore, for any maintenance operation, the robot and the system should be put into the power-off state. If necessary, a lock should be in place in order to prevent any other person from turning on the robot and/or the system. In case maintenance needs to be executed in the power-on state, the emergency stop button must be pressed.
- (3) If it becomes necessary to enter the robot operating space while the power is on, press the emergency stop button on the operator box or operator panel, or the teach pendant before entering the range. The maintenance worker must indicate that maintenance work is in progress and be careful not to allow other people to operate the robot carelessly.
- (4) When entering the area enclosed by the safety fence, the worker must check the whole robot system in order to make sure no dangerous situations exist. In case the worker needs to enter the safety area whilst a dangerous situation exists, extreme care must be taken, and whole robot system status must be carefully monitored.
- (5) Before the maintenance of the pneumatic system is started, the supply pressure should be shut off and the pressure in the piping should be reduced to zero.
- (6) Before the start of maintenance work, check that the robot and its peripheral equipment are all in the normal operating condition.
- (7) Do not operate the robot in the automatic operation while anybody is in the robot operating space.
- (8) When you maintain the robot alongside a wall or instrument, or when multiple users are working nearby, make certain that their escape path is not obstructed.
- (9) When a tool is mounted on the robot, or when any movable device other than the robot is installed, such as belt conveyor, pay careful attention to its motion.
- (10) If necessary, have a user who is familiar with the robot system stand beside the operator panel and observe the work being performed. If any danger arises, the user should be ready to press the EMERGENCY STOP button at any time.
- (11) When replacing a part, please contact your local FANUC representative. If a wrong procedure is followed, an accident may occur, causing damage to the robot and injury to the user.
- (12) When replacing or reinstalling components, take care to prevent foreign material from entering the system.
- (13) When handling each unit or printed circuit board in the controller during inspection, turn off the circuit breaker to protect against electric shock.
If there are two cabinets, turn off the both circuit breaker.
- (14) A part should be replaced with a part recommended by FANUC. If other parts are used, malfunction or damage would occur. Especially, a fuse that is not recommended by FANUC should not be used. Such a fuse may cause a fire.
- (15) When restarting the robot system after completing maintenance work, make sure in advance that there is no person in the operating space and that the robot and the peripheral equipment are not abnormal.
- (16) When a motor or brake is removed, the robot arm should be supported with a crane or other equipment beforehand so that the arm would not fall during the removal.
- (17) Whenever grease is spilled on the floor, it should be removed as quickly as possible to prevent dangerous falls.
- (18) The following parts are heated. If a maintenance user needs to touch such a part in the heated state, the user should wear heat-resistant gloves or use other protective tools.
 - Servo motor
 - Inside the controller
 - Reducer
 - Gearbox
 - Wrist unit

- (19) Maintenance should be done under suitable light. Care must be taken that the light would not cause any danger.
- (20) When a motor, reducer, or other heavy load is handled, a crane or other equipment should be used to protect maintenance workers from excessive load. Otherwise, the maintenance workers would be severely injured.
- (21) The robot should not be stepped on or climbed up during maintenance. If it is attempted, the robot would be adversely affected. In addition, a misstep can cause injury to the worker.
- (22) When performing maintenance work in high place, secure a footstep and wear safety belt.
- (23) After the maintenance is completed, spilled oil or water and metal chips should be removed from the floor around the robot and within the safety fence.
- (24) When a part is replaced, all bolts and other related components should put back into their original places. A careful check must be given to ensure that no components are missing or left not mounted.
- (25) In case robot motion is required during maintenance, the following precautions should be taken :
 - Foresee an escape route. And during the maintenance motion itself, monitor continuously the whole robot system so that your escape route will not become blocked by the robot, or by peripheral equipment.
 - Always pay attention to potentially dangerous situations, and be prepared to press the emergency stop button whenever necessary.
- (26) The robot should be periodically inspected. (Refer to the robot mechanical manual and controller maintenance manual.) A failure to do the periodical inspection can adversely affect the performance or service life of the robot and may cause an accident
- (27) After a part is replaced, a test execution should be given for the robot according to a predetermined method. (See TESTING section of "Controller operator's manual".) During the test execution, the maintenance worker should work outside the safety fence.

4 SAFETY OF THE TOOLS AND PERIPHERAL EQUIPMENT

4.1 PRECAUTIONS IN PROGRAMMING

- (1) Use a limit switch or other sensor to detect a dangerous condition and, if necessary, design the program to stop the robot when the sensor signal is received.
- (2) Design the program to stop the robot when an abnormality occurs in any other robots or peripheral equipment, even though the robot itself is normal.
- (3) For a system in which the robot and its peripheral equipment are in synchronous motion, particular care must be taken in programming so that they do not interfere with each other.
- (4) Provide a suitable interface between the robot and its peripheral equipment so that the robot can detect the states of all devices in the system and can be stopped according to the states.

4.2 PRECAUTIONS FOR MECHANISM

- (1) Keep the component cells of the robot system clean, operate the robot where insulated from the influence of oil, water, and dust.
- (2) Don't use unconfirmed liquid for cutting fluid and cleaning fluid.
- (3) Adopt limit switches or mechanical stoppers to limit the robot motion, and avoid the robot from collisions against peripheral equipment or tools.
- (4) Observe the following precautions about the mechanical unit cables. Failure to follow precautions may cause problems.
 - Use mechanical unit cable that have required user interface.
 - Do not add user cable or hose to inside of the mechanical unit.
 - Please do not obstruct the movement of the mechanical unit when cables are added to outside of mechanical unit.
 - In the case of the model that a cable is exposed, please do not perform remodeling (Adding a protective cover and fix an outside cable more) obstructing the behavior of the outcrop of the cable.
 - When installing user peripheral equipment on the robot mechanical unit, please pay attention that the device does not interfere with the robot itself.
- (5) The frequent power-off stop for the robot during operation causes the trouble of the robot. Please avoid the system construction that power-off stop would be operated routinely. (Refer to bad case example.) Please perform power-off stop after reducing the speed of the robot and stopping it by hold stop or cycle stop when it is not urgent. (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type.)

(Bad case example)

 - Whenever poor product is generated, a line stops by emergency stop and power-off of the robot is incurred.
 - When alteration is necessary, safety switch is operated by opening safety fence and power-off stop is incurred for the robot during operation.
 - An operator pushes the emergency stop button frequently, and a line stops.
 - An area sensor or a mat switch connected to safety signal operates routinely and power-off stop is incurred for the robot.
 - Power-off stop is regularly incurred due to an inappropriate setting for Dual Check Safety (DCS).
- (6) Power-off stop of Robot is executed when collision detection alarm (SRVO-050) etc. occurs. Please try to avoid unnecessary power-off stops. It may cause the trouble of the robot, too. So remove the causes of the alarm.

5 SAFETY OF THE ROBOT MECHANICAL UNIT

5.1 PRECAUTIONS IN OPERATION

- (1) When operating the robot in the jog mode, set it at an appropriate speed so that the operator can manage the robot in any eventuality.
- (2) Before pressing the jog key, be sure you know in advance what motion the robot will perform in the jog mode.

5.2 PRECAUTIONS IN PROGRAMMING

- (1) When the operating spaces of robots overlap, make certain that the motions of the robots do not interfere with each other.
- (2) Be sure to specify the predetermined work origin in a motion program for the robot and program the motion so that it starts from the origin and terminates at the origin. Make it possible for the operator to easily distinguish at a glance that the robot motion has terminated.

5.3 PRECAUTIONS FOR MECHANISMS

Keep the robot operation area clean, and operate the robot in an environment free of grease, water, and dust.

5.4 PROCEDURE TO MOVE ARM WITHOUT DRIVE POWER IN EMERGENCY OR ABNORMAL SITUATIONS

For emergency or abnormal situations (e.g. persons trapped in or pinched by the robot), brake release unit can be used to move the robot axes without drive power.

Please refer to controller maintenance manual and mechanical unit operator's manual for using method of brake release unit and method of supporting robot.

6 SAFETY OF THE END EFFECTOR

6.1 PRECAUTIONS IN PROGRAMMING

- (1) To control the pneumatic, hydraulic and electric actuators, carefully consider the necessary time delay after issuing each control command up to actual motion and ensure safe control.
- (2) Provide the end effector with a limit switch, and control the robot system by monitoring the state of the end effector.

7 STOP TYPE OF ROBOT (R-30iA, R-30iA Mate)

The following three robot stop types exist:

Power-Off Stop (Category 0 following IEC 60204-1)

Servo power is turned off and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.

The following processing is performed at Power-Off stop.

- An alarm is generated and servo power is turned off.
- The robot operation is stopped immediately. Execution of the program is paused.

Frequent Power-Off stop of the robot during operation can cause mechanical problems of the robot.

Avoid system designs that require routine or frequent Power-Off stop conditions.

Controlled stop (Category 1 following IEC 60204-1)

The robot is decelerated until it stops, and servo power is turned off.

The following processing is performed at Controlled stop.

- The alarm "SRVO-199 Controlled stop" occurs along with a decelerated stop. Execution of the program is paused.
- An alarm is generated and servo power is turned off.

Hold (Category 2 following IEC 60204-1)

The robot is decelerated until it stops, and servo power remains on.

The following processing is performed at Hold.

- The robot operation is decelerated until it stops. Execution of the program is paused.

WARNING

- 1 The stopping distance and time of Controlled stop is longer than those of Power-Off stop. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time is necessary when Controlled stop is used. Please refer to the operator's manual of a particular robot model for the data of stopping distance and time.
- 2 In multi arm system, the longest stopping distance and time of Controlled Stop among each robot are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the multi arm system.
- 3 In the system which has extended axis, the longer stopping distance and time of Controlled Stop among robot and extended axis are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the system which has extended axis. Please refer to the extended axis setup procedure of the controller operator's manual for considering the stopping distance and time of the extended axis.
- 4 In case of Controlled stop, motor power shutdown is delayed for a maximum of 2 seconds. In this case, a risk assessment for the whole robot system is necessary, including the 2 seconds delay.

When the emergency stop button is pressed or the FENCE is open, the stop type of robot is Power-Off stop or Controlled stop. The configuration of stop type for each situation is called *stop pattern*. The stop pattern is different according to the controller type or option configuration.

There are the following 3 Stop patterns.

Stop pattern	Mode	Emergency stop button	External Emergency stop	FENCE open	SVOFF input	Servo disconnect
A	AUTO	P-Stop	P-Stop	C-Stop	C-Stop	P-Stop
	T1	P-Stop	P-Stop	-	C-Stop	P-Stop
	T2	P-Stop	P-Stop	-	C-Stop	P-Stop
B	AUTO	P-Stop	P-Stop	P-Stop	P-Stop	P-Stop
	T1	P-Stop	P-Stop	-	P-Stop	P-Stop
	T2	P-Stop	P-Stop	-	P-Stop	P-Stop
C	AUTO	C-Stop	C-Stop	C-Stop	C-Stop	C-Stop
	T1	P-Stop	P-Stop	-	C-Stop	P-Stop
	T2	P-Stop	P-Stop	-	C-Stop	P-Stop

P-Stop: Power-Off stop

C-Stop: Controlled stop

-: Disable

The following table indicates the Stop pattern according to the controller type or option configuration.

Option	R-30iA				R-30iA Mate		
	Standard (Single)	Standard (Dual)	RIA type	CE type	Standard	RIA type	CE type
Standard	B (*)	A	A	A	A (**)	A	A
Stop type set (Stop pattern C) (A05B-2500-J570)	N/A	N/A	C	C	N/A	C	C

(*) R-30iA standard (single) does not have servo disconnect.

(**) R-30iA Mate Standard does not have servo disconnect, and the stop type of SVOFF input is Power-Off stop.

The stop pattern of the controller is displayed in "Stop pattern" line in software version screen. Please refer to "Software version" in operator's manual of controller for the detail of software version screen.

"Controlled stop by E-Stop" option

When "Stop type set (Stop pattern C) (A05B-2500-J570) option is specified, the stop type of the following alarms becomes Controlled stop but only in AUTO mode. In T1 or T2 mode, the stop type is Power-Off stop which is the normal operation of the system.

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open. (R-30iA controller)
SRVO-194 Servo disconnect	Servo disconnect input (SD4-SD41, SD5-SD51) is open. (R-30iA controller)
SRVO-218 Ext.E-stop/Servo Disconnect	External emergency stop input (EES1-EES11, EES2-EES21) is open. (R-30iA Mate controller)
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.

Controlled stop is different from Power-Off stop as follows:

- In Controlled stop, the robot is stopped on the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Controlled stop, physical impact is less than Power-Off stop. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End Of Arm Tool) should be minimized.
- The stopping distance and stopping time of Controlled stop is longer than the stopping distance and stopping time of Power-Off stop, depending on the robot model and axis. Please refer to the operator's manual of a particular robot model for the data of stopping distance and stopping time.

For the R-30iA or R-30iA Mate, this function is available only in CE or RIA type hardware.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.



WARNING

The stopping distance and stopping time of Controlled stop are longer than the stopping distance and stopping time of Power-Off stop. A risk assessment for the whole robot system, which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

8 STOP TYPE OF ROBOT (R-30iB, R-30iB Mate)

There are following four types of Stopping Robot.

Power-Off Stop (Category 0 following IEC 60204-1)

Servo power is turned off, and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.

“**Power-Off stop**” performs following processing.

- An alarm is generated, and then the servo power turns off. Instantly the robot stops.
- Execution of the program is paused.

Frequent Power-Off stop of the robot during operation can cause mechanical problems of the robot.

Avoid system designs that require routine or frequent Power-Off stop conditions.

Controlled stop (Category 1 following IEC 60204-1)

The robot is decelerated until it stops, and servo power is turned off.

“**Controlled stop**” performs following processing.

- The alarm “**SRVO-199 Controlled stop**” occurs along with a decelerated stop. The program execution is paused.
- An alarm is generated, and then the servo power turns off.

Smooth stop (Category 1 following IEC 60204-1)

The robot is decelerated until it stops, and servo power is turned off.

“**Smooth stop**” performs following processing.

- The alarm “**SRVO-289 Smooth Stop**” occurs along with a decelerated stop. The program execution is paused.
- An alarm is generated, and then the servo power turns off.

- In Smooth stop, the robot decelerates until it stops with the deceleration time shorter than Controlled stop.

Hold (Category 2 following IEC 60204-1)

The robot is decelerated until it stops, and servo power remains on.

“**Hold**” performs following processing.

- The robot operation is decelerated until it stops. Execution of the program is paused.

WARNING

- 5 The stopping distance and time of Controlled stop and Smooth stop are longer than those of Power-Off stop. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time is necessary when Controlled stop or Smooth Stop is used. Please refer to the operator's manual of a particular robot model for the data of stopping distance and time.
- 6 In multi arm system, the longest stopping distance and time of Controlled Stop or Smooth Stop among each robot are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the multi arm system.
- 7 In the system which has extended axis, the longer stopping distance and time of Controlled Stop or Smooth Stop among robot and extended axis are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the system which has extended axis. Please refer to the extended axis setup procedure of the controller operator's manual for considering the stopping distance and time of the extended axis.
- 8 When Smooth stop occurs during deceleration by Controlled stop, the stop type of robot is changed to Power-Off Stop.
When Smooth stop occurs during deceleration by Hold, the stop type of robot is changed to Power-Off Stop.
- 9 In case of Controlled stop or Smooth Stop, motor power shutdown is delayed for a maximum of 2 seconds. In this case, a risk assessment for the whole robot system is necessary, including the 2 seconds delay.

When the emergency stop button is pressed or the FENCE is open, the stop type of robot is Power-Off stop, Controlled stop, or Smooth stop. The configuration of stop type for each situation is called *stop pattern*. The stop pattern is different according to the option configuration.

There are the following 3 Stop patterns.

Stop pattern	Mode	Emergency stop button	External Emergency stop	FENCE open	SVOFF input	Deadman switch (*)
A	AUTO	P-Stop	P-Stop	C-Stop	C-Stop	-
	T1	P-Stop	P-Stop	-	C-Stop	P-Stop
	T2	P-Stop	P-Stop	-	C-Stop	P-Stop
C	AUTO	C-Stop	C-Stop	C-Stop	C-Stop	-
	T1	P-Stop	P-Stop	-	C-Stop	P-Stop
	T2	P-Stop	P-Stop	-	C-Stop	P-Stop
D	AUTO	S-Stop	S-Stop	C-Stop	C-Stop	-
	T1	S-Stop	S-Stop	-	C-Stop	S-Stop
	T2	S-Stop	S-Stop	-	C-Stop	S-Stop

P-Stop: Power-Off stop

C-Stop: Controlled stop

S-Stop: Smooth stop

-: Disable

(*) The stop pattern of NTED input is same as Deadman switch.

The following table indicates the Stop pattern according to the controller type or option configuration.

Option	R-30iB/ R-30iB Mate
Standard	A(**)
Controlled stop by E-Stop (A05B-2600-J570)	C(**)
Smooth E-Stop (A05B-2600-J651)	D(**)

(**)R-30iB Mate does not have SVOFF input.

The stop pattern of the controller is displayed in "Stop pattern" line in software version screen. Please refer to "Software version" in operator's manual of controller for the detail of software version screen.

"Controlled stop by E-Stop" option

When "Controlled stop by E-Stop" (A05B-2600-J570) option is specified, the stop type of the following alarms become Controlled stop but only in AUTO mode. In T1 or T2 mode, the stop type is Power-Off stop which is the normal operation of the system.

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.

Controlled stop is different from **Power-Off stop** as follows:

- In Controlled stop, the robot is stopped on the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Controlled stop, physical impact is less than Power-Off stop. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End Of Arm Tool) should be minimized.
- The stopping distance and time of Controlled stop is longer than those of Power-Off stop, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.

**WARNING**

The stopping distance and time of Controlled stop are longer than those of Power-Off stop. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

"Smooth E-Stop Function" option

When "Smooth E-Stop Function" (A05B-2600-J651) option is specified, the stop type of the following alarms becomes Smooth stop in all operation modes (AUTO, T1 and T2 mode).

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-003 Deadman switch released	Both deadman switches on Teach pendant are released.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-037 IMSTP input (Group: %d)	IMSTP input (*IMSTP signal for a peripheral device interface) is OFF.
SRVO-232 NTED input	NTED input (NTED1-NTED11, NTED2-NTED21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.
SRVO-410 DCS SSO NTED input	In DCS Safe I/O connect function, SSO[5] is OFF.
SRVO-419 DCS PROFIsafe comm. error	PROFINET Safety communication error occurs.

Smooth stop is different from **Power-Off stop** as follows:

- In Smooth stop, the robot is stopped along the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Smooth stop, physical impact is less than Power-Off stop. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End Of Arm Tool) should be minimized.
- The stopping distance and time of Smooth stop is longer than those of Power-Off stop, depending on the robot model and axis.

Smooth stop is different from **Controlled stop** as follows:

- The stopping distance and time of Smooth stop is normally shorter than those of Controlled stop, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.

**WARNING**

The stopping distance and time of Smooth stop are longer than those of Power-Off stop. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

9 STOP TYPE OF ROBOT (R-30iB Plus, R-30iB Mate Plus)

There are following three types of Stop Category.

Stop Category 0 following IEC 60204-1 (Power-off Stop)

Servo power is turned off, and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.

“**Stop Category 0**” performs following processing.

- An alarm is generated, and then the servo power turns off. Instantly the robot stops.
- Execution of the program is paused.

Frequent Category 0 Stop of the robot during operation can cause mechanical problems of the robot.

Avoid system designs that require routine or frequent Category 0 Stop conditions.

Stop Category 1 following IEC 60204-1 (Controlled Stop, Smooth Stop)

The robot is decelerated until it stops, and servo power is turned off.

“**Stop Category 1**” performs following processing.

- The alarm "**SRVO-199 Controlled stop**" or "**SRVO-289 Smooth Stop**" occurs along with a decelerated stop. The program execution is paused.
- An alarm is generated, and then the servo power turns off.

In Smooth stop, the robot decelerates until it stops with the deceleration time shorter than Controlled stop.

The stop type of Stop Category 1 is different according to the robot model or option configuration. Please refer to the operator's manual of a particular robot model.

Stop Category 2 following IEC 60204-1 (Hold)

The robot is decelerated until it stops, and servo power remains on.

“**Stop Category 2**” performs following processing.

- The robot operation is decelerated until it stops. Execution of the program is paused.

⚠ WARNING

- 1 The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time is necessary when Stop Category 1 is used. Please refer to the operator's manual of a particular robot model for the data of stopping distance and time.
- 2 In multi arm system, the longest stopping distance and time of Stop Category 1 among each robot are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the multi arm system.
- 3 In the system which has extended axis, the longer stopping distance and time of Stop Category 1 among robot and extended axis are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the system which has extended axis. Please refer to the extended axis setup procedure of the controller operator's manual for considering the stopping distance and time of the extended axis.
- 4 When Stop Category 1 occurs during deceleration by Stop Category 2, the stop type of robot is changed to Stop Category 0.
- 5 In case of Stop Category 1, motor power shutdown is delayed for a maximum of 2 seconds. In this case, a risk assessment for the whole robot system is necessary, including the 2 seconds delay.

When the emergency stop button is pressed or the FENCE is open, the stop type of robot is Stop Category 0 or Stop Category 1. The configuration of stop type for each situation is called *stop pattern*. The stop pattern is different according to the option configuration.

There are the following 3 Stop patterns.

Stop pattern	Mode	Emergency stop button	External Emergency stop	FENCE open	SVOFF input	Deadman switch (*)
A	AUTO	Category 0	Category 0	Category 1	Category 1	-
	T1	Category 0	Category 0	-	Category 1	Category 0
	T2	Category 0	Category 0	-	Category 1	Category 0
C	AUTO	Category 1	Category 1	Category 1	Category 1	-
	T1	Category 0	Category 0	-	Category 1	Category 0
	T2	Category 0	Category 0	-	Category 1	Category 0
D	AUTO	Category 1	Category 1	Category 1	Category 1	-
	T1	Category 1	Category 1	-	Category 1	Category 1
	T2	Category 1	Category 1	-	Category 1	Category 1

Category 0: Stop Category 0

Category 1: Stop Category 1

-: Disable

(*) The stop pattern of NTED input is same as Deadman switch.

The following table indicates the Stop pattern according to the controller type or option configuration.

The case R651 is specified.

Option	R-30iB Plus/ R-30iB Mate Plus
Standard	C(**)
Old Stop Function (A05B-2670-J680)	A(**)
All Smooth Stop Function (A05B-2670-J651)	D(**)

The case R650 is specified.

Option	R-30iB Plus/ R-30iB Mate Plus
Standard	A(**)
Stop Category 1 by E-Stop (A05B-2670-J521)	C(**)
All Smooth Stop Function (A05B-2670-J651)	D(**)

(**)R-30iB Mate Plus does not have SVOFF input.

The stop pattern of the controller is displayed in "Stop pattern" line in software version screen. Please refer to "Software version" in operator's manual of controller for the detail of software version screen.

"Old Stop Function" option

When "Old Stop Function" (A05B-2670-J680) option is specified, the stop type of the following alarms becomes Stop Category 0 in AUTO mode.

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.

Stop Category 0 is different from **Stop Category 1** as follows:

- In Stop Category 0, servo power is turned off, and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.
- The stopping distance and time of Stop Category 0 is shorter than those of Stop Category 1, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.

"All Smooth Stop Function" option

When "All Smooth Stop Function" (A05B-2670-J651) option is specified, the stop type of the following alarms becomes Stop Category 1 in all operation modes (AUTO, T1 and T2 mode).

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-003 Deadman switch released	Both deadman switches on Teach pendant are released.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-037 IMSTP input (Group: %d)	IMSTP input (*IMSTP signal for a peripheral device interface) is ON.
SRVO-232 NTED input	NTED input (NTED1-NTED11, NTED2-NTED21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.
SRVO-410 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[5] is OFF.
SRVO-419 DCS PROFIsafe comm. error	PROFINET Safety communication error occurs.

Stop Category 1 is different from **Stop Category 0** as follows:

- In Stop Category 1, the robot is stopped along the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.

- In Stop Category 1, physical impact is less than Stop Category 0. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End of Arm Tool) should be minimized.
- The stopping distance and time of Stop Category 1 is longer than those of Stop Category 0, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.



WARNING

The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

"Stop Category 1 by E-Stop" option

When "Stop Category 1 by E-Stop" (A05B-2670-J521) option is specified, the stop type of the following alarms become Category 1 Stop but only in AUTO mode. In T1 or T2 mode, the stop type is Category 0 Stop which is the normal operation of the system.

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.

Stop Category 1 is different from **Stop Category 0** as follows:

- In Stop Category 1, the robot is stopped along the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Stop Category 1, physical impact is less than Stop Category 0. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End of Arm Tool) should be minimized.
- The stopping distance and time of Stop Category 1 is longer than those of Stop Category 0, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.



WARNING

The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

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1 PREFACE

Robot Initial Setup function is the function to enter some settings for robot initial setup on personal computer (PC), and load that settings to Robot controller. Robot Initial Setup function consists the following 2 parts.

- FANUC ROBOGUIDE Robot integration setup tool (A08B-9410-J804)
This is the software on PC to enter some settings for robot initial setup. In this manual, this function is called “Robot Integration Setup Tool”.
- R-30iA/R-30iA Mate controller Robot Initial Setting Load function (A05B-2500-J980)
This is the function in R-30iA/R-30iA Mate controller to load the setting entered on PC.
In order to use this function, it is required that teach pendant of Robot controller is *i*Pendant.
- R-30iB/R-30iB Mate controller Robot Initial Setup Interface function (A05B-2600-J947)
This is the function in R-30iB/R-30iB Mate controller to load the setting entered on PC.

There is the following option for FANUC ROBOGUIDE Robot Integration Setup Tool.

- FANUC ROBOGUIDE Servo gun integration setup tool (A08B-9410-J805)
This is the software on PC to enter some settings for servo gun setup in addition to robot initial setup.
In this manual, this function is called “Servo Gun Integration Setup Tool”.

In order to load settings created by Servo Gun Integration Setup Tool, it is required the following option in R-30iA controller.

- R-30iA controller Servo gun setup package (A05B-2500-J979)

In order to load settings created by Servo Gun Integration Setup Tool, it is required the following option in R-30iB controller.

- R-30iB controller Servo gun setup Interface package (A05B-2600-J890)

In this chapter, the functional overview of Robot Initial Setup function and the system requirement are described.

1.1 FUNCTIONAL OVERVIEW

This software is used to enter several settings for robot initial setup. The settings entered on PC are filed and loaded to Robot controller. In this PC tool, the following settings can be entered on PC.

<Function setting>

- Tool Frame
- User Frame
- Jog Frame
- Frame Number
- Air Pressure Abnormal Detection
- Hand Broken Detection
- Reference Position
- Soft Float
- Payload

- System Configuration
- Register
- Register Comment
- Position Register Comment
- Macro
- User Alarm

<I/O setting>

- I/O Configuration
- I/O Comment
- I/O Polarity, Complementary

The setting data entered on PC tool is output to a file. Then, the file is transferred to Robot controller by the memory card or the communication. Robot controller is read setting data entered on PC by Initial Setting Load function.

NOTE

About soft Float setting, Robot Integration Setup Tool does not support High Performance Soft Float (A05B-2500-J898).

NOTE

In R-30iA/R-30iA Mate/R-30iB/R-30iB Mate controller, soft float function is option. In case that soft float option is not available in R-30iA/R-30iA Mate /R-30iB/R-30iB Mate controller, the setting for soft float is not loaded.

NOTE

For detail of each setting data, please refer to operator's manual of R-30iA/R-30iA Mate/R-30iB/R-30iB Mate controller.

In case that Servo Gun Integration Setup Tool option is installed in PC and Servo Gun setup package is installed in R-30iA controller or Servo Gun setup interface package installed in R-30iB controller, the following settings for servo gun can be set by this function.

- Servo Gun function setting
- Servo Gun schedule setting
- Servo Gun axis setting

Robot Integration Setup Tool and Servo Gun Integration Setup Tool have the following features.

1. Input/Output function of setting file.
The setting items entered on PC can be output as the setting file (PRM file and DT file). You can do setting operation easily by loading output file to Robot controller. And you can confirm and edit the contents of the setting file that has been output from PC or Robot controller.
2. Save setting data to the file described by XML format.
In PC tool, you can save the setting data to the file described XML format. This XML file has all data entered on PC. You can reload and edit this XML file by this PC tool. This XML file is not the file to load to Robot controller.

1.2 SYSTEM REQUIREMENT

Robot Integration Setup Tool and Servo Gun Integration Setup Tool can be executed on the following operating system.

- Microsoft Windows XP®
- Microsoft Windows Vista®
- Microsoft Windows 7®

This manual does not describe the basic, common operations of Windows®. If you use Windows® for the first time, please read the manual for Windows® first to learn its basic operations.

System requirements of PC for Robot Integration Setup Tool and Servo Gun Integration Setup Tool are as follows.

Table 1.2 System requirements

Operating System	Windows XP® with Service Pack 2 or later (Japanese, English) Windows Vista® with Service Pack 1 or later (Japanese, English) Windows 7® (Japanese, English)
CPU	Pentium 4 (3.0GHz or over recommended)
Memory	Minimum: 256MB of system memory Recommended: Over 1GB
Hard Disk	30 MB of free hard disk space
Display	1024 x 768 screen resolution

Note that Microsoft® Virtual PC and VMware® Workstation are not supported.

* Windows, Windows XP, Windows Vista, Windows 7 are registered trademark of Microsoft in United States.

* VMware is trademarks or registered trademarks of VMware, Inc. in the United States and/or various jurisdictions.

* Pentium® is registered trademark of Intel in United States.

* VS-FlexGrid Pro 8.0J Copyright (C) 2001-2007 ComponentOne LLC.

2 INSTALL AND UNINSTALL

The procedure to install and uninstall this software is as follows.

2.1 INSTALLATION REQUIREMENTS

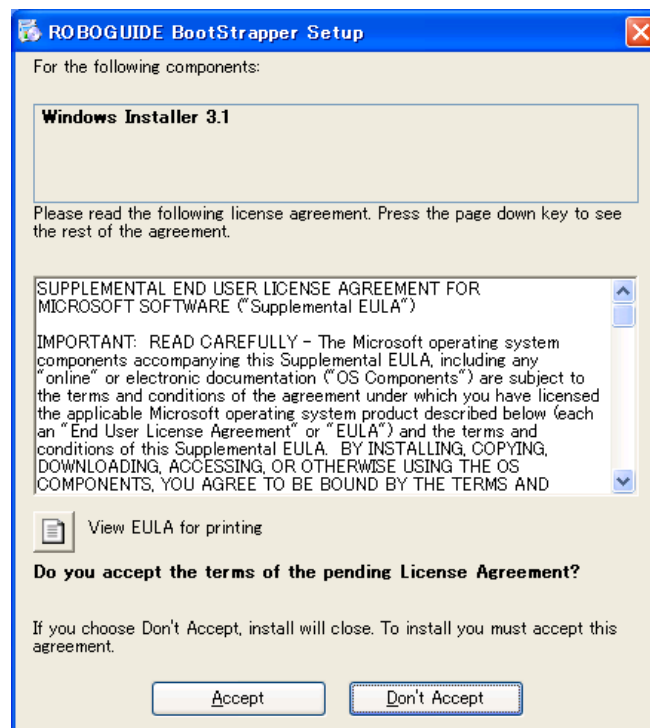
- Robot Integration Setup Tool V7.70 (or later) installer cannot replace its V1.0.0 installation. Please uninstall Robot Integration Setup Tool V1.0.0 prior to V7.70 installation. (Refer to 2.3.2 Uninstall)
- Servo Gun Integration Setup Tool requires exactly the same version of Robot Integration Setup Tool is installed on the target PC.

2.2 ROBOT INTEGRATION SETUP TOOL INSTALL

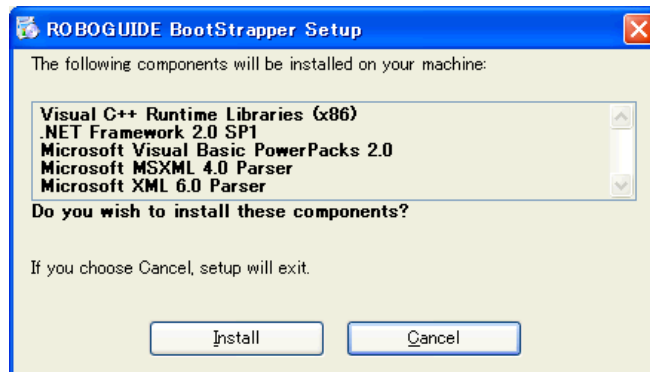
Insert the CD-ROM and the installer launched automatically. (If not, launch setup.exe in the CD-ROM.) The installer might show different screens depending on installed components. Please follow the guidance on the screen.

2.2.1 Requisites

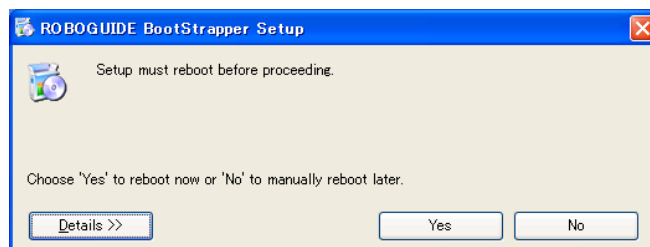
- (1) Before you install, you must have the pre-requisite software installed. If all pre-requisites are installed, please skip this section and follow the steps from 2.2.2.
Following screen is showed. Follow the navigation to proceed.



- (2) Pre-requisites to be installed are shown. Click "Install".



- (3) This might requires reboot your PC.



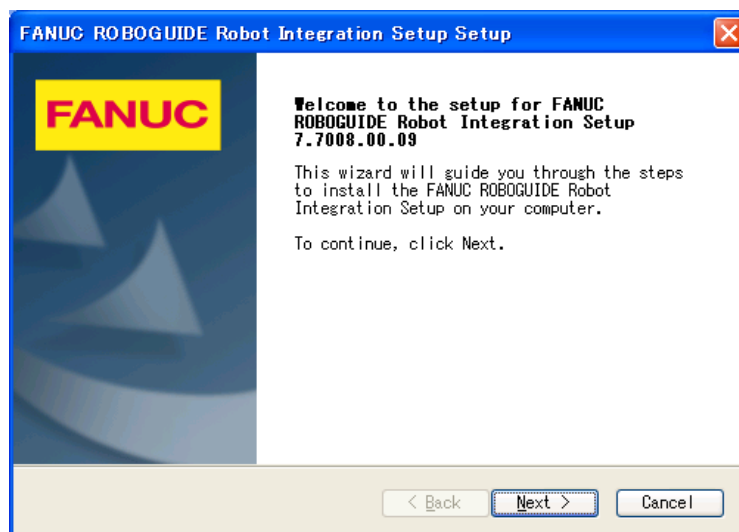
- (4) Click "Yes" to proceed.



- (5) After power-cycle, the setup will automatically continue. If not, launch setup.exe in the CD-ROM.

2.2.2 Robot Integration Setup Tool

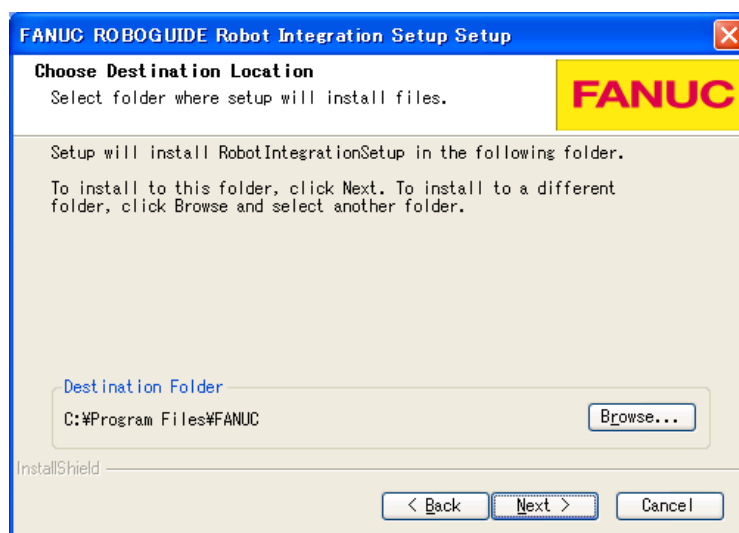
- (1) If all pre-requisites are installed, the following screen will be displayed.



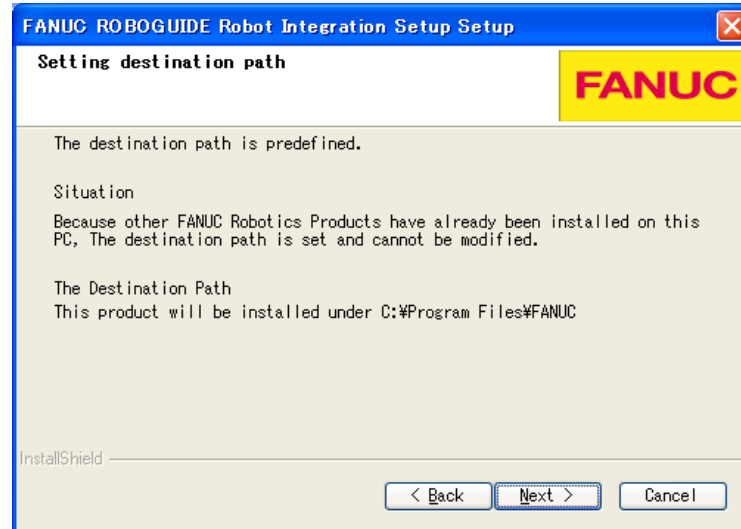
If the previous version, V1.0.0, of Robot Integration Setup Tool exists on the target PC, the following dialog will pop up. Please uninstall this previous version first.



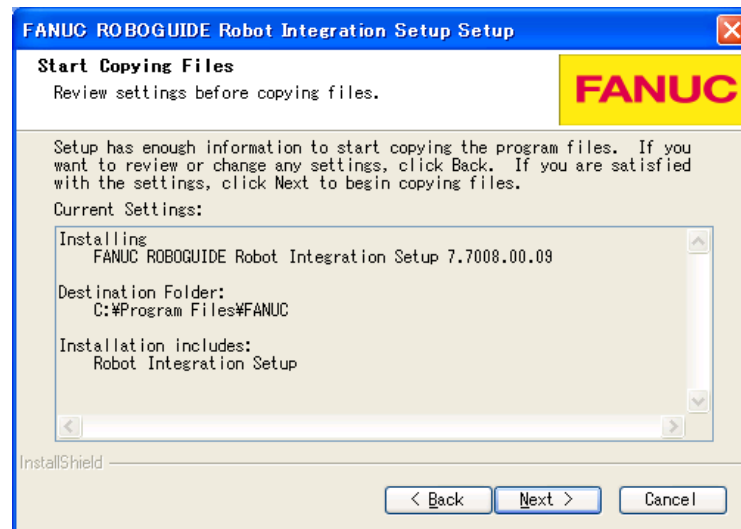
- (2) Click "Next>" to proceed.



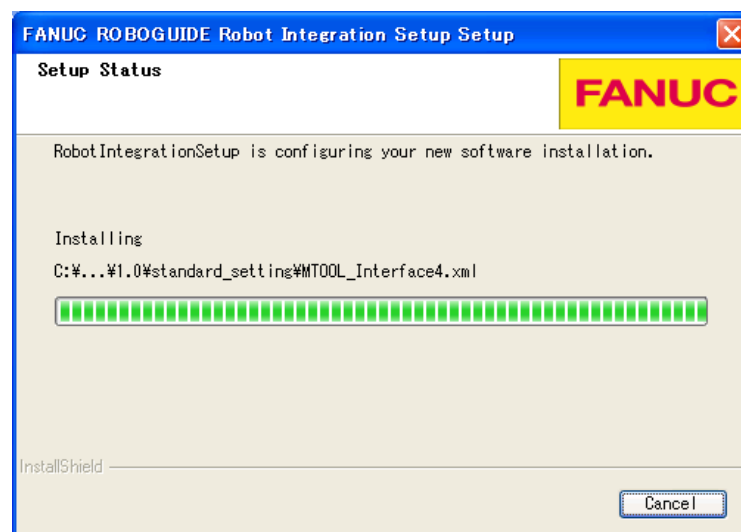
Select destination location if you want. If any of FANUC PC products are installed, this destination is fixed and cannot be changed. In this case the following screen is showed instead.



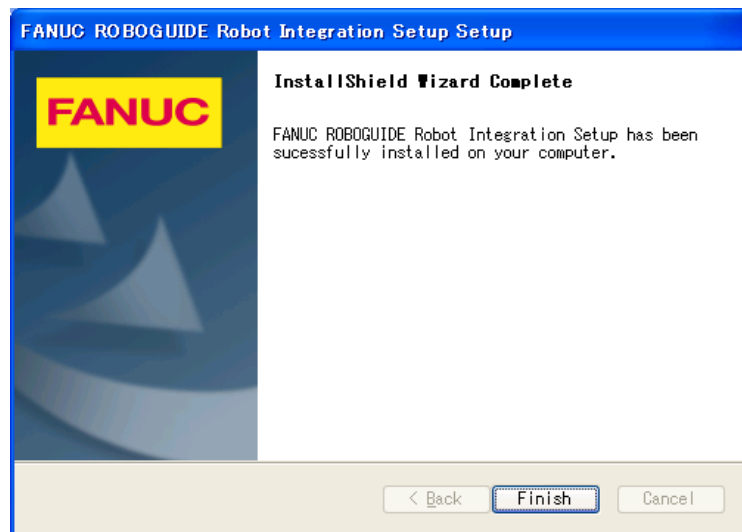
- (3) Click "Next>". Confirmation screen is showed.



- (4) Click "Next>" to proceed.



- (5) The following screen is showed at the end of installation.



- (6) Click "Finish".

2.2.3 Uninstall

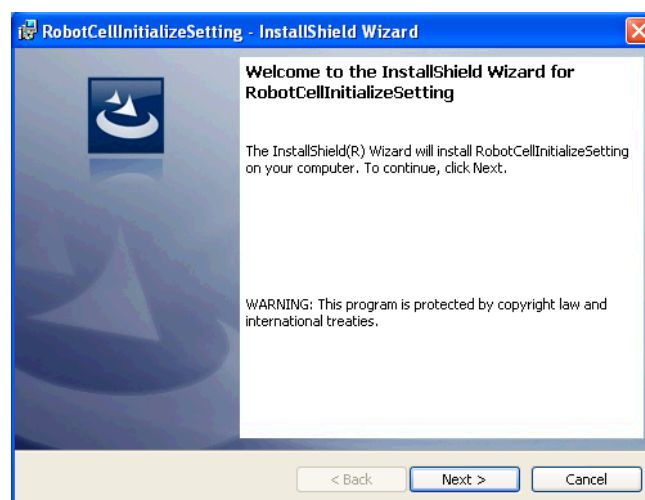
To uninstall this product, use "Add or Remove Programs" in Control Panel and select "FANUC ROBOGUIDE Robot Integration Setup".

2.3 ROBOT INTEGRATION SETUP TOOL (V1.0.0)

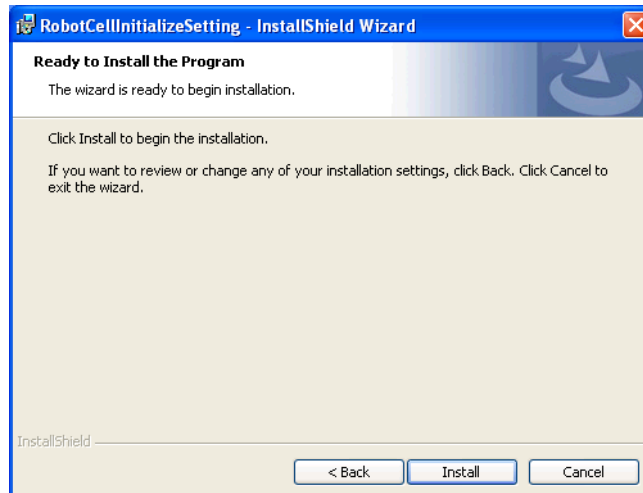
2.3.1 Install

Installation of Robot Integration Setup Tool V1.0.0 will be described here.

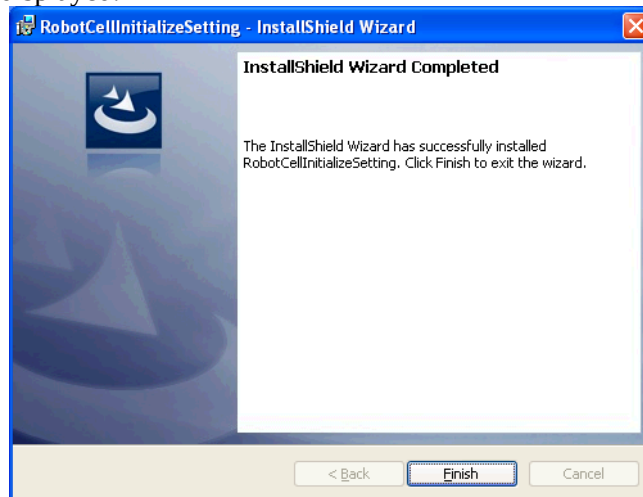
- (1) Insert the CD-ROM. The following screen is displayed. Install this software according to the instruction on the screen. (It is possible to display the different screen by OS or the setting on PC.)



- (2) Click 'Next' button in this screen.



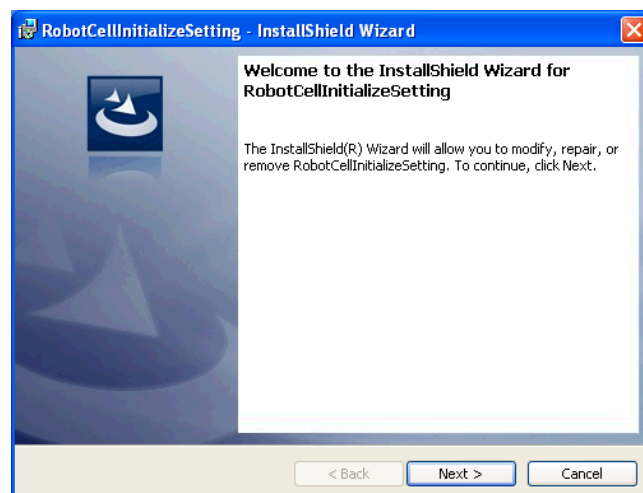
- (3) Click 'Install' button to start the software install. After a while, software install finishes, and the following screen is displayed.



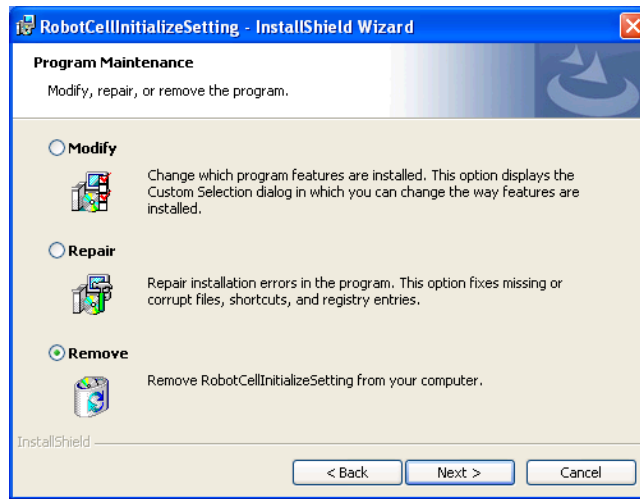
- (4) Click 'Finish' button to finish the software install operation.

2.3.2 Uninstall

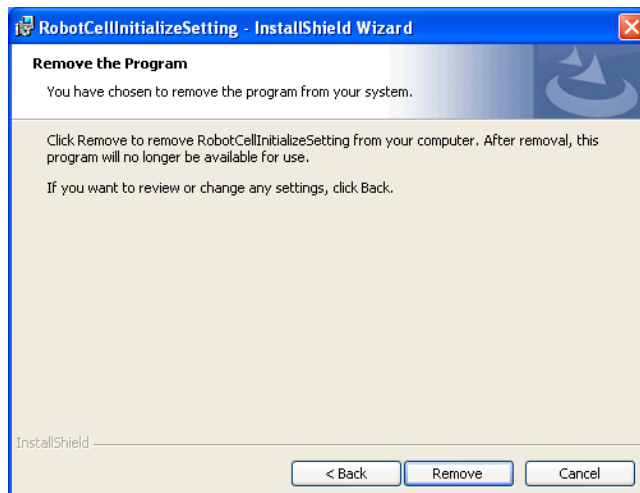
- (1) Run Setup.exe in CD ROM.
The following screen is displayed. Uninstall the software according to the instruction on the screen.



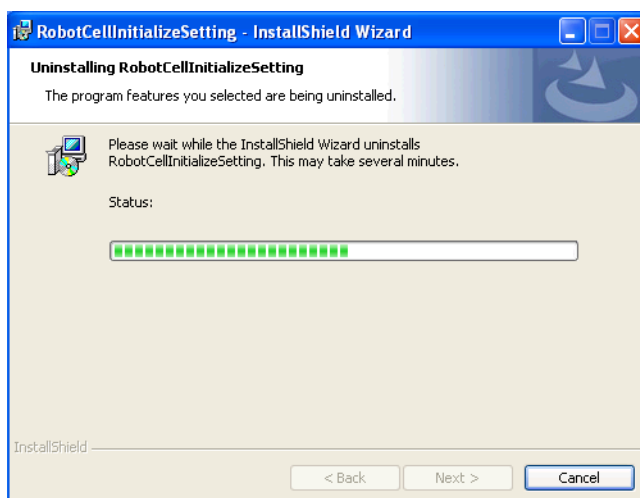
- (2) Click 'Next' button. The following screen is displayed.



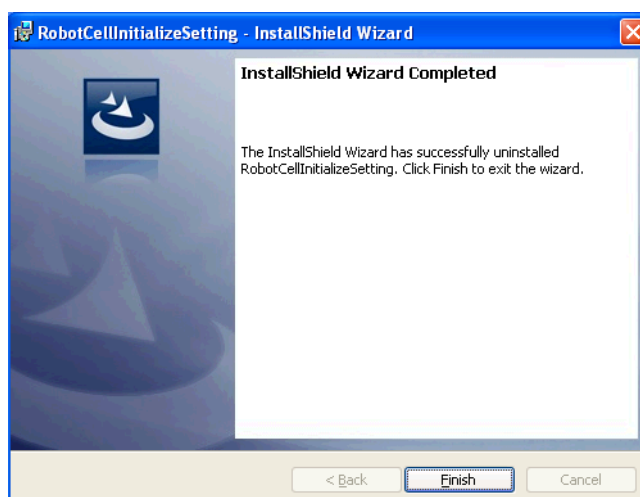
- (3) Check 'Remove' and click 'Next' Button in this screen.



- (4) Click 'Remove' button to start the software uninstall.



- (5) After the following screen is displayed, click 'Finish' button to finish the software uninstall operation.

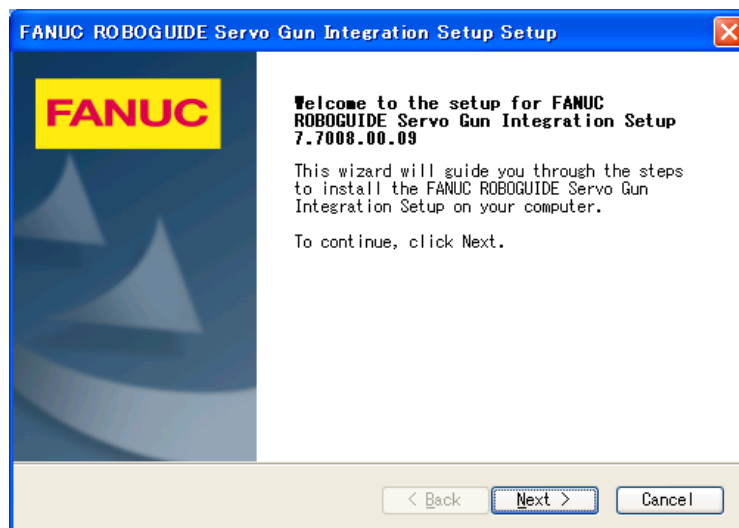


2.4 SERVO GUN INTEGRATION SETUP TOOL

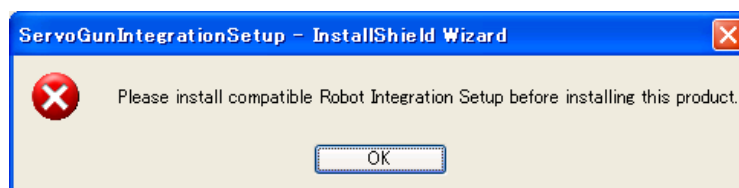
2.4.1 Install

Servo Gun Integration Setup Tool requires the Robot Integration Setup Tool is installed.

- (1) Insert the CD-ROM and launch setup.exe inside. Please follow the navigation. (Screen may be different depending on your OS or configuration)



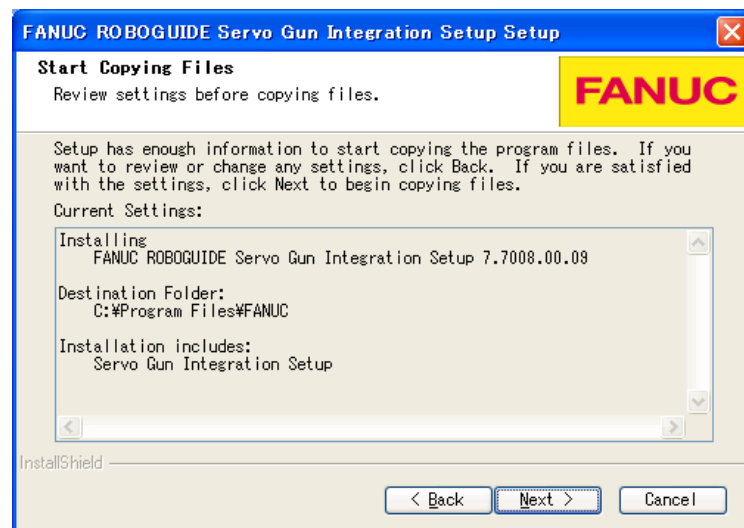
If the exactly same version of Robot Integration Setup Tool is not found on the target PC, the following dialog will popup. Install the Robot Integration Setup Tool first.



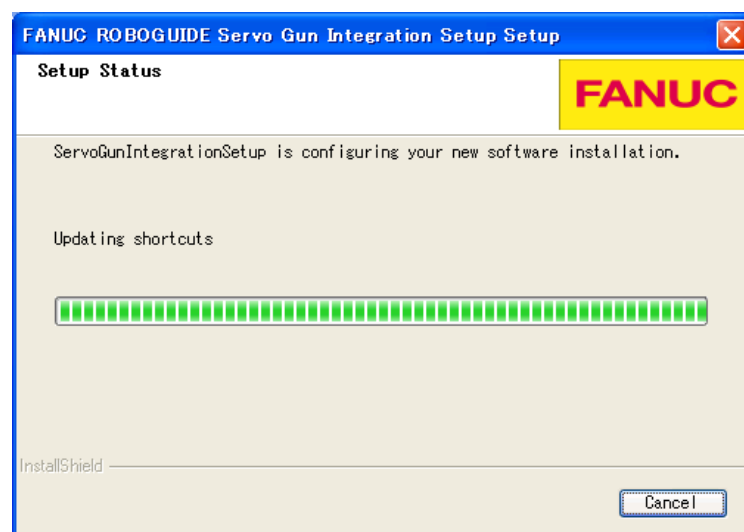
- (2) Click "Next>". The destination will be displayed.



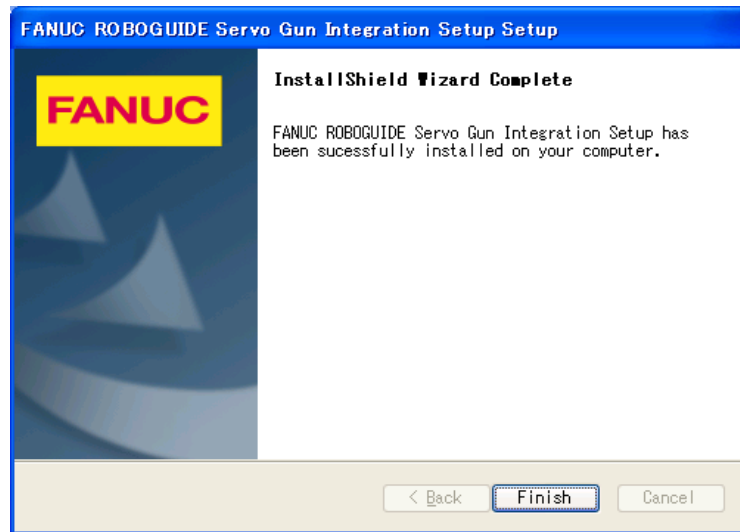
- (3) Click "Next>". Confirmation screen is showed.



- (4) Click "Next>" to proceed.



- (5) The following screen is showed at the end of installation.



- (6) Click "Finish".

2.4.2 Uninstall

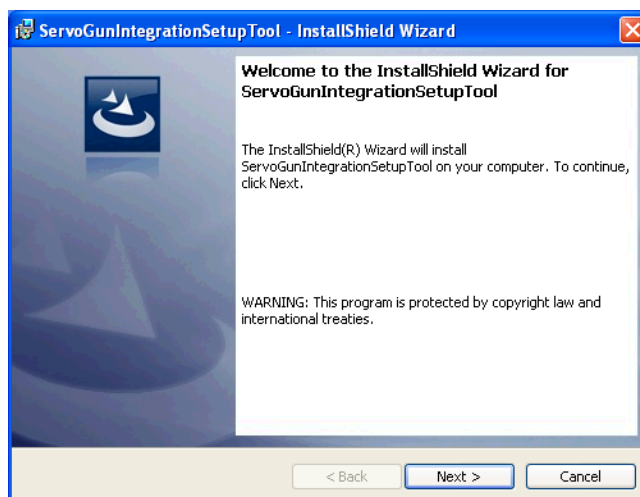
To uninstall this product, use "Add or Remove Programs" in Control Panel and select "FANUC ROBOGUIDE Servo Gun Integration Setup".

2.5 SERVO GUN INTEGRATION SETUP TOOL (V1.0.0)

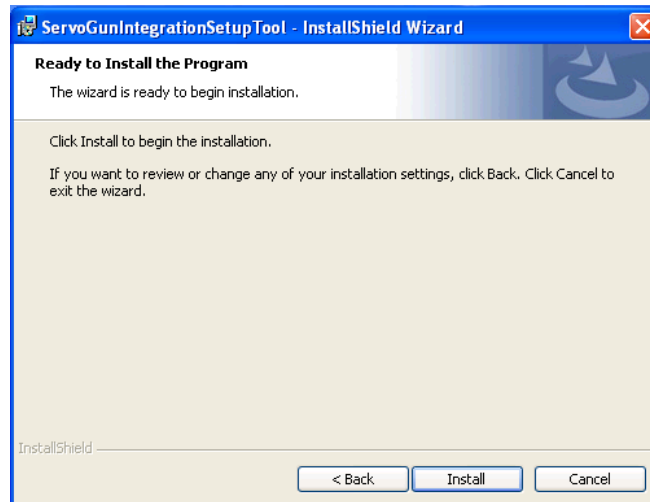
2.5.1 Install

To Install Servo Gun Integration Setup Tool, Servo Gun Integration Setup Tool option (A08B-9410-J805) is needed, and it is required to install Robot Integration Setup Tool in advance.

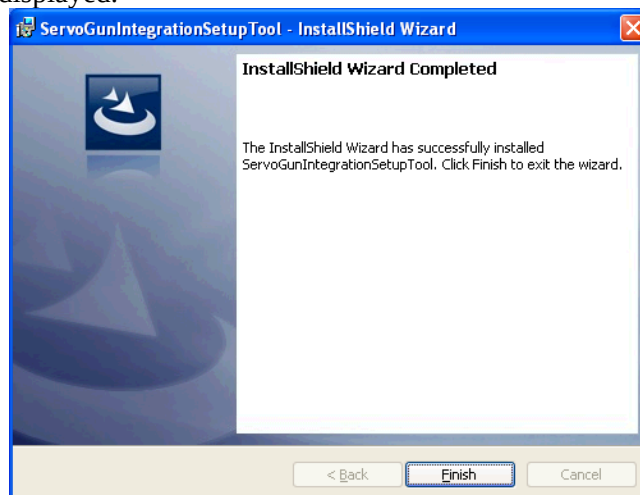
- (1) Run Setup.exe for Servo Gun Integration Setup Tool in CD ROM.
The following screen is displayed. Install this software according to the instruction on the screen.
(It is possible to display the different screen by OS or the setting on PC.)



- (2) Click 'Next' button in this screen.



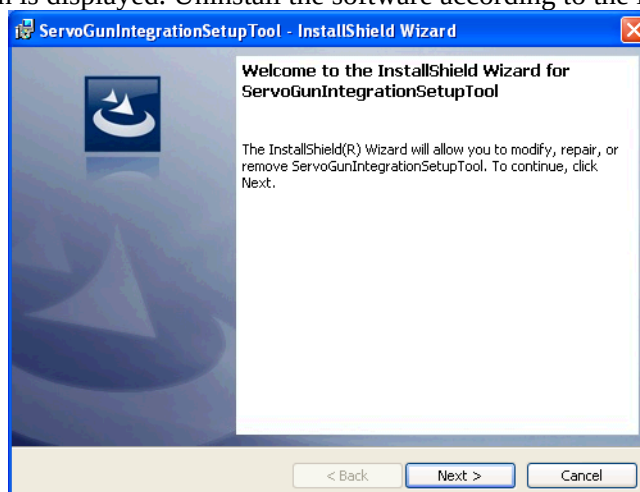
- (3) Click 'Install' button to start the software install. After a while, software install finishes, and the following screen is displayed.



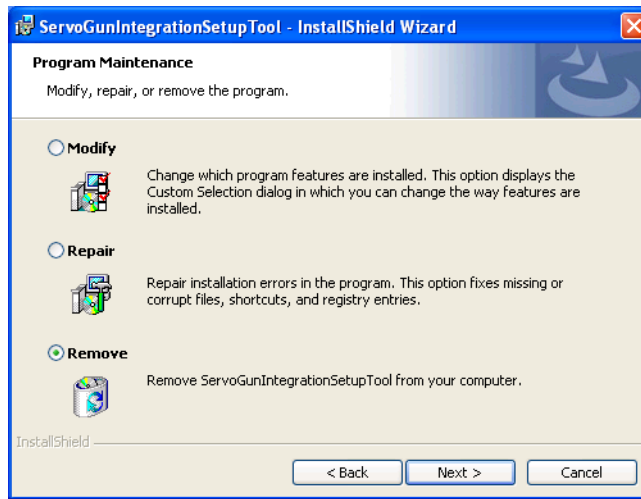
- (4) Click 'Finish' button to finish the software install operation.

2.5.2 Uninstall

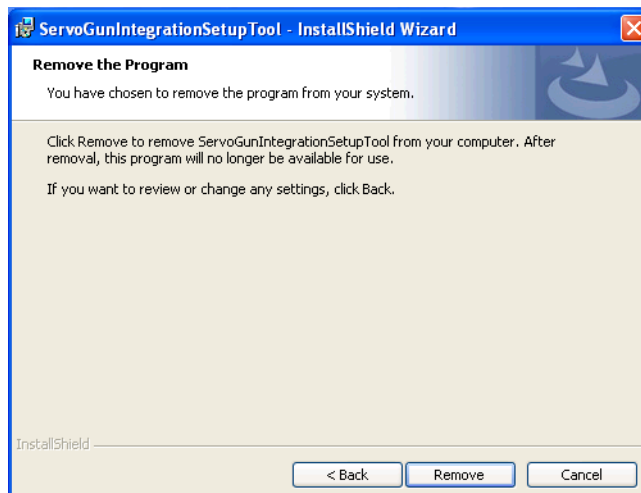
- (1) Run Setup.exe for Servo Gun Integration Setup Tool in CD ROM. The following screen is displayed. Uninstall the software according to the instruction on the screen.



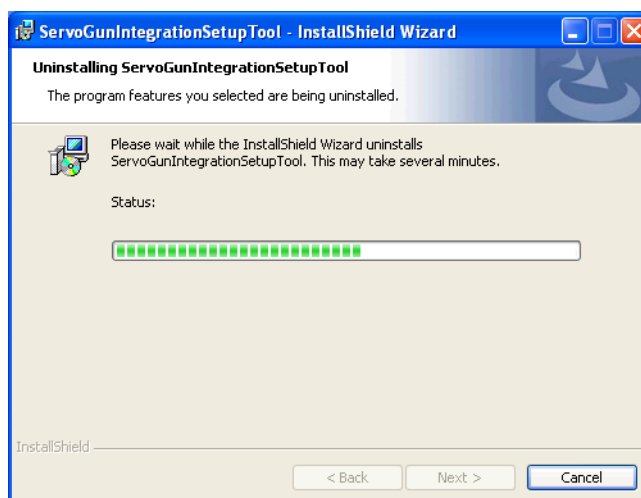
- (2) Click 'Next' button. The following screen is displayed.



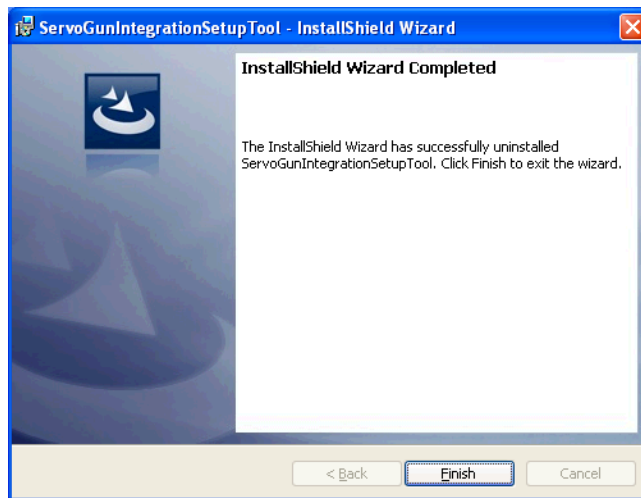
- (3) Check 'Remove' and click 'Next' Button in this screen.



- (4) Click 'Remove' button to start the software uninstall.



- (5) After the following screen is displayed, click 'Finish' button to finish the software uninstall operation.



2.6 LICENSING

You need registration to use this product. Please open license manager ([Start] -> Fanuc Robotics -> License Manager) and send displayed Part number, software code 1 and software code2 to FANUC sales. You will get registration key REG1 and REG2, then input them in the license screen to register.

You can use this product for 30 days without registration.

3 ROBOT INITIAL SETUP OPERATION

In this chapter, the operations for Robot Integration Setup Tool on PC are described.

3.1 OPERATION FLOW

The operation flow for robot initial setting is as follows.

1. In the main screen of Robot Integration Setup Tool, set the configuration of the robot cell.
In Robot Integration Setup Tool, set the data with respect to each robot cell. In the main screen, set the cell name and the configuration of a robot cell. For details, refer to 3.2 MAIN SCREEN.
2. In the detail screen, set the data for each machine in the robot cell.
In the detail screen, set the data for each machine in the robot cell. In case that the machine type is Robot controller, there are 'Function setting' and 'I/O setting' in the detail screen. In case that the machine type is another, there is only 'I/O setting' in the detail screen. For details, refer to 3.3 DETAIL SCREEN.
3. Output the settings to the initial setting file on the PC tool.
Output the setting for Robot controller to the initial setting file in order to load the setting to Robot controller. The output operation of the setting data can be executed in the main screen. For details, refer to 3.2.4.1 Output initial setting file.

3.2 MAIN SCREEN

In the main screen, set the following data of the robot cell.

Table 3.2 Setup Items in Main Screen

Items	Contents
Robot cell name	Set the robot cell name without fail, because the output file is handled by the cell name.
Robot cell comment	Set the comment about the robot cell. This setting can be omitted.
Machines in the robot cell	Set the machines in the robot cell. And set the machine type for each machine. In case that the machine type is Robot controller, set the number of group.

The operations in the main screen are as follows.

1. Add new line to the table by 'Add' button (6), then set the machine name (9) and select the machine type (10).
2. In case that the machine type is Robot controller, select the number of the group (11).
3. Click the 'Detail' button (8) to display the detail screen (3.3 DETAIL SCREEN) for the selected line.

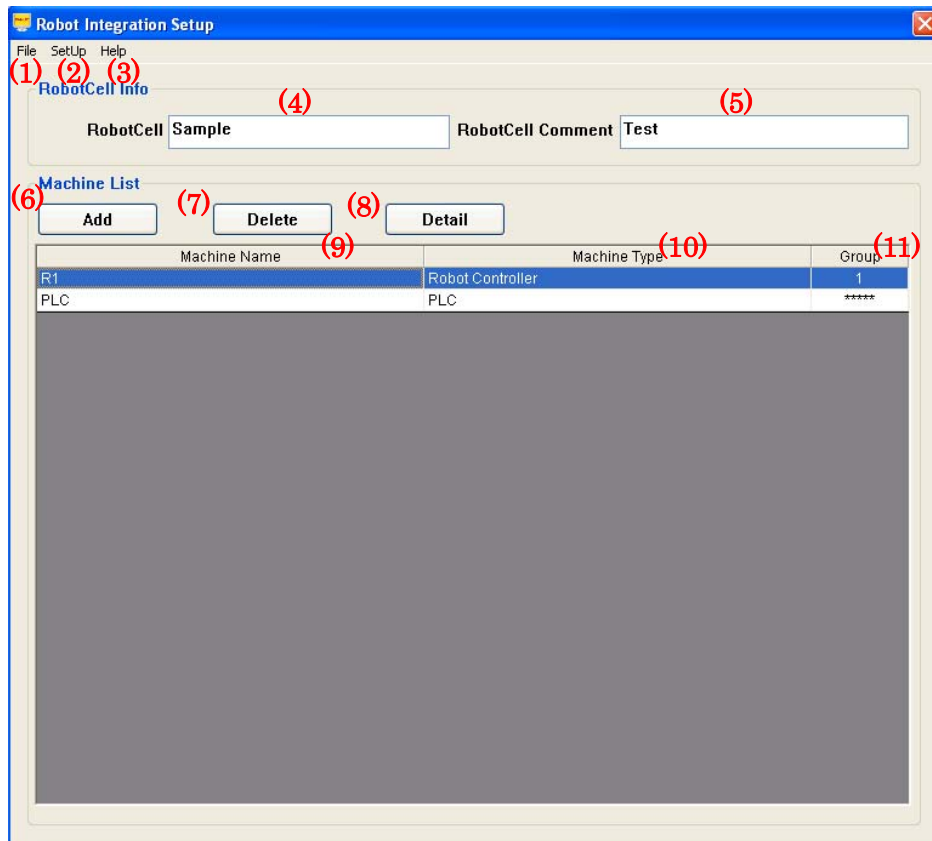


Fig. 3.2 Main screen

Followings are the descriptions for each part in the main screen.

- (1) 'File' menu
 - New All machines in the table are deleted and new robot cell is created.
 - Open Saved file is opened. Refer to 3.2.1 Open.
 - OverWrite Setting data are saved. Refer to 3.2.2 Overwrite.
 - Save Setting data are saved to other file. Refer to 3.2.3 Save.
 - Output Setting data are output to initial setting file. Refer to 3.2.4 Output.
 - Input Initial setting file is loaded. Refer to 3.2.5 Input.
 - Convert CSV Cannot use.
 - Import Standard Setting Registered setting are loaded. Refer to 3.2.7 Import Standard setting.
 - Servo Gun Servo Gun Integration Setup Tool is started. Refer to 3.2.8 Start Servo Gun Integration Setup Tool.
 - End Robot Integration Setup Tool is finished.
- (2) 'Setup' menu
 - Initial Setting File Directory Directory to output initial setting file is designated. Refer to 3.2.9 Initial Setting File Directory.
 - Work File Directory Directory to output work file is designated. Refer to 3.2.10 Work File Directory.
 - Change Language The language is changed. Refer to 3.2.11 Change Language.
 - Equipment Type Edit Equipment type is added and removed. Refer to 3.2.12 Machine Type Edit.
- (3) 'Help' menu
 - Version Version information is displayed. Refer to 3.2.13 Version information.
- (4) 'RobotCell Name' input box
 - Input Robot Cell Name
- (5) 'RobotCell Comment' input box

- Input the comment for Robot Cell
- (6) 'Add' button
 - Add new machine in Robot Cell by this button.
- (7) 'Delete' button
 - Remove the selected machine in Robot Cell.
- (8) 'Detail' button
 - Display the detail screen for the selected machine.
- (9) 'Machine Name'
 - Input the machine name.
- (10) 'Machine Type'
 - Select machine type from 'Robot Controller', 'PLC', 'Machine Tool', 'Jig', and 'Hand'. Machine type can be added in Equipment Type Edit screen in Setup menu.
- (11) 'Group'
 - In case that the machine type is Robot controller, set the number of group. According to the setting of the number of group, the detail screen is changed. In case that the machine type is another, it is not need to set the number of group.

3.2.1 Open

1. Select 'Open' in File menu. (1)
2. 'Open' window is displayed.
3. Select XML file for Robot Integration Setup Tool. (2)
4. Click 'Open' button to start loading the selected file. (3)

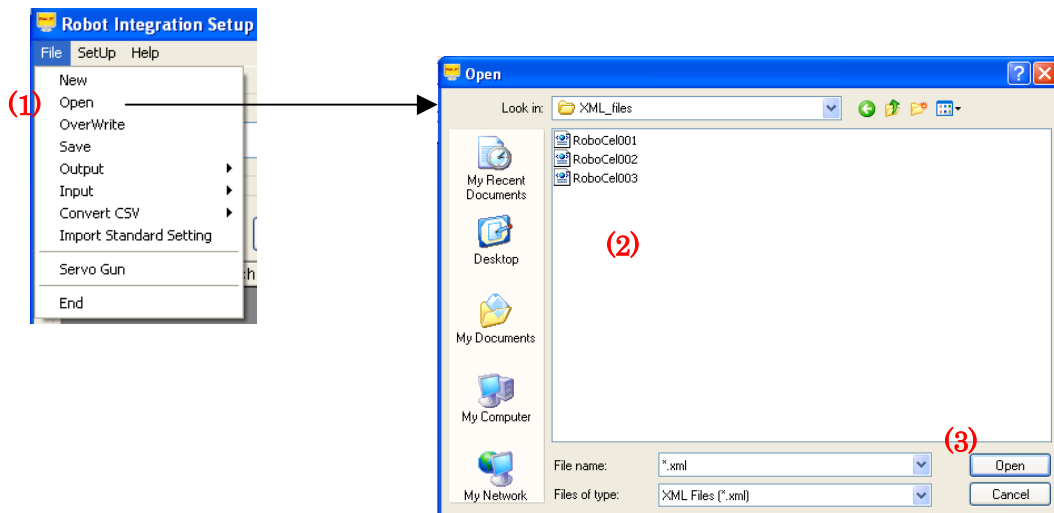


Fig. 3.2.1 Open

3.2.2 Overwrite

1. Select 'OverWrite' in File menu. (1)
2. In case that the saved file exists, the dialog to confirm overwriting the existing file (2) is displayed. In case that the saved file is not exist, the dialog to confirm creating new file (3) is displayed.
3. Save the setting data to the file.

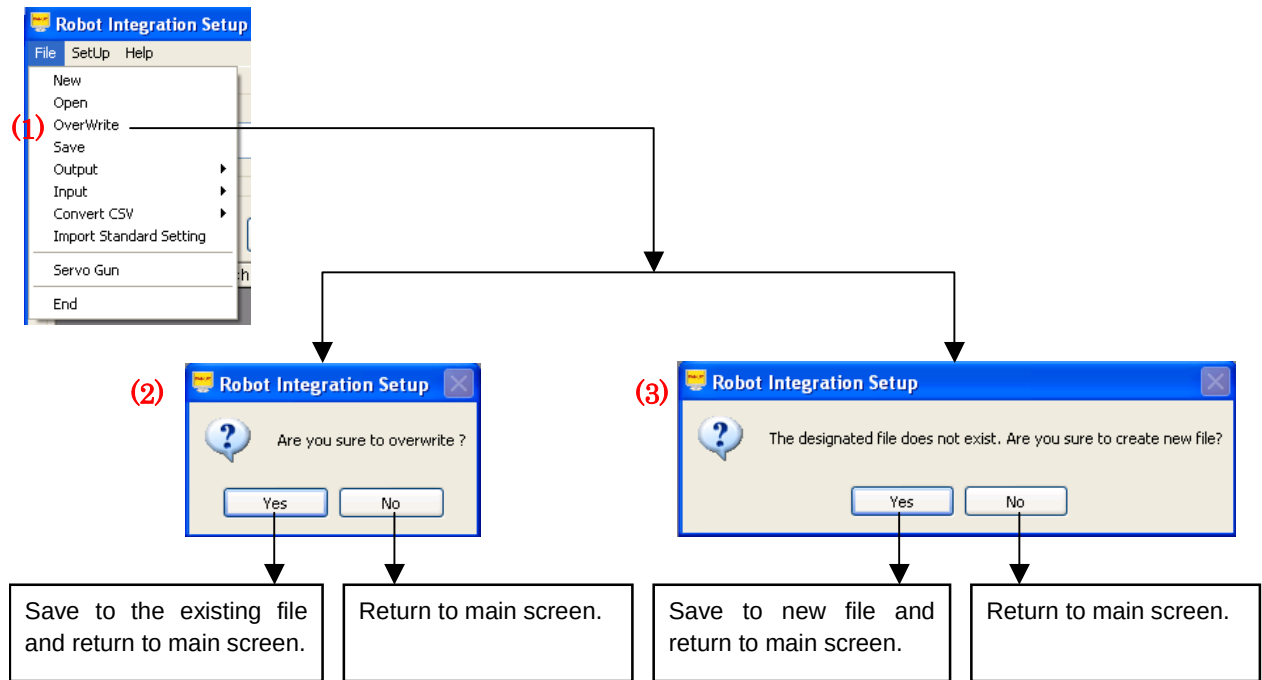


Fig. 3.2.2 Overwrite

3.2.3 Save

1. Select 'Save' in File menu. (1)
2. 'Save As' window is displayed.
3. Input new file name to 'File name' input box. (2)
4. Click 'Save' button. (3)

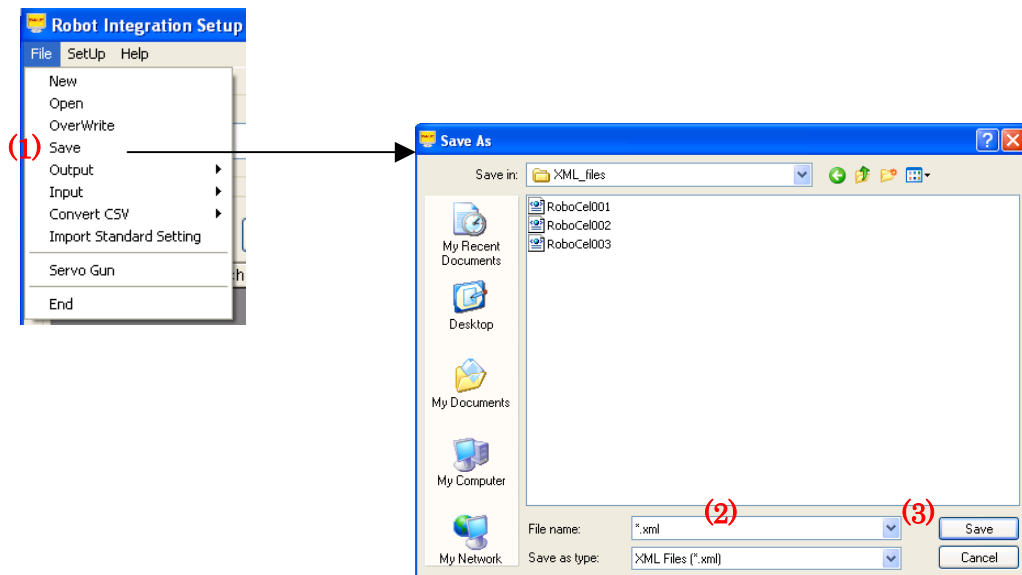


Fig. 3.2.3 Save

3.2.4 Output

There are two types as the output file. One is the initial setting file and another is the work file.

- Initial setting file is the ASCII format file. This file is used to load the data set on PC tool to Robot controller.
- Work file is the CSV format file. This file is used to refer the I/O setting in order to describe the document of the work instruction or the operation manual for the system.

3.2.4.1 Output initial setting file

1. Select 'Output' in File menu and select 'Initialization File Output' in the Output menu. (1)
2. Initial setting file is output to the directory designated in 3.2.9 Initial Setting File Directory. (2)

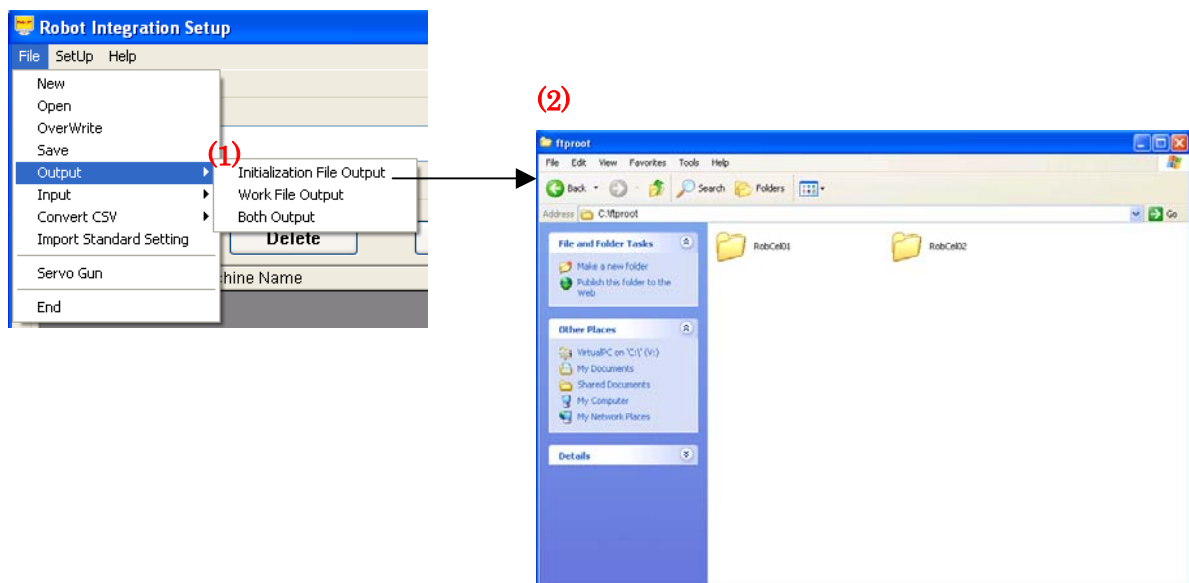


Fig. 3.2.4.1 (a) Initial setting file output

The initial setting file is handled with 'Robot Cell Name' set in main screen. In case that the file named same Robot Cell Name is already exists, the initial setting file is overwritten. In this case, the following dialog is displayed. If you want not to overwrite the file, click 'No' to abort output operation, and change the Robot Cell Name in main menu.

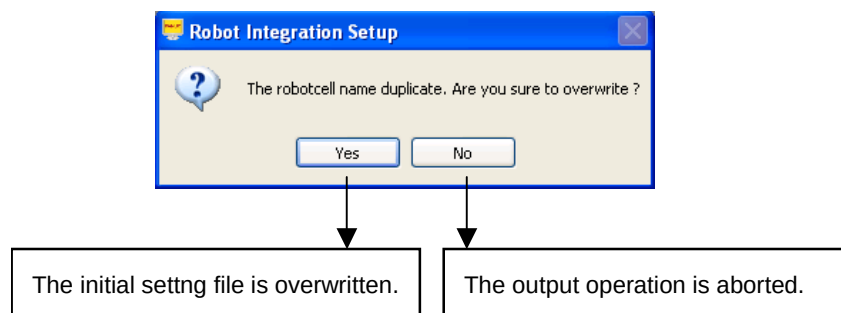


Fig. 3.2.4.1 (b) Initialization file output (Overwrite)

3.2.4.2 Output work file

1. Select 'Output' in File menu and select 'Work File Output' in the Output menu. (1)
2. Work File is output to the directory designated in 3.2.10 Work File Directory. (2)

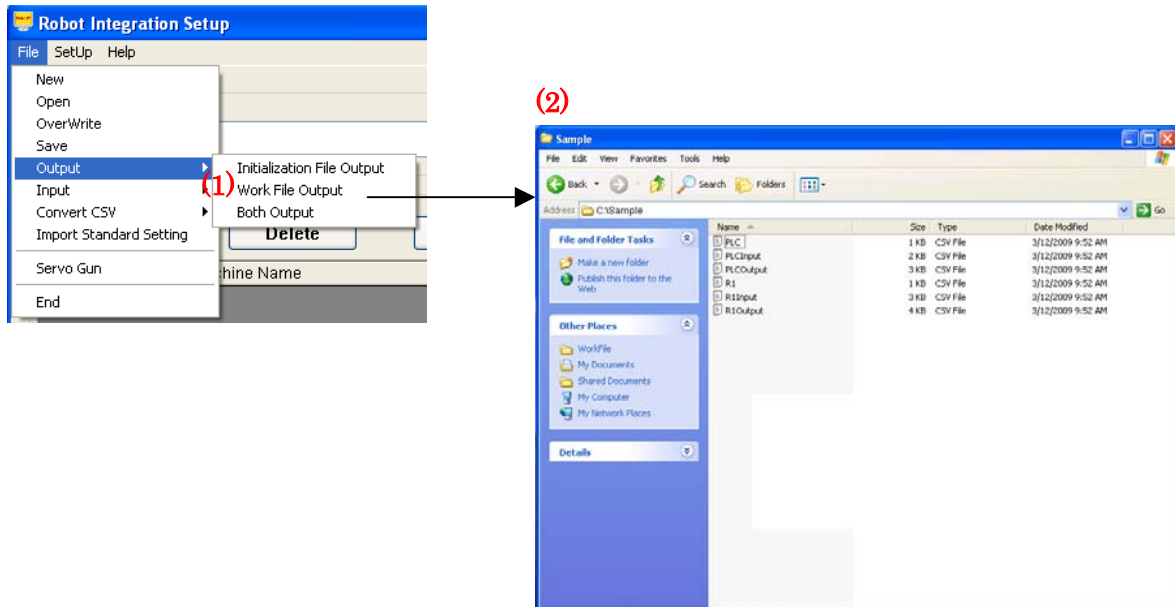


Fig. 3.2.4.2 Work file output

3.2.4.3 Both output

1. Select 'Output' in File menu and select 'Both Output' in the Output menu.
2. Both 3.2.4.1 Output initial setting file and 3.2.4.2 Output work file are executed.

3.2.5 Input

Both the initial setting file and the work file are able to input to Robot Integration Setup Tool.

3.2.5.1 Input initial setting file

1. Select 'Input' in File menu and select 'Initialization File Reading' in Input menu. (1)
2. 'Open' window is displayed. (2)
3. Select Initial Setting File (.PRM) that you want to input.
4. Initial setting file is read. (3)

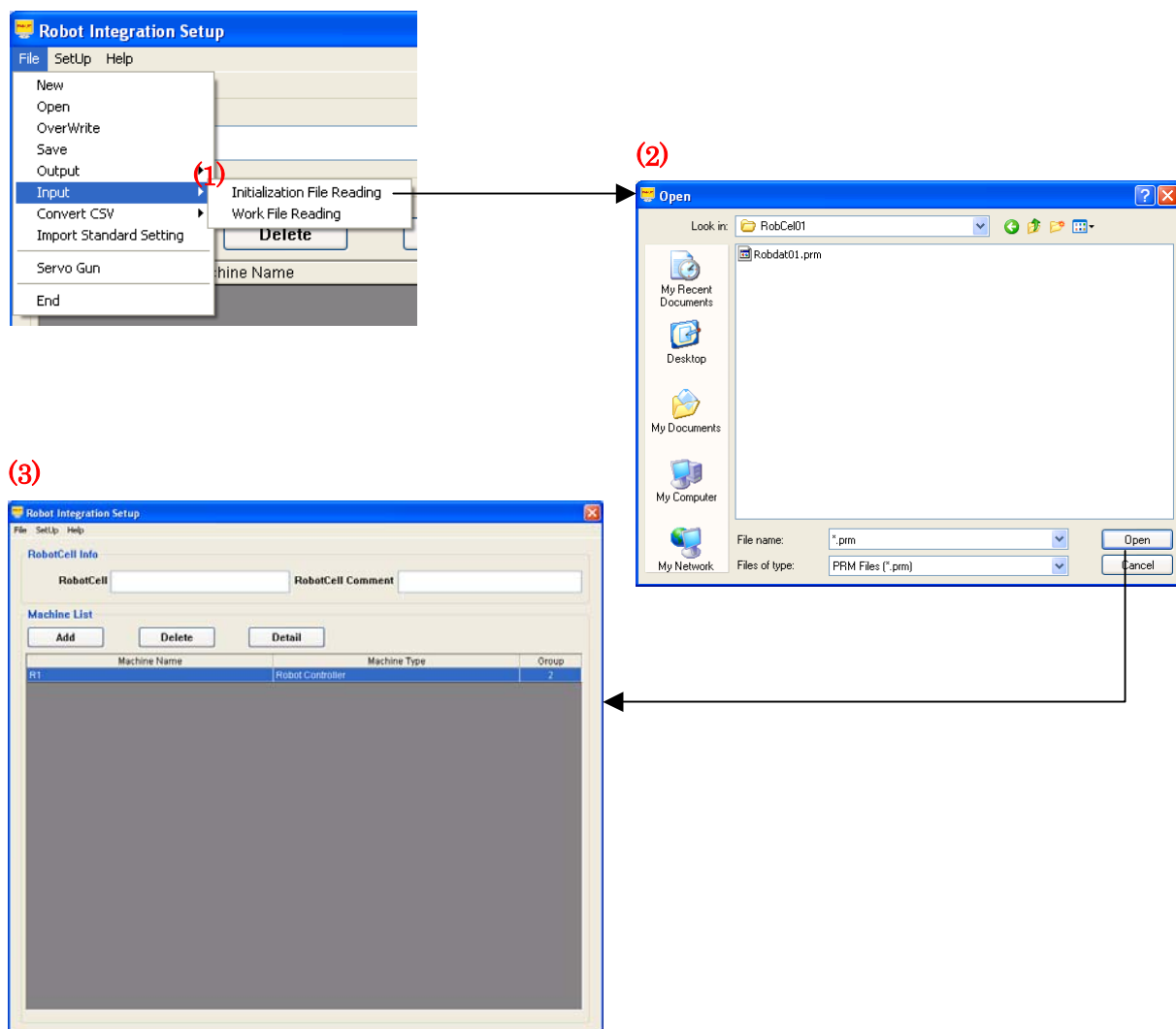


Fig. 3.2.5.1 Initial setting file input

3.2.5.2 Input work file

1. Select 'Input' in File menu and select 'Work File Reading' in Input menu. (1)
2. 'Open' window is displayed. (2)
3. Select work file that you want to input, and click 'Open' button.
4. 'Confirmation of Work File' screen is displayed. (3)
5. When you want to input another work file, click 'Select File' button. (4)
6. 'Open' window is displayed again. Select new work file to input.
7. If 'Close' button is clicked, 'Confirmation of Work File' screen is closed. (5)

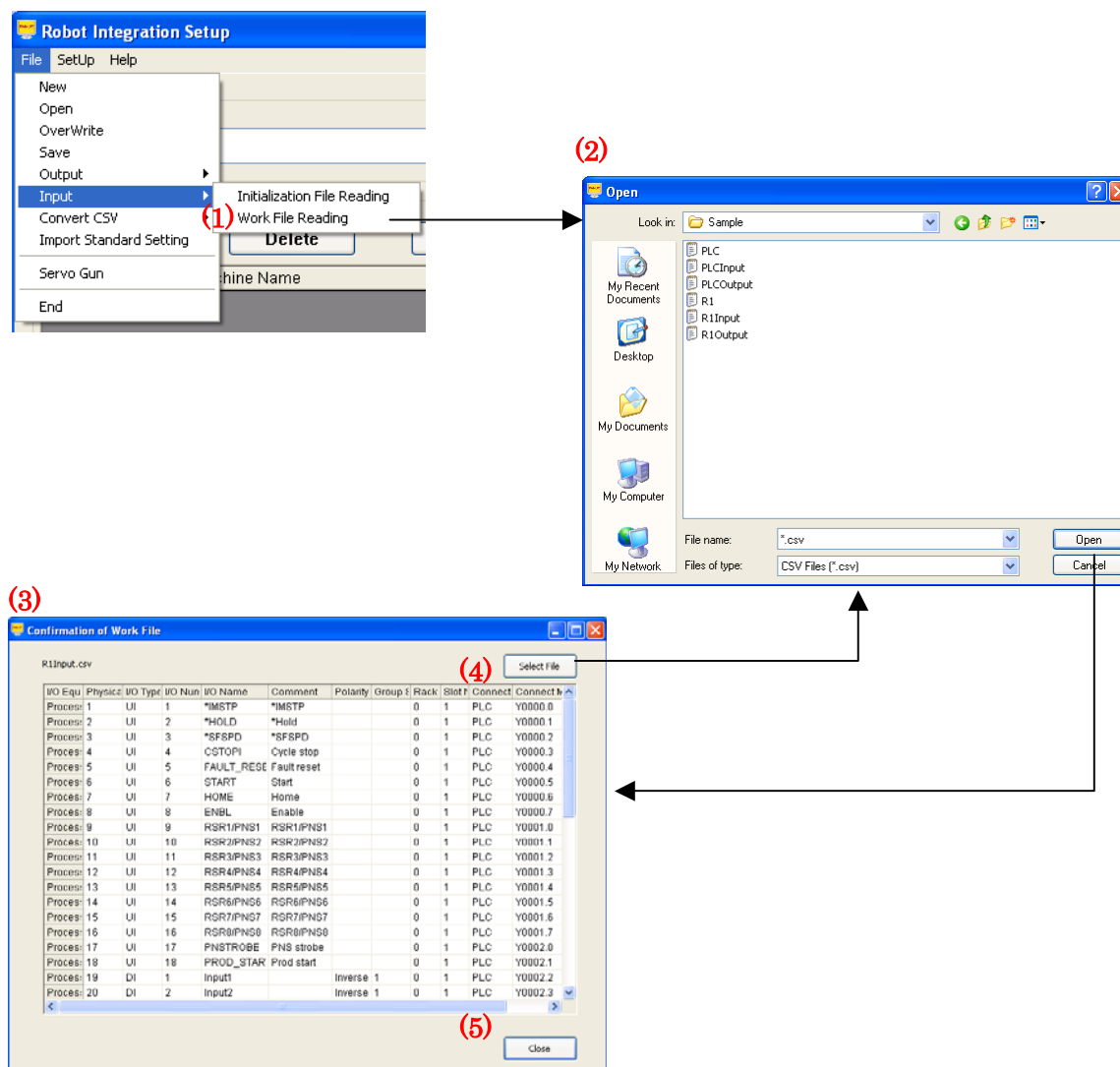


Fig. 3.2.5.2 Work file reading

3.2.6 CSV File Conversion

Import from CSV file is not currently supported.

3.2.7 Import Standard Setting

The standard settings for Machine tool connection function are registered in Robot Integration Setup Tool. The standard setting for Machine tool connection is the setting for the system that Robot and Machine tool are connected using PMC function in Robot controller. In case that same system is set up, this standard setting is useful. As standard setting, following settings are registered.

Table 3.2.7 Standard setting

Cell Name	Cell Comment	Content
Standard 1	MTOOL Connection (One CNC)	I/O setting for Machine Tool Connection Function. Robot connects with 1 CNC by FL-Net.
Standard 2	MTOOL Connection (Four CNC)	I/O setting for Machine Tool Connection Function. Robot connects with 4 CNC by FL-Net.

1. Select 'Import Standard Setting' in File menu. (1)
2. 'Select standard setting' window is displayed. (2)
3. Select the standard setting that you want to import, and click 'OK' button.(3)

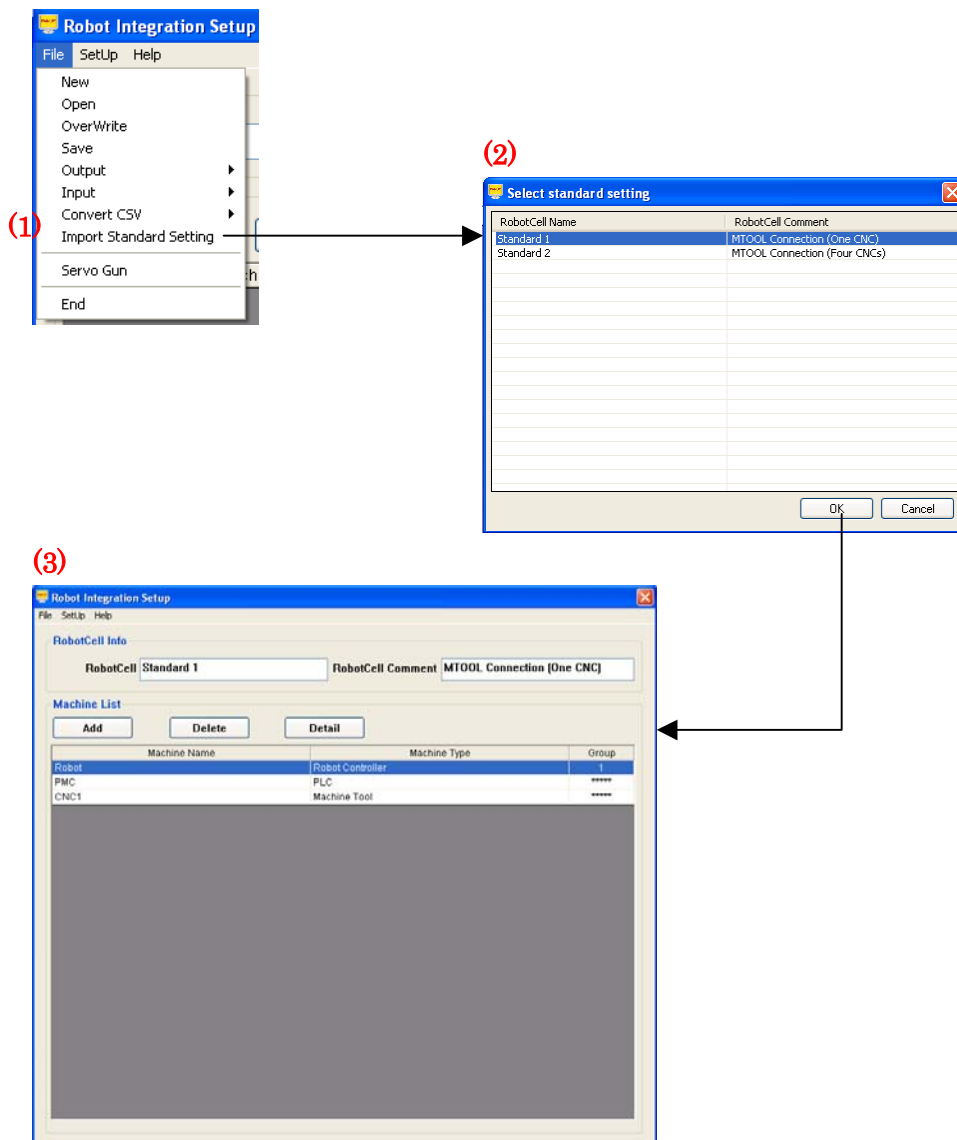


Fig. 3.2.7 Import Standard Setting

3.2.8 Start Servo Gun Integration Setup Tool

In case that Servo Gun Integration Setup Tool option (A08B-9410-J805) is installed in PC, you can start Servo Gun Integration Setup Tool in this menu.

1. Select 'Servo Gun' in File menu. (1)
2. Servo Gun Integration Setup Tool is started. (2)
(In case that the option is not installed, 'Servo Gun' cannot be selected in the menu.)

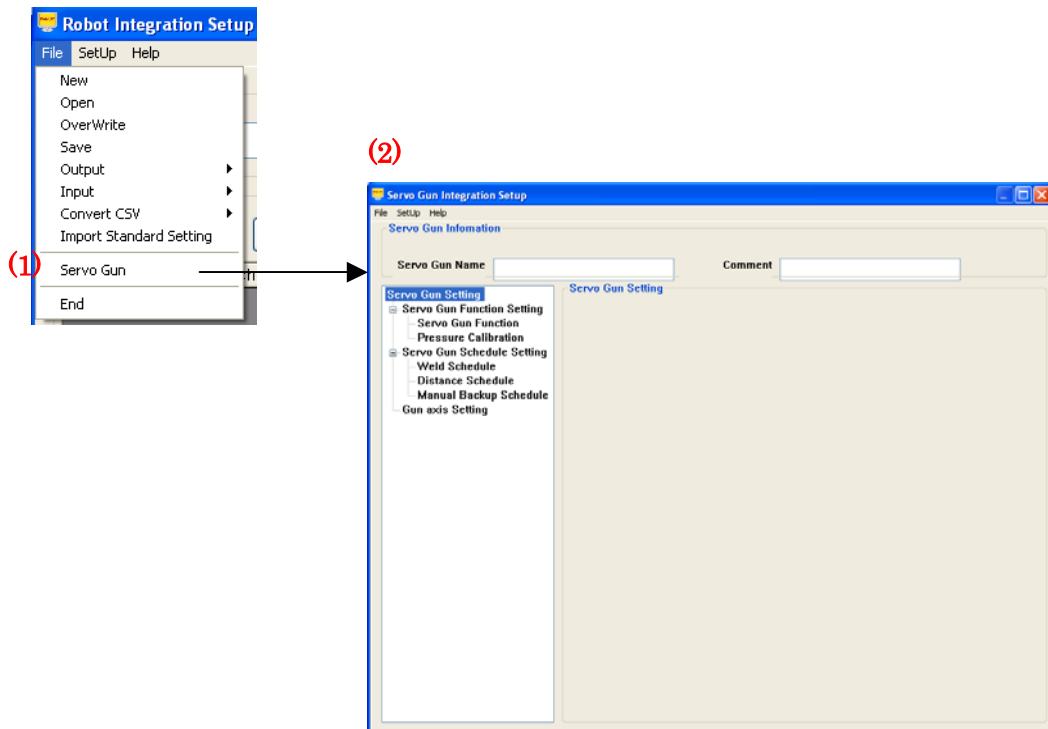


Fig. 3.2.8 Start Servo Gun Integration Setup Tool

3.2.9 Initial Setting File Directory

1. Select 'Initial Setting File Directory' in Setup menu. (1)
2. The dialog to input directory to save the initial setting file is displayed.
3. Input path in the input box (2) or select path in 'Browser for Folder' (5) displayed by '...' button (3).
4. Click 'OK' button. (4)

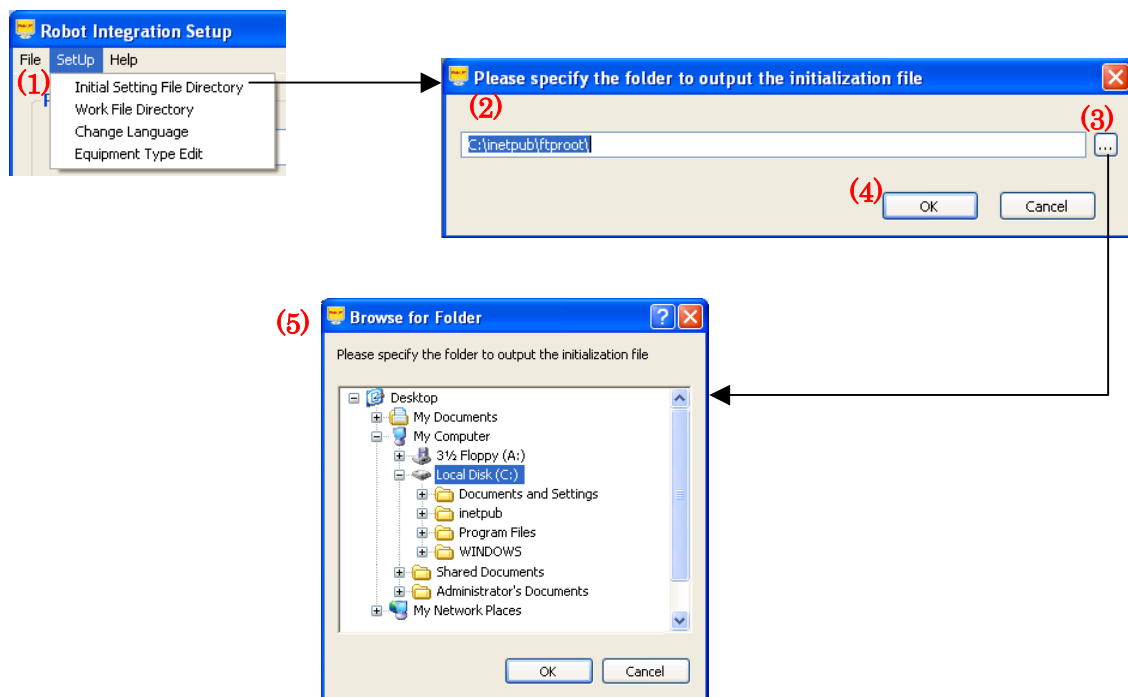


Fig. 3.2.9 Initial Setting file directory

3.2.10 Work File Directory

1. Select 'Work File Directory' in Setup menu. (1)
2. The dialog to input directory to save the work file is displayed.
3. Input path in the input box (2) or select path in 'Browse for Folder' (5) displayed by '...' button (3).
4. Click 'OK' button. (4)

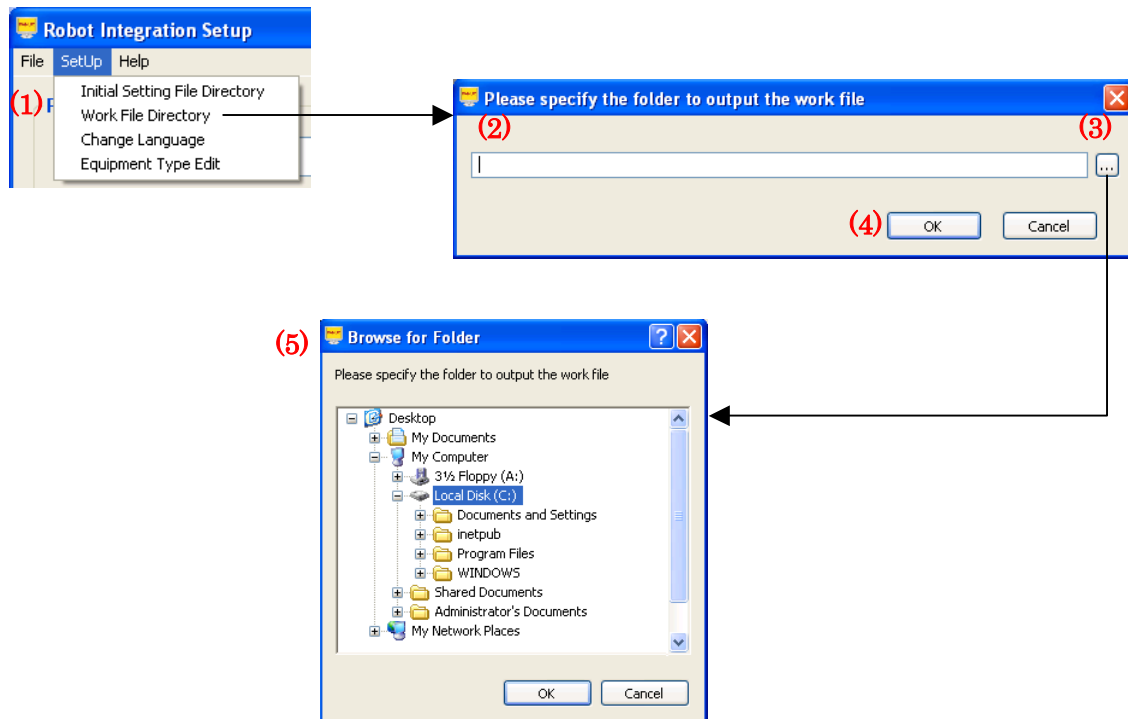


Fig. 3.2.10 Work file directory

3.2.11 Change Language

1. Select 'Change Language' in Setup menu. (1)
2. 'Change Language' dialog is displayed. (2)
3. Check the language and click 'OK' button. (3)
4. The language in the screen is changed to the selected language. (4)

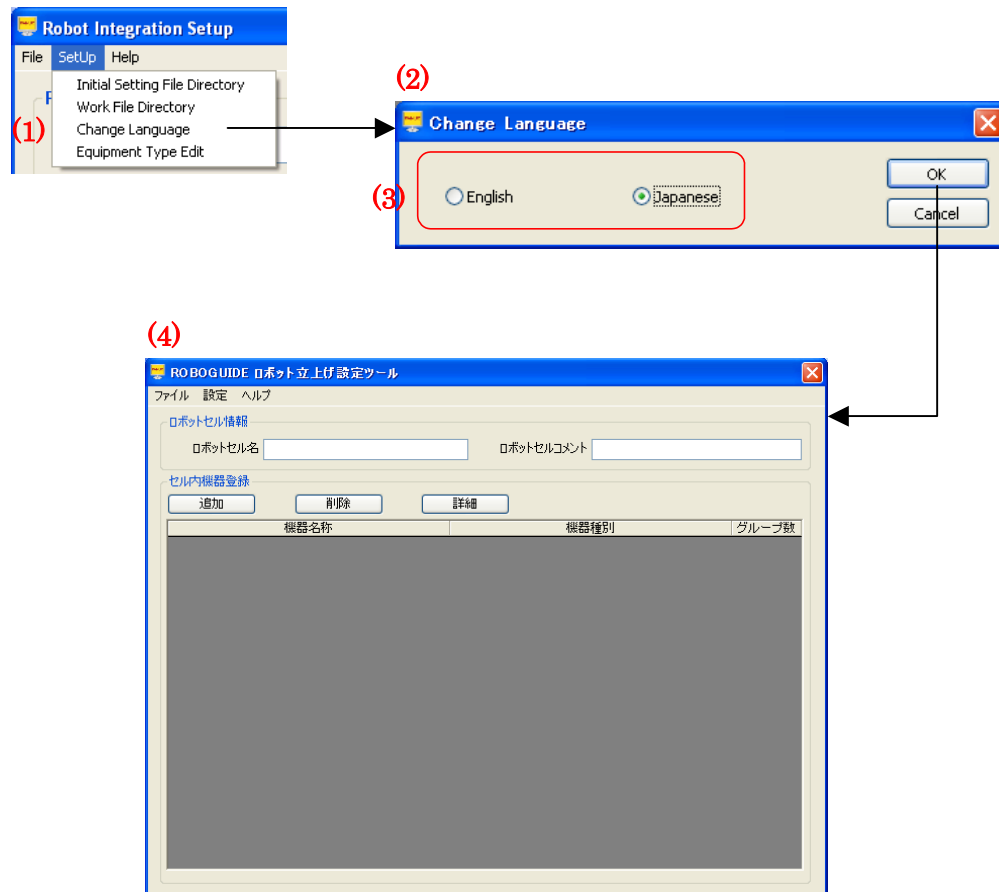


Fig. 3.2.11 Change language

3.2.12 Machine Type Edit

1. Select 'Equipment Type Edit' in Setup menu. (1)
2. In order to add new machine type, input the machine type in 'Add Machine Type (Japanese)' input box (3) in Japanese, then input the machine type in 'Add Machine Type (English)' input box (4) in English. Then, Click 'Add' button (5).
3. In order to delete the machine type, select the machine type in the list box (7), then click 'Delete' button (6).
4. The result of editing the machine type is reflected in the list box (7).
5. Click 'OK' button (8) to store the result of editing. If 'Cancel' (9) button is clicked before 'OK' button, the result of editing is canceled.

It is possible to add new machine type by either Japanese name or English name. If the Japanese name is not input, the machine type is displayed as '*****' in Japanese screen.

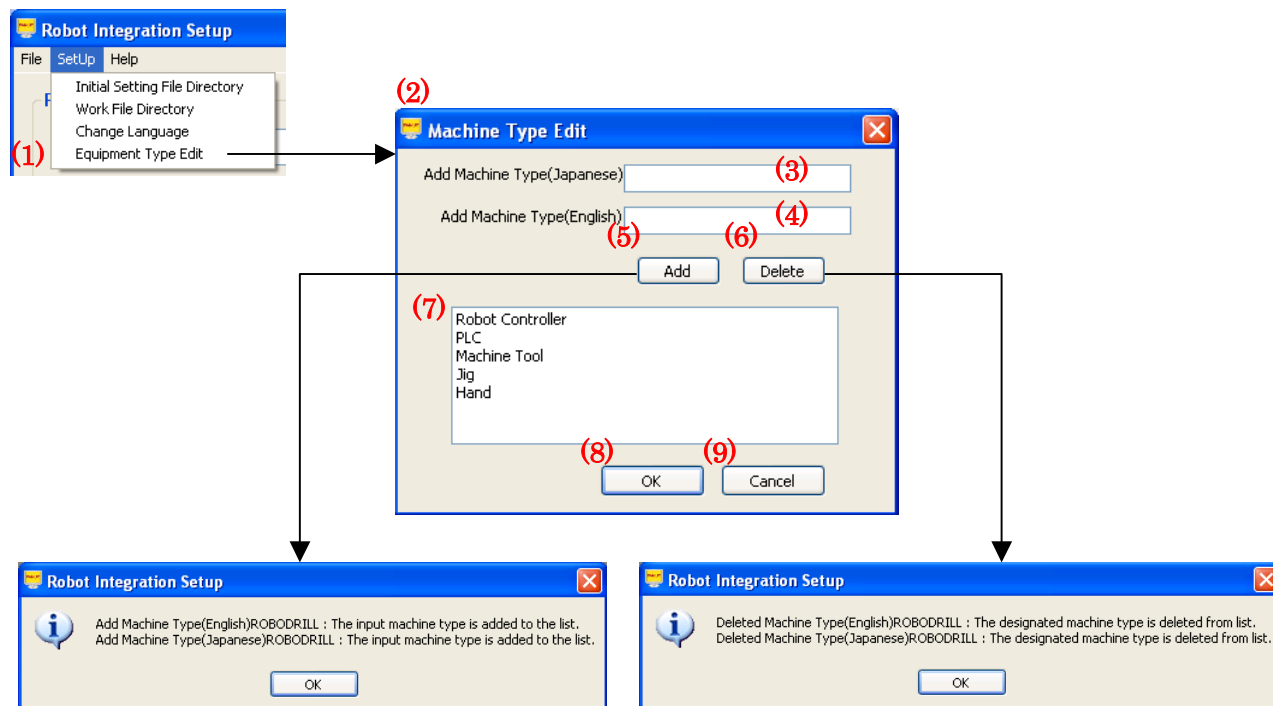


Fig. 3.2.12 Equipment type edit

3.2.13 Version Information

Select 'Version' in Help menu to display 'Version information' screen.

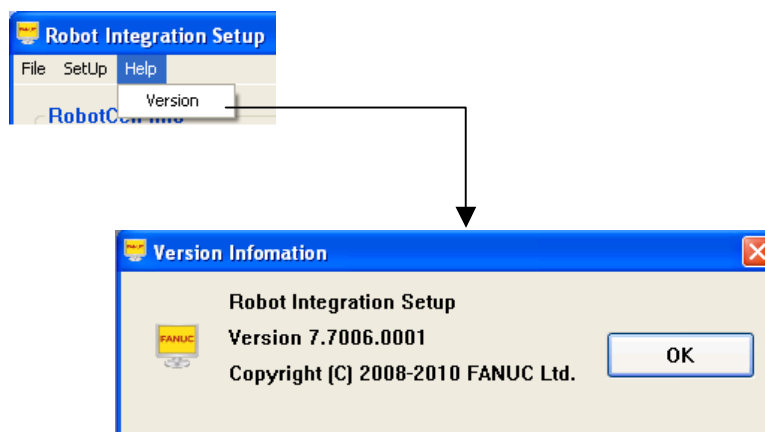


Fig. 3.2.13 Version

3.3 DETAIL SCREEN

Detail screen varies depending on Machine type. In case that Machine type is Robot Controller, the detail screen consists of Function setting and I/O setting. In the case of other Machine type, there is only I/O setting. And, the setting items in I/O setting vary depending on Machine type.

The operations in detail screen are as follows.

1. Select the setting item in the Setting Item Tree (1).
2. Input data in Data Input Field (4).
3. Click 'OK' button (12).

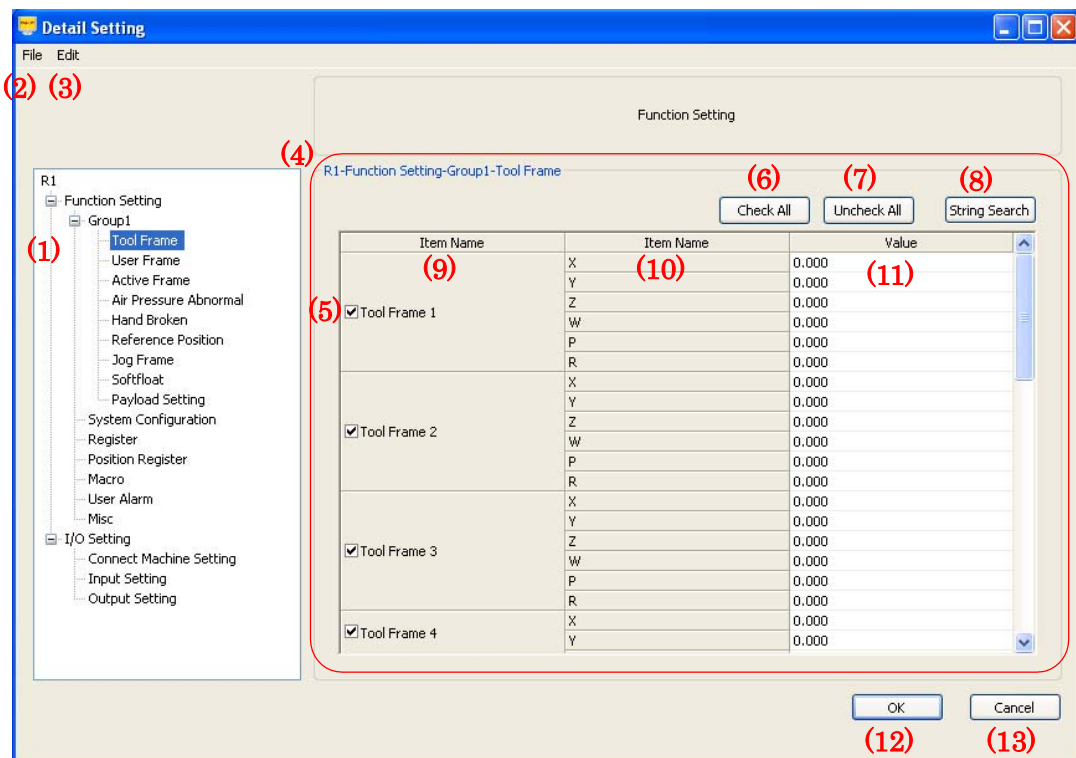


Fig. 3.3 Detail screen

Followings are the description for each part in the detail screen.

- (1) Setting Item Tree
 - The tree view of the setting items is displayed.
 - The setting data is displayed in Data input field (4) by selecting an item in the tree.
- (2) 'File' menu
 - Save The data are saved to the file. Refer to 3.2.2 Overwrite in main screen.
- (3) 'Edit' menu
 - Cut The data set in the selected cell is cut.
 - Copy The data is copied to the selected cell.
 - Paste The data in the clipboard is pasted.
 - Delete The data in the selected cell is deleted.
- (4) Data input Field
 - The data in the setting item selected in Setting Item Tree are displayed. Input or change the data in this field.
- (5) Check Box
 - The data checked in the check box are output to the initial setting file.

- Uncheck the check box of the data that you want not to load to Robot controller.
- (6) 'Check All' button
 - All check boxes in the selected Data Input Field are checked.
- (7) 'Uncheck All' button
 - All check boxes in the selected Data Input Field are unchecked.
- (8) 'String Search' button
 - The string search is executed. Refer to 3.3.5 String Search Window.
- (9) 'Item Name' column in Data Input Filed
 - This column in the Data input Field is displayed item name. This column cannot be changed.
- (10) 'Item Name' column in Data Input Filed
 - This column in the Data input Field is displayed item name. This column cannot be changed.
- (11) 'Value' column in the Data Input Field
 - Set the data in this column.
- (12) 'OK' button
 - When 'OK' button is clicked, all setting data are saved and main screen is displayed.
- (13) 'Cancel' button
 - When 'Cancel' button is clicked, the data set and changed in the detail screen is canceled and main screen is displayed.

3.3.1 In case that Machine Type is Robot Controller

Followings are the description for Data Input Field in case that Machine type is Robot Controller. In case that Machine type is Robot Controller, the detail screen consists of Function setting and I/O setting.

Function Setting

3.3.1.1 Tool frame, User frame

Input the frame data X, Y, Z, and W, P, R.

Item Name	Item Name	Value
☒ Tool Frame 1	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
	P	0.000
☒ Tool Frame 2	R	0.000
	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
☒ Tool Frame 3	P	0.000
	R	0.000
	X	0.000
	Y	0.000
	Z	0.000

Fig. 3.3.1.1 (a) Tool Frame

Item Name	Item Name	Value
☒ User Frame 1	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
	P	0.000
	R	0.000
☒ User Frame 2	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
	P	0.000
	R	0.000
☒ User Frame 3	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
	P	0.000
	R	0.000

Fig. 3.3.1.1 (b) User Frame

3.3.1.2 Active frame

Input active frame number for tool frame, user frame, and jog frame.

Item Name	Item Name	Value
☒ Active User Frame		0
☒ Active Tool Frame		1
☒ Active Jog Frame		1

Fig. 3.3.1.2 Active Frame

3.3.1.3 Air pressure abnormal, Hand broken

Select Enable or Disable in the pull down list.

In the item Air Pressure Abnormal, select enable if you want to enable the detection of Air Pressure Abnormal.

In the item Hand Broken, select enable if you want to enable the detection of Hand Broken.

Item Name	Item Name	Value
☒ Air Pressure(Group1)	(*PPABN)Detection	FALSE

Fig. 3.3.1.3 (a) Air pressure

Item Name	Item Name	Value
☒ Hand Broken(Group1)		TRUE

Fig. 3.3.1.3 (b) Hand broken

3.3.1.4 Reference position

Set data for the reference position.

Item Name	Item Name	Value
	Comment	
	Enable/Disable	FALSE
	I/O Type	DO
	I/O number	0
	Reference position data(J1)	0.000
	Reference position data(J2)	0.000
	Reference position data(J3)	0.000
	Reference position data(J4)	0.000
	Reference position data(J5)	0.000
	Reference position data(J6)	0.000
	Reference position data(J7)	0.000
	Reference position data(J8)	0.000
	Reference position data(J9)	0.000
	Tolerance(J1)	0.000
	Tolerance(J2)	0.000
	Tolerance(J3)	0.000
	Tolerance(J4)	0.000
	Tolerance(J5)	0.000
	Tolerance(J6)	0.000
	Tolerance(J7)	0.000

Fig. 3.3.1.4 Reference position

3.3.1.5 Jog frame

Input the jog frame data X, Y, Z, and W, P, R.

Item Name	Item Name	Value
	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
	P	0.000
	R	0.000
	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
	P	0.000
	R	0.000
	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
	P	0.000
	R	0.000

Fig. 3.3.1.5 Jog frame

3.3.1.6 Soft Float

Input the data for soft float function. Setting items vary depending on the setting of “Soft float type”. When “Soft float type” is changed, the setting items are changed automatically.

Note

- Robot Integration Setup Tool does not support High Performance Soft Float (A05B-2500-J898).
- In R-30iA/R-30iA Mate/R-30iB/R-30iB Mate controller, soft float function is option. In case that soft float option is not available in R-30iA/R-30iA Mate/R-30iB/R-30iB Mate controller, the setting for soft float is not loaded.

The screenshot shows the 'R1-Function Setting-Group1-Softfloat' window. It has a 'Schedule 1' checkbox which is checked. The table lists the following settings:

Item Name	Item Name	Value
	Comment	
	Soft float type	Cartesian
	Soft float Enable	FALSE
	Soft float Coordinate System	World Frame
	X Soft ratio	0
	X Soft Limiter	0
	Y Soft ratio	0
	Y Soft Limiter	0
	Z Soft ratio	0
	Z Soft Limiter	0
	X Spin Soft ratio	0
	X Spin Soft Limiter	0
	Y Spin Soft ratio	0
	Y Spin Soft Limiter	0
	Z Spin Soft ratio	0
	Z Spin Soft Limiter	0

Fig. 3.3.1.6 (a) Soft Float (Cartesian)

The screenshot shows the 'R1-Function Setting-Group1-Softfloat' window. It has a 'Schedule 1' checkbox which is checked. The table lists the following settings:

Item Name	Item Name	Value
	Comment	
	Soft float type	Joint
	Soft float delay time	0
	Axis 1 Soft ratio	0
	Axis 1 Soft float	FALSE
	Axis 2 Soft ratio	0
	Axis 2 Soft float	FALSE
	Axis 3 Soft ratio	0
	Axis 3 Soft float	FALSE
	Axis 4 Soft ratio	0
	Axis 4 Soft float	FALSE
	Axis 5 Soft ratio	0
	Axis 5 Soft float	FALSE
	Axis 6 Soft ratio	0
	Axis 6 Soft float	FALSE
	Axis 7 Soft ratio	0
	Axis 7 Soft float	FALSE
	Axis 8 Soft ratio	0
	Axis 8 Soft float	FALSE
	Axis 9 Soft ratio	0

Fig. 3.3.1.6 (b) Soft Float (Joint)

3.3.1.7 Payload

Input the data for Payload.

Item Name	Item Name	Value
☒ Payload Number		Not Selected
	Comment	
	Payload (kg)	0.00
	Payload Center X (cm)	0.00
☒ Schedule 1	Payload Center Y (cm)	0.00
	Payload Center Z (cm)	0.00
	Payload Inertia X (kgf*cm*s^2)	0.00
	Payload Inertia Y (kgf*cm*s^2)	0.00
	Payload Inertia Z (kgf*cm*s^2)	0.00
	Comment	
	Payload (kg)	0.00
	Payload Center X (cm)	0.00
	Payload Center Y (cm)	0.00
	Payload Center Z (cm)	0.00
☒ Schedule 2	Payload Inertia X (kgf*cm*s^2)	0.00
	Payload Inertia Y (kgf*cm*s^2)	0.00
	Payload Inertia Z (kgf*cm*s^2)	0.00

Fig. 3.3.1.7 Payload

3.3.1.8 System configuration

Input the data about system configuration.

Item Name	Item Name	Value
☒ Use HOT START		TRUE
☒ I/O power fail recovery		RECOVER ALL
☒ Autoexec program for COLD start		
☒ Autoexec program for HOT start		
☒ HOT START done signal		0
☒ Restore selected program		TRUE
☒ Enable UI signals		Enable
	START for CONTINUE only	FALSE
☒ Shell setup	CSTOPI for ABORT	FALSE
	Abort all programs by CSTOPI	FALSE
	PROD START depend on PNSTROBE	Disable
☒ Detect FAULT RESET signal		Fall
☒ WAIT timeout		30.00
☒ RECEIVE timeout		30.00
☒ Return to top of program		TRUE
☒ Original program name(F1)		RSR
☒ Original program name(F2)		PNS
☒ Original program name(F3)		STYLE
☒ Original program name(F4)		JOB
☒ Original program name(F5)		TEST

Fig. 3.3.1.8 System configuration

Following items in the System Configuration cannot be set by this function.

- WJNT for default motion
- Reset CHAIN FAILURE detection
- UOP auto assignment

3.3.1.9 Register, Position register

In register setting, input value and comment for register. In position register setting, input comment for position register.

R1-Function Setting-Register

Check All Uncheck All String Search

Register Number	Value	Comment
☒ 1	0.00	
☒ 2	0.00	
☒ 3	0.00	
☒ 4	0.00	
☒ 5	0.00	
☒ 6	0.00	
☒ 7	0.00	
☒ 8	0.00	
☒ 9	0.00	
☒ 10	0.00	
☒ 11	0.00	
☒ 12	0.00	
☒ 13	0.00	
☒ 14	0.00	
☒ 15	0.00	
☒ 16	0.00	
☒ 17	0.00	
☒ 18	0.00	
☒ 19	0.00	

Fig. 3.3.1.9 (a) Register

R1-Function Setting-Position Register

Check All Uncheck All String Search

Register Number	Comment
☒ 1	
☒ 2	
☒ 3	
☒ 4	
☒ 5	
☒ 6	
☒ 7	
☒ 8	
☒ 9	
☒ 10	
☒ 11	
☒ 12	
☒ 13	
☒ 14	
☒ 15	
☒ 16	
☒ 17	
☒ 18	
☒ 19	
☒ 20	

Fig. 3.3.1.9 (b) Position register

3.3.1.10 Macro

Set Macro Name, Program Name, Assign Signal Type, and Assign Signal Number for Macro command.

Item Name	Macro Name	Program Name	Assign Type	Assign Number
☒ Macro 01	Macro1		MF	1
☒ Macro 02	Macro2		MF	2
☒ Macro 03	Macro3		MF	3
☒ Macro 04	Macro4		MF	11
☒ Macro 05	Macro5		MF	12

Fig. 3.3.1.10 Macro

The macro that is set as the system level macro in Robot controller cannot modify by this function.

3.3.1.11 User alarm

Set the severity of user alarm and the message of user alarm.

About setting of the severity, select the severity from 'WARN', 'STOP.L', 'ABORT.L', 'STOP.G', and 'ABORT.G' by pull down list.

About setting of the message, input string data.

Item Name	Value
☒ UserAlarm01 Severity	STOP.L
☒ UserAlarm02 Severity	STOP.L
☒ UserAlarm03 Severity	STOP.L
☒ UserAlarm04 Severity	STOP.L
☒ UserAlarm05 Severity	STOP.L
☒ UserAlarm06 Severity	STOP.L
☒ UserAlarm07 Severity	STOP.L
☒ UserAlarm08 Severity	STOP.L
☒ UserAlarm09 Severity	STOP.L
☒ UserAlarm10 Severity	STOP.L
☒ UserAlarm01 Message	
☒ UserAlarm02 Message	
☒ UserAlarm03 Message	
☒ UserAlarm04 Message	
☒ UserAlarm05 Message	
☒ UserAlarm06 Message	
☒ UserAlarm07 Message	
☒ UserAlarm08 Message	
☒ UserAlarm09 Message	
☒ UserAlarm10 Message	

Fig. 3.3.1.11 User alarm

3.3.1.12 Miscellaneous

Set the following two items.

- Output BATLTM signal when the backup battery for the pulse coder is abnormal. (Enable/Disable)
- Recover override when safety fence is closed. (Enable/Disable)

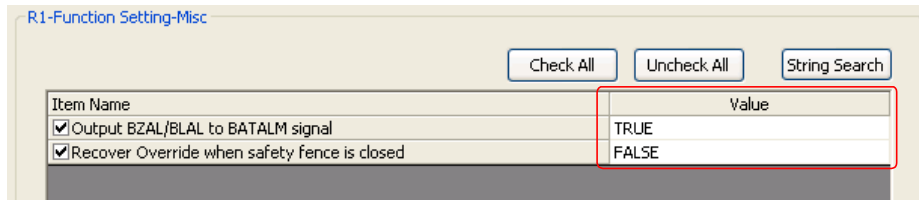


Fig. 3.3.1.12 Miscellaneous Setting

I/O setting

In I/O setting, set 'Connect Machine Setting' at first. Then, set 'Input Setting' and 'Output Setting' according to the number of input signals and output signals set in 'Connect Machine Setting'. In case that the items in 'Connect Machine Setting' are not set, 'Input Setting' and 'Output Setting' cannot be set.

3.3.1.13 Connect machine setting

1. Select the connected machine name by pull down list in 'Connect Machine' (1). Only the machine name set in main screen can be set in this item.
2. Select 'I/O equipment' by pull down list (2), and set 'Input Point' (3), 'Output Point' (4), 'Input Start Physical No.' (5), 'Output Start Physical No.' (6), 'Rack No.' (7), 'Slot No.' (8), and 'Connect Method' (9).
3. To set 'Peripheral I/O', double-click the 'Peripheral I/O' cell (10). In case that the 'Peripheral I/O' is set, '*' is marked in the cell. 'Peripheral I/O' can be set in only one 'I/O module'.
4. If you want not to output I/O setting to the initial setting file, uncheck the check box 'File Out' (11).

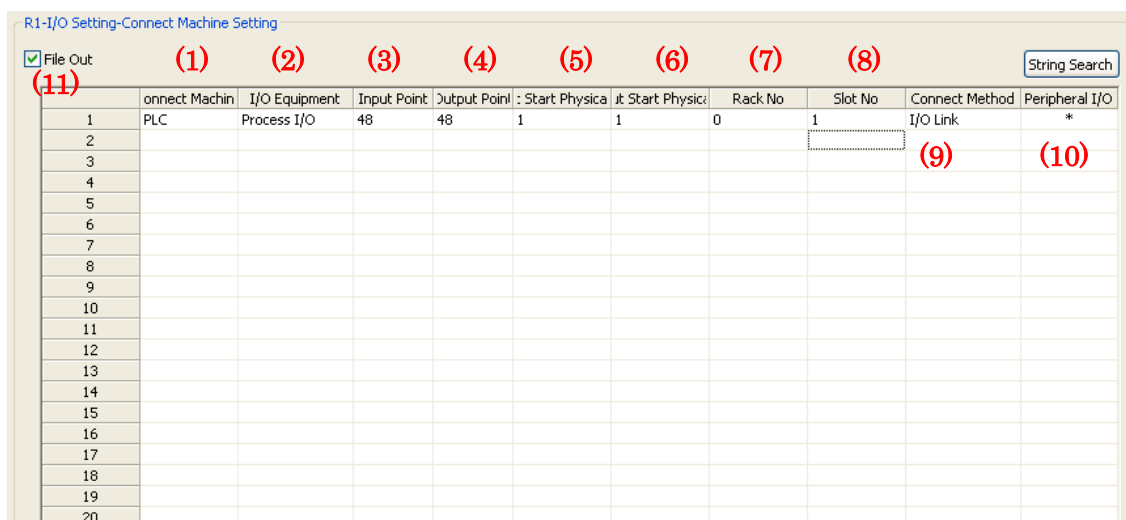


Fig. 3.3.1.13 Connect machine setting

3.3.1.14 Input setting

1. 'I/O Type' (1) and 'I/O Number' (2) are set automatically. But they can be changed. Select 'I/O Type' (1) by pull down list, and change 'I/O number' if required.
2. Input the signal name in 'I/O Name' cell (3).
3. Input comment for the signal in 'Comment' cell (4).

- Select 'Polarity' (5) from 'Normal', 'Inverse' by pull down list.
In case that the signal type is UI, the polarity of the signal cannot be set.
- When Group Input is used, set the GI number in 'GI' cell (6).
The same GI number is set only to the serial signals.
- 'Rack No.', 'Slot No.', 'Connect Machine', and 'Connect Machine Signal Name' (7) are set by Connect Machine Setting and setting in the connected machine automatically.

R1-I/O Setting-Input Setting

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	
I/O Equipment	Physical No	I/O Type	I/O Number	I/O Name	Comment	Polarity	Group Setting	Rack No Slot No Connect Machine ect Machine Signal
Process I/O	1	UI	1	*IMSTP	*IMSTP			0 1 PLC Y0000.0
	2	UI	2	*HOLD	*Hold			0 1 PLC Y0000.1
	3	UI	3	*SFSPD	*SFSPD			0 1 PLC Y0000.2
	4	UI	4	CSTOPI	Cycle stop			0 1 PLC Y0000.3
	5	UI	5	FAULT_RESET	Fault reset			0 1 PLC Y0000.4
	6	UI	6	START	Start			0 1 PLC Y0000.5
	7	UI	7	HOME	Home			0 1 PLC Y0000.6
	8	UI	8	ENBL	Enable			0 1 PLC Y0000.7
	9	UI	9	RSR1/PNS1	RSR1/PNS1			0 1 PLC Y0001.0
	10	UI	10	RSR2/PNS2	RSR2/PNS2			0 1 PLC Y0001.1
	11	UI	11	RSR3/PNS3	RSR3/PNS3			0 1 PLC Y0001.2
	12	UI	12	RSR4/PNS4	RSR4/PNS4			0 1 PLC Y0001.3
	13	UI	13	RSR5/PNS5	RSR5/PNS5			0 1 PLC Y0001.4
	14	UI	14	RSR6/PNS6	RSR6/PNS6			0 1 PLC Y0001.5
	15	UI	15	RSR7/PNS7	RSR7/PNS7			0 1 PLC Y0001.6
	16	UI	16	RSR8/PNS8	RSR8/PNS8			0 1 PLC Y0001.7
	17	UI	17	PNSTROBE	PNS strobe			0 1 PLC Y0002.0
	18	UI	18	PROD_START	Prod start			0 1 PLC Y0002.1
	19	DI	1	Input1	Input 1	Inverse	1	0 1 PLC Y0002.2
	20	DI	2	Input2	Input 2	Inverse	1	0 1 PLC Y0002.3

Fig. 3.3.1.14 Input setting

3.3.1.15 Output setting

- 'I/O Type' (1) and 'I/O Number' (2) are set automatically. But they can be changed. Select 'I/O Type' (1) by pull down list, and change 'I/O number' if required.
- Input the signal name in 'I/O Name' cell (3).
- Input comment for the signal in 'Comment' cell (4).
- Select 'Polarity' (5) from 'Normal', 'Inverse' by pull down list.
In case that the signal type is UO, the polarity of the signal cannot be set.
- Select 'Complementary' (6) from 'FALSE', 'TRUE' by pull down list.
- When Group Output is used, set the GO number in 'GO' cell (7).
The same GO number is set only to the serial signals.
- 'Rack No.', 'Slot No.', 'Connect Machine', and 'Connect Machine Signal Name' (8) are set by Connect Machine Setting and setting in the connected machine automatically.

R1-I/O Setting-Output Setting

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
I/O Equipment	Physical No	I/O Type	I/O Number	I/O Name	Comment	Polarity	Complement	Group Setting
Process I/O	1	UO	1	CMDENBL	Cmd enabled			0 1 PLC X0000.0
	2	UO	2	SYSRDY	System ready			0 1 PLC X0000.1
	3	UO	3	PROGRUN	Prg running			0 1 PLC X0000.2
	4	UO	4	PAUSED	Prg paused			0 1 PLC X0000.3
	5	UO	5	HELD	Motion held			0 1 PLC X0000.4
	6	UO	6	FAULT	Fault			0 1 PLC X0000.5
	7	UO	7	ATPERCH	At perch			0 1 PLC X0000.6
	8	UO	8	TPENBL	TP enabled			0 1 PLC X0000.7
	9	UO	9	BATALM	Batt Alarm			0 1 PLC X0001.0
	10	UO	10	BUSY	Busy			0 1 PLC X0001.1
	11	UO	11	ACK1/SNO1	ACK1/SNO1			0 1 PLC X0001.2
	12	UO	12	ACK2/SNO2	ACK2/SNO2			0 1 PLC X0001.3
	13	UO	13	ACK3/SNO3	ACK3/SNO3			0 1 PLC X0001.4
	14	UO	14	ACK4/SNO4	ACK4/SNO4			0 1 PLC X0001.5
	15	UO	15	ACK5/SNO5	ACK5/SNO5			0 1 PLC X0001.6
	16	UO	16	ACK6/SNO6	ACK6/SNO6			0 1 PLC X0001.7
	17	UO	17	ACK7/SNO7	ACK7/SNO7			0 1 PLC X0002.0
	18	UO	18	ACK8/SNO8	ACK8/SNO8			0 1 PLC X0002.1
	19	UO	19	SNACK	SNACK			0 1 PLC X0002.2
	20	UO	20	RESERVED	Reserved			0 1 PLC X0002.3

Fig. 3.3.1.15 Output setting

3.3.2 In case that Machine Type is PLC

Followings are the description for Data Input Field in case that the machine type is PLC. In case that Machine type is PLC, there is only I/O setting in detail screen.

I/O Setting

In I/O setting, set 'Connect Machine Setting' at first. Then, set 'Input Setting' and 'Output Setting' according to the number of input signals and output signals set in 'Connect Machine Setting'. In case that 'Connect Machine Setting' is not set, 'Input Setting' and 'Output Setting' cannot be set.

3.3.2.1 Connect machine setting

1. Select the connected machine name by pull down list in 'Connect Machine' (1). Only the machine name set in main screen can be set in this item.
2. Select 'I/O Equipment' (2) by pull down list, and set 'Input Point' (3), 'Output Point' (4), 'Input Start Physical No.' (5), 'Output Start Physical No.' (6), 'Rack No.' (7), 'Slot No.' (8), and 'Connect Method' (9). In case that the machine type is PLC, 'Peripheral I/O' (10) cannot be set.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Connect Machine	I/O Equipment	Input Point	Output Point	Input Start Physical No.	Output Start Physical No.	Rack No.	Slot No.	Connect Method	I/O Link
1	R1	I/O Adapter	48	48	1	1	0	1	I/O Link	I/O Link
2										
3										
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
15										
16										
17										
18										
19										
20										

Fig. 3.3.2.1 Connection machine setting

3.3.2.2 Input setting, Output setting

1. Input the signal name to 'I/O name' (1).
2. Input the description of this signal to 'Contents' (2).
3. 'I/O Equipment', 'Rack No.', 'Slot No.', 'Connect Machine', and 'Connect Machine Signal Name' (3) are set by Connect Machine Setting and setting in the connected machine automatically.

PLC-I/O Setting-Input Setting

(1)	(2)	(3)					
Number	I/O Name	Contents	I/O Equipment	Rack No	Slot No	Connect Machine	Connect Machine Signal Name
1	X0000.0	PLC_IN_001	I/O Adapter	0	1	R1	CMDBNBL
2	X0000.1	PLC_IN_002	I/O Adapter	0	1	R1	SVSRDY
3	X0000.2	PLC_IN_003	I/O Adapter	0	1	R1	PROGRUN
4	X0000.3	PLC_IN_004	I/O Adapter	0	1	R1	PAUSED
5	X0000.4	PLC_IN_005	I/O Adapter	0	1	R1	HELD
6	X0000.5	PLC_IN_006	I/O Adapter	0	1	R1	FAULT
7	X0000.6	PLC_IN_007	I/O Adapter	0	1	R1	ATPERCH
8	X0000.7	PLC_IN_008	I/O Adapter	0	1	R1	TPENBL
9	X0001.0	PLC_IN_009	I/O Adapter	0	1	R1	BATALM
10	X0001.1	PLC_IN_010	I/O Adapter	0	1	R1	BUSY
11	X0001.2	PLC_IN_011	I/O Adapter	0	1	R1	ACK1/SNO1
12	X0001.3	PLC_IN_012	I/O Adapter	0	1	R1	ACK2/SNO2
13	X0001.4	PLC_IN_013	I/O Adapter	0	1	R1	ACK3/SNO3
14	X0001.5	PLC_IN_014	I/O Adapter	0	1	R1	ACK4/SNO4
15	X0001.6	PLC_IN_015	I/O Adapter	0	1	R1	ACK5/SNO5
16	X0001.7	PLC_IN_016	I/O Adapter	0	1	R1	ACK6/SNO6
17	X0002.0	PLC_IN_017	I/O Adapter	0	1	R1	ACK7/SNO7
18	X0002.1	PLC_IN_018	I/O Adapter	0	1	R1	ACK8/SNO8
19	X0002.2	PLC_IN_019	I/O Adapter	0	1	R1	SNACK
20	X0002.3	PLC_IN_020	I/O Adapter	0	1	R1	RESERVED

Fig. 3.3.2.2 (a) Input setting

PLC-I/O Setting-Output Setting

(1)	(2)	(3)					
Number	I/O Name	Contents	I/O Equipment	Rack No	Slot No	Connect Machine	Connect Machine Signal Name
1	Y0000.0	PLC_OUT_001	I/O Adapter	0	1	R1	*IMSTP
2	Y0000.1	PLC_OUT_002	I/O Adapter	0	1	R1	*HOLD
3	Y0000.2	PLC_OUT_003	I/O Adapter	0	1	R1	*SFSPD
4	Y0000.3	PLC_OUT_004	I/O Adapter	0	1	R1	CSTOPI
5	Y0000.4	PLC_OUT_005	I/O Adapter	0	1	R1	FAULT_RESET
6	Y0000.5	PLC_OUT_006	I/O Adapter	0	1	R1	START
7	Y0000.6	PLC_OUT_007	I/O Adapter	0	1	R1	HOME
8	Y0000.7	PLC_OUT_008	I/O Adapter	0	1	R1	ENBL
9	Y0001.0	PLC_OUT_009	I/O Adapter	0	1	R1	RSR1/PNS1
10	Y0001.1	PLC_OUT_010	I/O Adapter	0	1	R1	RSR2/PNS2
11	Y0001.2	PLC_OUT_011	I/O Adapter	0	1	R1	RSR3/PNS3
12	Y0001.3	PLC_OUT_012	I/O Adapter	0	1	R1	RSR4/PNS4
13	Y0001.4	PLC_OUT_013	I/O Adapter	0	1	R1	RSR5/PNS5
14	Y0001.5	PLC_OUT_014	I/O Adapter	0	1	R1	RSR6/PNS6
15	Y0001.6	PLC_OUT_015	I/O Adapter	0	1	R1	RSR7/PNS7
16	Y0001.7	PLC_OUT_016	I/O Adapter	0	1	R1	RSR8/PNS8
17	Y0002.0	PLC_OUT_017	I/O Adapter	0	1	R1	PNSTROBE
18	Y0002.1	PLC_OUT_018	I/O Adapter	0	1	R1	PROD_START
19	Y0002.2	PLC_OUT_019	I/O Adapter	0	1	R1	Input1
20	Y0002.3	PLC_OUT_020	I/O Adapter	0	1	R1	Input2

Fig. 3.3.2.2 (b) Output setting

3.3.3 In case that Machine Type is Machine Tool, Jig, Hand

Followings are the description for Data Input Field in case that the machine type is Machine Tool or Jig or Hand. In case that Machine type is Machine Tool or Jig or Hand, there is only I/O setting in detail screen. In case that machine type is new machine type added in Machine Type Edit, the setting items are same as Machine Tool.

I/O Setting

In I/O setting, set 'Connect Machine Setting' at first. Then, set 'Input Setting' and 'Output Setting' according to the number of input signals and output signals set in 'Connect Machine Setting'. In case that 'Connect Machine Setting' is not set, 'Input Setting' and 'Output Setting' cannot be set.

3.3.3.1 Connect machine setting

1. Select the connected machine name by pull down list in 'Connect Machine' (1). Only the machine name set in main screen can be set in this item.
2. Select 'I/O Equipment' (2) by pull down list, and set 'Input Point' (3), 'Output Point' (4), 'Input Start Physical No.' (5), 'Output Start Physical No.' (6) 'Rack No.' (7), 'Slot No.' (8), and 'Connect Method' (9). In case that the machine type is Machine Tool or Jig or Hand, 'Peripheral I/O' (10) cannot be set.

Machine-I/O Setting-Connect Machine Setting

☒ File Out

(1) (2) (3) (4) (5) (6) (7) (8)

String Search

	Connect Machine	I/O Equipment	Input Point	Output Point	ut Start Physical	ut Start Physical	Rack No	Slot No	Connect Method	Peripheral I/O
1	PLC	I/O Adapter	32	32	1	1	1	1	I/O Link	
2									(9)	(10)
3										
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
15										
16										
17										
18										
19										
20										

Fig. 3.3.3.1 Connection machine setting

3.3.3.2 Input setting, Output setting

1. Input the signal address to 'Address' (1).
2. Input the signal name to 'I/O Name' (2).
3. 'Connect Machine', and 'Connect Machine Signal Name' (3) are set by Connect Machine Setting and setting in the connected machine automatically.

Machine-I/O Setting-Input Setting

(1) (2) (3)

String Search

Number	Address	I/O Name	Connect Machine	Connect Machine Signal Name
1	X0000.0	D1Start	PLC	D1START
2	X0000.1	D1Abort	PLC	D1ABORT
3	X0000.2	D1Home	PLC	D1HOME
4	X0000.3	D1Door	PLC	D1DOOR
5	X0000.4	D1P1Sel	PLC	D1P1SEL
6	X0000.5	D1P2Sel	PLC	D1P2SEL
7	X0000.6		PLC	UnSetting
8	X0000.7		PLC	UnSetting
9	X0001.0		PLC	UnSetting
10	X0001.1		PLC	UnSetting
11	X0001.2		PLC	UnSetting
12	X0001.3		PLC	UnSetting
13	X0001.4		PLC	UnSetting
14	X0001.5		PLC	UnSetting
15	X0001.6		PLC	UnSetting
16	X0001.7		PLC	UnSetting
17	X0002.0		PLC	UnSetting
18	X0002.1		PLC	UnSetting
19	X0002.2		PLC	UnSetting
20	X0002.3		PLC	UnSetting

Fig. 3.3.3.2 (a) Input setting

Machine-I/O Setting-Output Setting

(1) (2) (3)

String Search

Number	Address	I/O Name	Connect Machine	Connect Machine Signal Name
1	Y0000.0		PLC	D1BUSY
2	Y0000.1		PLC	D1DOPEN
3	Y0000.2		PLC	D1DCLOS
4	Y0000.3		PLC	D1HOMEP
5	Y0000.4		PLC	D1AUTO
6	Y0000.5		PLC	D1ALARM
7	Y0000.6		PLC	UnSetting
8	Y0000.7		PLC	UnSetting
9	Y0001.0		PLC	UnSetting
10	Y0001.1		PLC	UnSetting
11	Y0001.2		PLC	UnSetting
12	Y0001.3		PLC	UnSetting
13	Y0001.4		PLC	UnSetting
14	Y0001.5		PLC	UnSetting
15	Y0001.6		PLC	UnSetting
16	Y0001.7		PLC	UnSetting
17	Y0002.0		PLC	UnSetting
18	Y0002.1		PLC	UnSetting
19	Y0002.2		PLC	UnSetting
20	Y0002.3		PLC	UnSetting

Fig. 3.3.3.2 (b) Output setting

3.3.4 Check Box

In Function setting of detail screen, all setting items have check box (1). When the initial setting file output operation is done, the items checked in its check box are output to the initial setting file.

When the each setting screen is displayed first, all check boxes in the screen are checked. In order to uncheck all check boxes in the screen, click 'Uncheck All' button (2). In order to check all check boxes in the screen, click 'Check All' button (3).

Item Name	Item Name	Value
☒ Tool Frame 1	X	0.000
	Y	0.000
	Z	0.000
	W	0.000
	P	0.000
	R	0.000
☒ Tool Frame 2	X	0.000
	Y	0.000
	Z	0.000
	W	0.000

Fig. 3.3.4 (a) Check box

In I/O setting of detail screen, there is only one check box in Connection Machine Setting screen. When this check box (1) is unchecked, all I/O settings are not output to the initial setting file

	onnect Machin	I/O Equipment	Input Point	Output Point	Start Physical	Start Physical	Rack No	Slot No	Connect Method	Peripheral I/O
1	PLC	Process I/O	48	48	1	1	0	1	I/O Link	*
2										
3										
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
15										
16										
17										
18										
19										
20										

Fig. 3.3.4 (b) Check box (I/O setting)

3.3.5 String Search Window

When 'String Search' button is clicked, the String Search window is displayed.

1. Click 'String Search' button (1). String Search window is displayed.
2. Input the search string to the input box (2) in String Search window.
3. Click 'OK' button (3).
4. As the result of string search, the cell that includes the search string is colored yellow. (4)

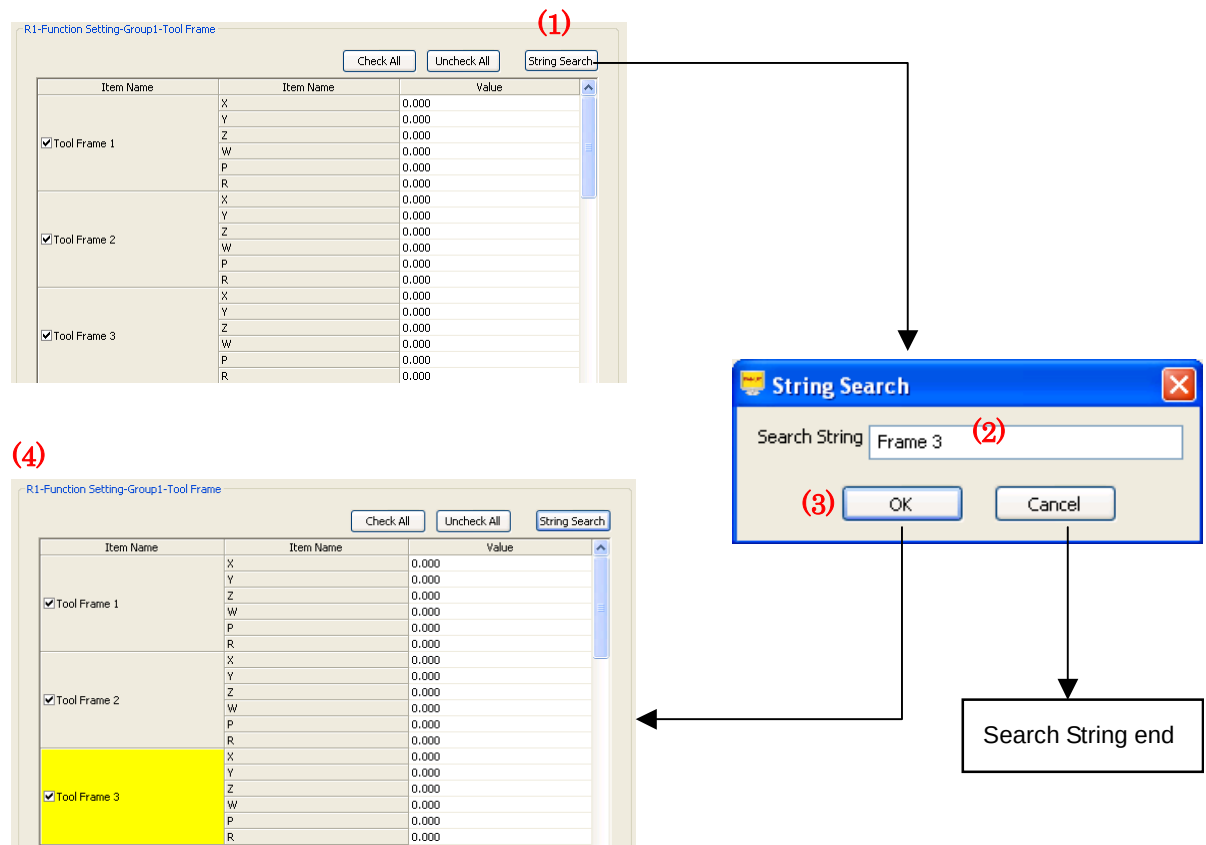


Fig. 3.3.5 String Search

4 SERVO GUN SETUP OPERATION (OPTION)

This chapter explains about operation of Servo Gun Integration Setup Tool.

4.1 OPERATION FLOW FOR SERVO GUN SETUP

Servo gun setup is performed by the following procedure.

1. Set up servo gun on main screen.
This software can set up per a kind of servo gun. Hardware parameter and function setting of servo gun are specified on main screen. Refer to 4.2 MAIN SCREEN for more detail.
2. Specify detail parameter on per detail setting screen.
Refer to 4.6 DETAIL SETTING SCREEN for more detail.
3. Output servo gun data to setup file to load into R-30iA controller.
Output operation is performed on main screen. Refer to 4.3.1 Output for more detail.

4.2 MAIN SCREEN

When Servo Gun Integration Setup Tool is started up via Robot integration setting tool, then the following screen appears.

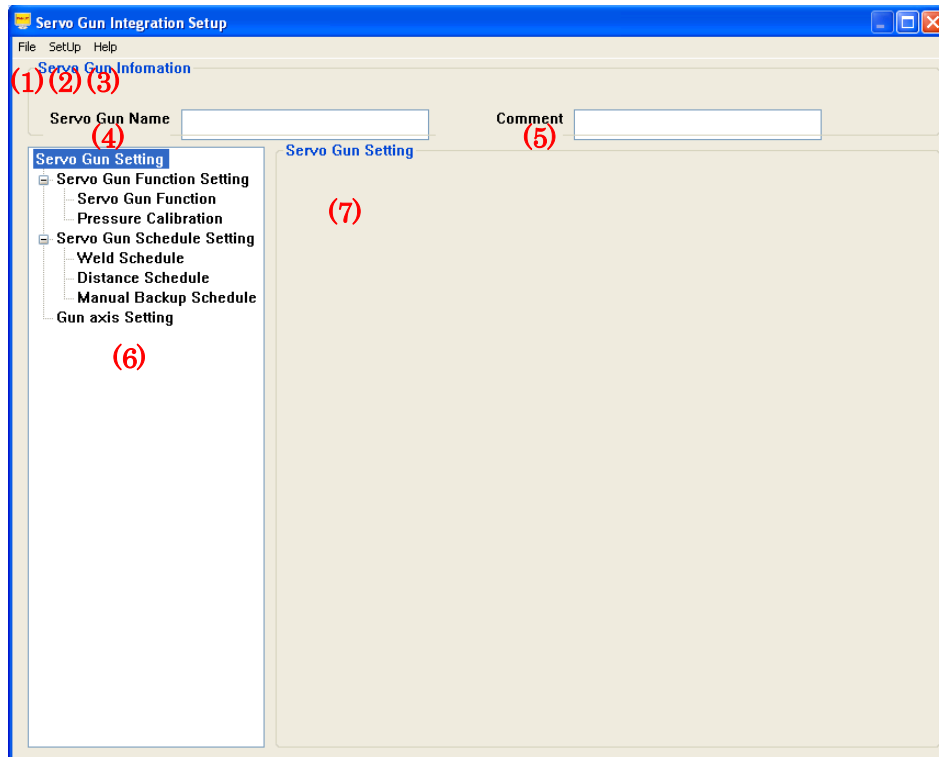


Fig. 4.2 Main screen

The followings are items on this screen.

- (1) "File" menu
 - "New" It creates new servo gun setup data.
 - "Open" It opens XML file that saves servo gun setup data.
 - "Save" It saves servo gun setup data into new XML file.
 - "OverWrite" It saves servo gun setup data into XML file that is opened with overwriting.
 - "Output" It outputs servo gun setup data on the tool into setup file.
 - "Input" It loads setup file that is created in R-30iA controller to input automatically.
 - "Convert CSV" Cannot use.
 - "End" It ends Servo Gun Integration Setup Tool.
- (2) "SetUp" menu
 - "Set Up File Address" It specifies address to output setup file.
 - "Change Language" It changes language of Servo Gun Integration Setup Tool.
 - "Unit" It changes unit of pressure force.
 - "Motor" It sets up to add a kind of motor.
 - "Motor File address" It specifies address of motor file that includes new motor.
- (3) "Help" menu
 - "Version" It shows software version.
- (4) "Servo Gun Name" edit box
 - Write servo gun name to set up.
- (5) "Comment" edit box
 - Write comment about servo gun name to set up.
- (6) "Servo gun setting" category list
 - "Servo Gun Function Setting" It performs setting of servo gun function.
 - "Servo Gun Schedule Setting" It performs setting of per servo gun schedule.
 - "Gun axis Setting" It performs setting of per servo gun axis.
- (7) Detail setting screen
 - "Servo Gun Setting" It displays setting items of selected setting category.

4.3 FILE MENU

Via File menu, it can perform file operation of Servo Gun Integration Setup Tool.

The following basic operation is same as Robot Integration Setup Tool.

- "New"
- "Open"
- "Save"
- "OverWrite"
- "End"

In this section, it explains about own file operation of Servo Gun Integration Setup Tool.

4.3.1 Output

Via Output menu, setup file is output to load into R-30iA controller.

1. Select "File"- "Set Up File Output"(1).
2. Setup files are output into SVGNPRMS folder on path that is specified by Set Up File Address. SVGNPRMS folder is created automatically, if it is not existed.
3. Setup files are created for per named servo gun. If a existed named servo gun has been output, a message whether the setup data is overwritten or not is prompted (2).

WARNING

In SVGNPRMS folder, setup files SVGNxxxx.prm, SVGNxxxx.dt are created.(xxxx is a number of creating order.)
Also, SVGNLIST.csv indicates name of servo gun that is saved in SVGNxxxx.prm.
Do not edit these files.

NOTE

When it loads setup file to R-30iA/R-30iB controller, copy SVGNPRMS folder that includes setup file to root on memory card.

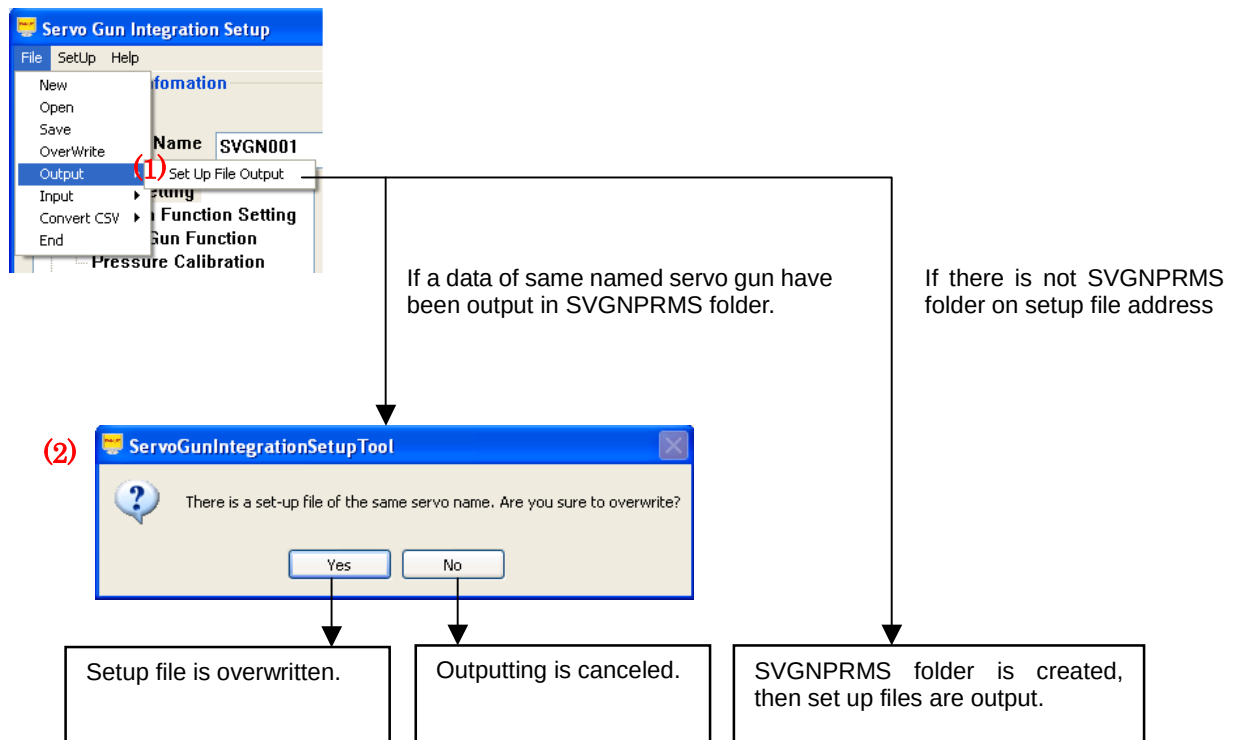


Fig. 4.3.1 Output

4.3.2 Input

Via Input menu, setup file SVGN0000.prm that is created in R-30iA/R-30iB controller can be load. About how to create SVGN0000.prm, refer to 5.2.6 Store Current Setting of Servo Gun.

1. Select "File"->"Input"->"Set Up File Input"(1)
2. "File Open" dialog(2) appears.
3. Select setup file (SVGN0000.prm).
4. Load setup file(3).
5. Setup data that is included in load setup file is applied to Servo Gun Integration Setup Tool.

NOTE

The servo gun is named as "SVGNO00". Change the default name.
Also servo gun axis setting cannot be loaded.

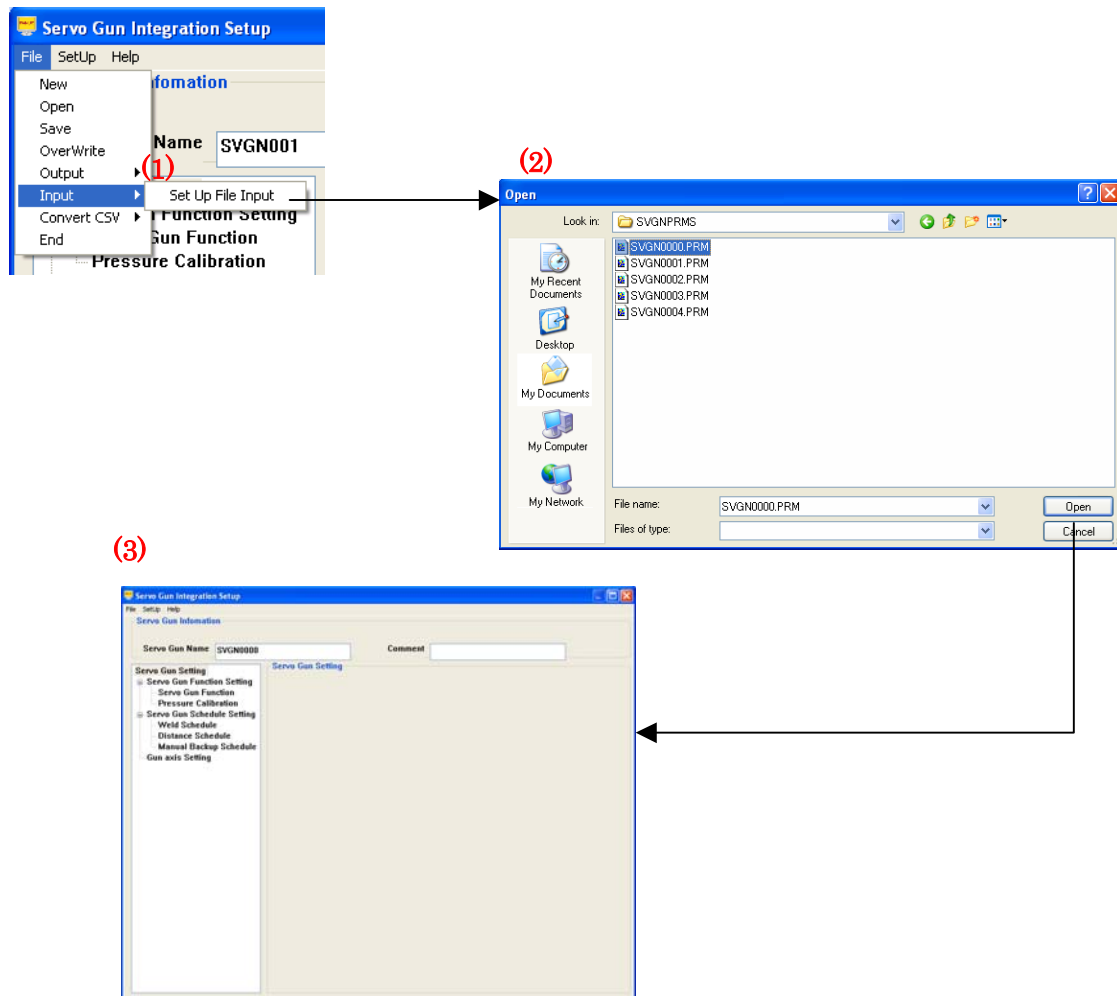


Fig. 4.3.2 Input

4.3.3 CSV File Conversion

The data input from CSV file cannot be used.

4.4 SETUP MENU

4.4.1 Set Up File Address

1. Select "SetUp"- "Set Up File Address" (1).
2. "Output initialize file path" dialog (2) appears.
3. Enter a path in box (2). Or click button (3), then open "Browse for folder" dialog (5) and select a path that you want to output.
4. Click "OK" button (4).

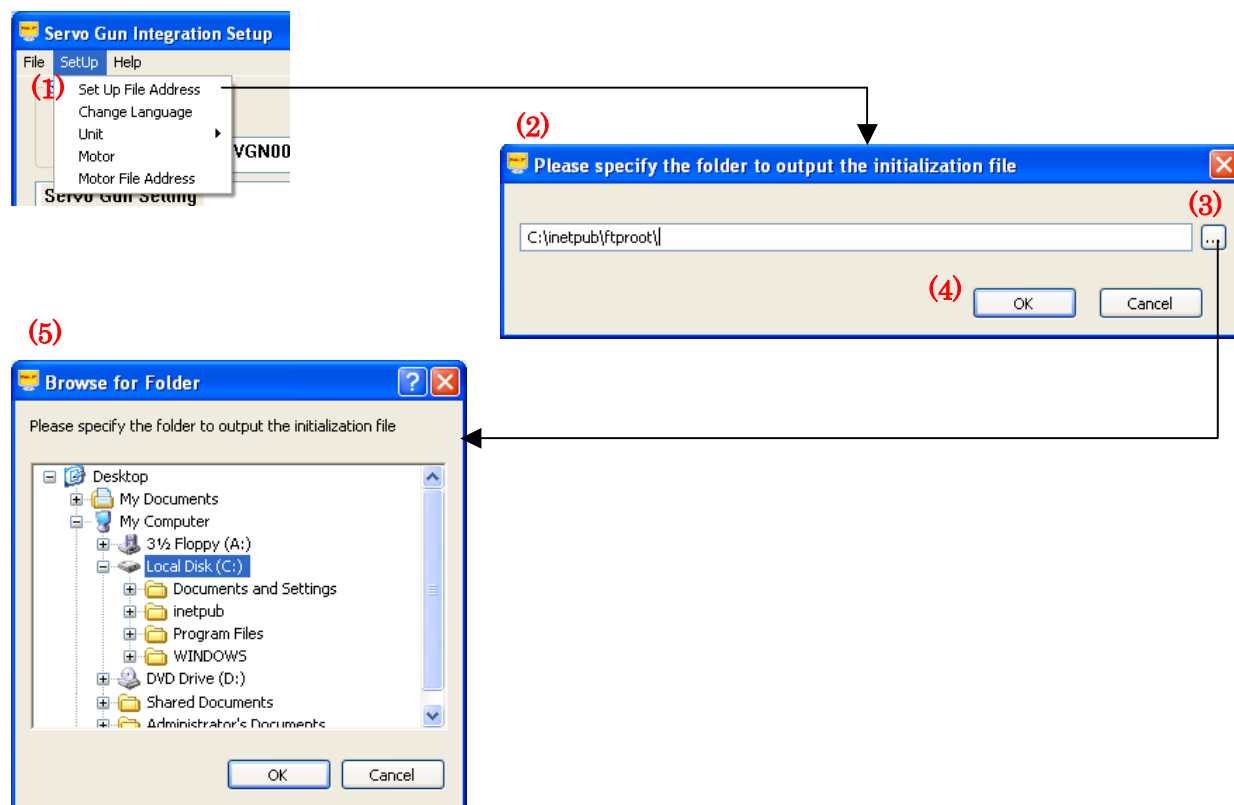


Fig. 4.4.1 Setup file folder

4.4.2 Change Language

1. Select "SetUp"- "Change Language" (1).
2. "Change Language" dialog (2) appears.
3. Select language via Radio button (3), then click "OK" button.
4. Language on screen is changed (4).

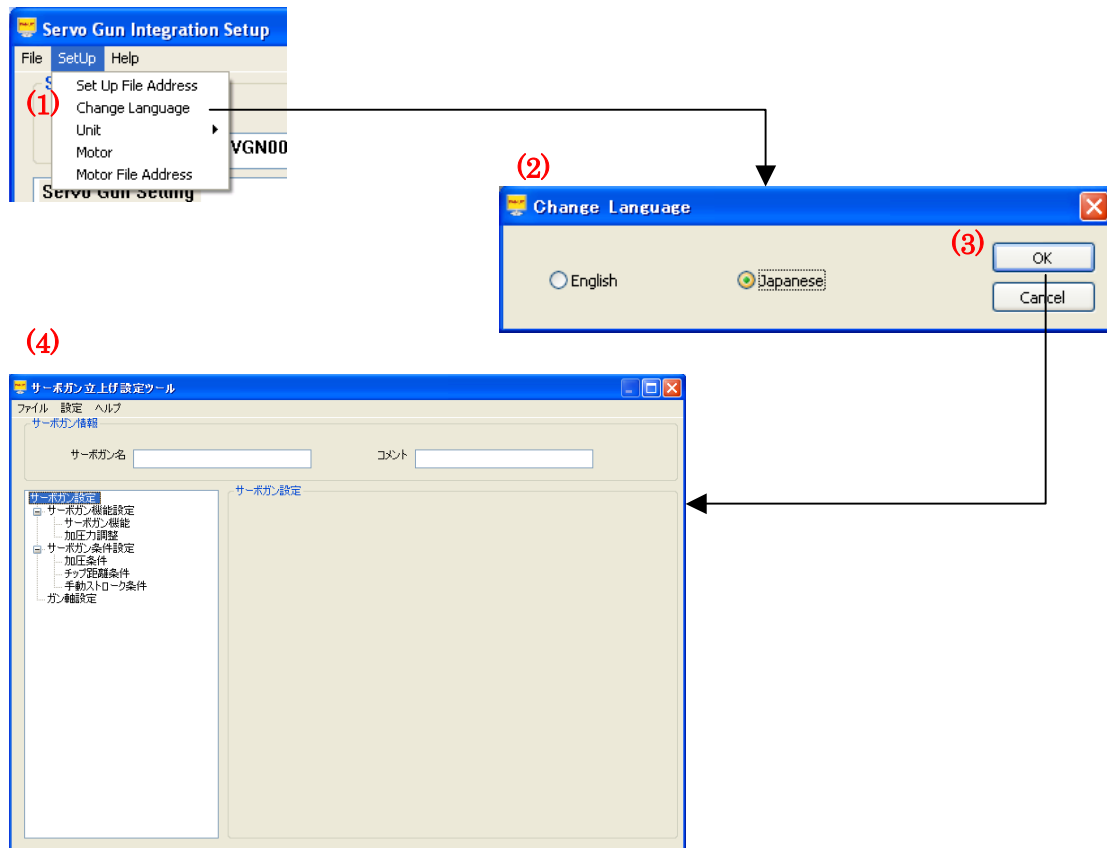


Fig. 4.4.2 Change language

4.4.3 Unit for Pressure Force

1. Click pressure unit that you want to use via “SetUp”-“Unit”(1) menu.
2. Pressure unit of max pressure of gun axis setting and pressure data of pressure schedule is changed.

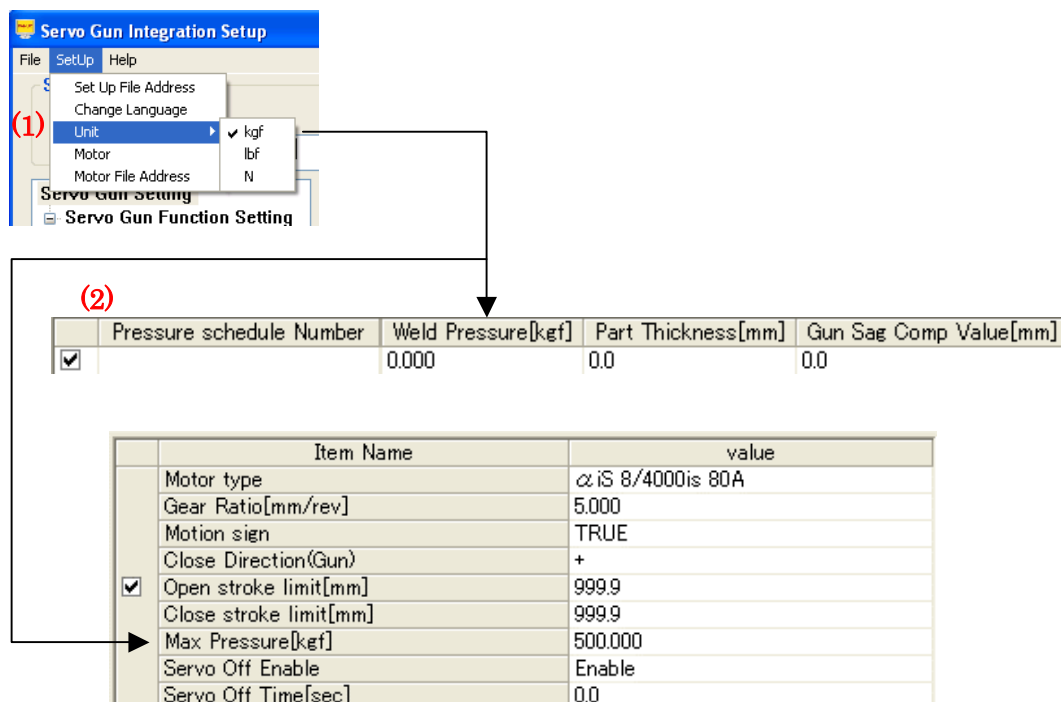


Fig. 4.4.2 Unit for pressure force

4.4.4 Motor Settings

You can select only general servomotor for servo gun via “Motor type” of servo gun axis setting. About special servomotor expect pre-provided servomotor on “Motor type” pull down menu, you have to add it in beforehand. Adding special servomotor* allows for you to select it via “Motor type” pull down menu.

* This product can add servomotor that is supported by R-30iA/R-30iB Servo gun axes option.

Start up motor setting screen by the following procedure.

1. Select “SetUp”-“Motor”(1)
2. “Motor Setting screen” (2) appears.

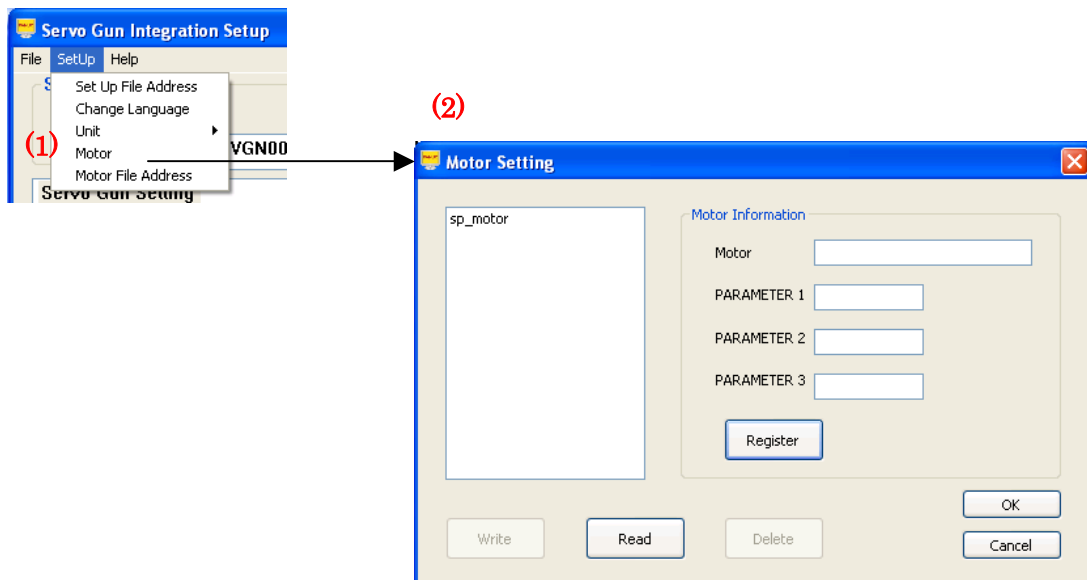


Fig. 4.4.4 Motor Setting

4.4.4.1 Motor setting screen

The following shows motor setting screen.

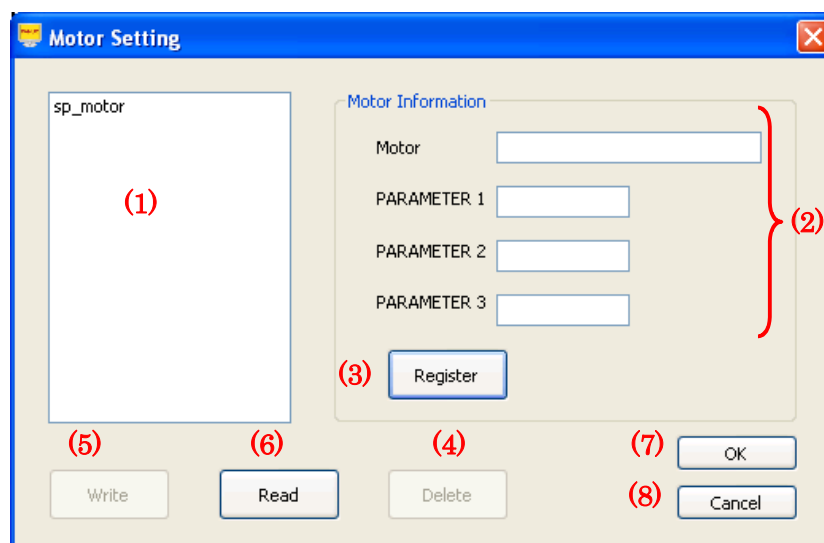


Fig. 4.4.4.1 Motor Setting screen

- (1) Motor list : It lists up motors that user add.
- (2) Motor information : It displays and edits about motor data.
- (3) Register button : This button registers Motor data on motor information. If the motor name is existed, the data is overwritten. If not, the data is added.
- (4) Delete button : This button deletes selected motor data on motor list.
- (5) Output button : This button outputs selected motor data on motor list to XML file. A number of motor can be output to one XML file.
- (6) Read button : This button loads XML file that include motor data. Also it can load XML file that created by other PC that Servo Gun Integration Setup Tool is installed.
- (7) OK button : This button saves setting on motor setting screen, and then returns to main screen.
- (8) Cancel button : This button does not save setting on motor setting screen, and then returns to main screen.

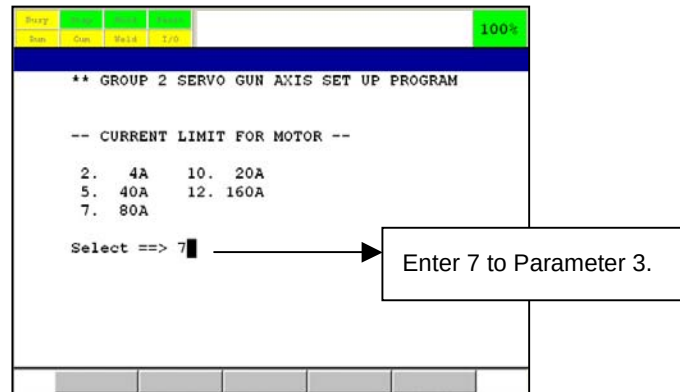
4.4.4.2 Motor addition

Motor data is added by the following procedure.

1. Enter motor name on motor information menu. (1)
2. Enter motor parameter 1-3 * on motor information menu. (2)

* Motor parameters are basically same parameter as number that is entered to identify Motor size, Motor type (rpm), and Motor current via servo gun axes setting of robot maintenance on R-30iA/R-30iB controller. The following sample shows setting of aiS8 4000/80A.

The first screenshot shows the 'MOTOR SIZE (alpha iS)' menu. It lists options: 60. aiS2, 61. aiS4, 62. aiS8, 63. aiS12, and 0. Next page. A callout box points to the '62' option with the text 'Enter 62 to Parameter 1.' The second screenshot shows the 'MOTOR TYPE' menu. It lists options: 1. /2000, 2. /3000, 11. /4000, 12. /5000, 13. /6000. A callout box points to the '11' option with the text 'Enter 11 to Parameter 2.'



Do not enter parameter that R-30iA/R-30iB servo gun axes option is not supported.
Please call FANUC, if you have unknown motor parameters.

3. Click Register button (3).
4. The named motor is added into motor list (4).
5. If the named motor exists in motor list, message whether motor data is overwritten or not (5) is prompted.

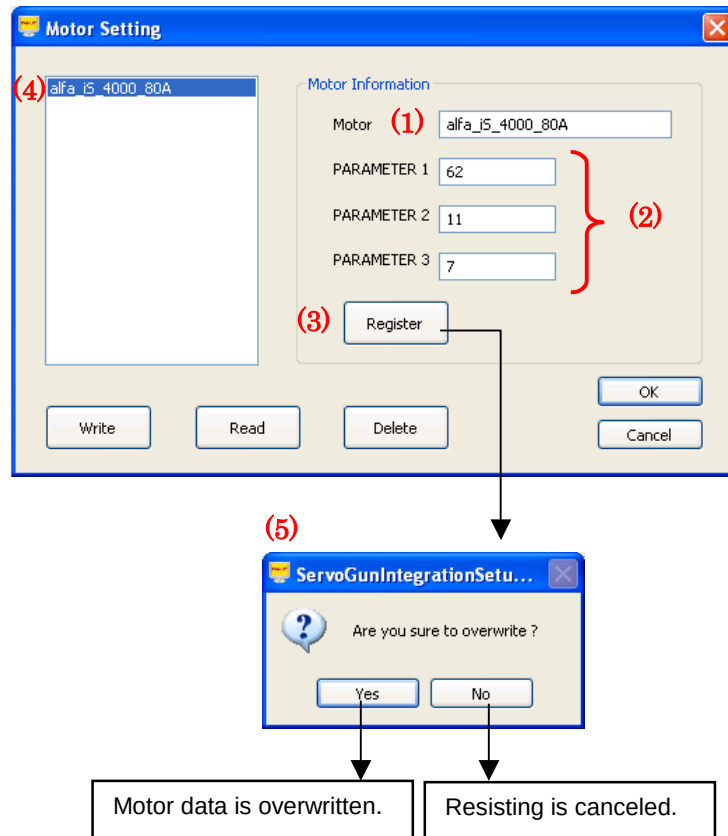


Fig. 4.4.4.2 Motor addition

4.4.4.3 Output motor setting file

It can output added motor setting to XML file.

1. Select motor that you want to output in motor list (1). If a number of motor are in the list, any number of motor that you want to output can be selected, then output to one XML file.
2. Click Write button (2).
3. File save dialog (3) appears.
4. Enter file name, then click Save button (4). If named XML file exists, it can be overwritten.

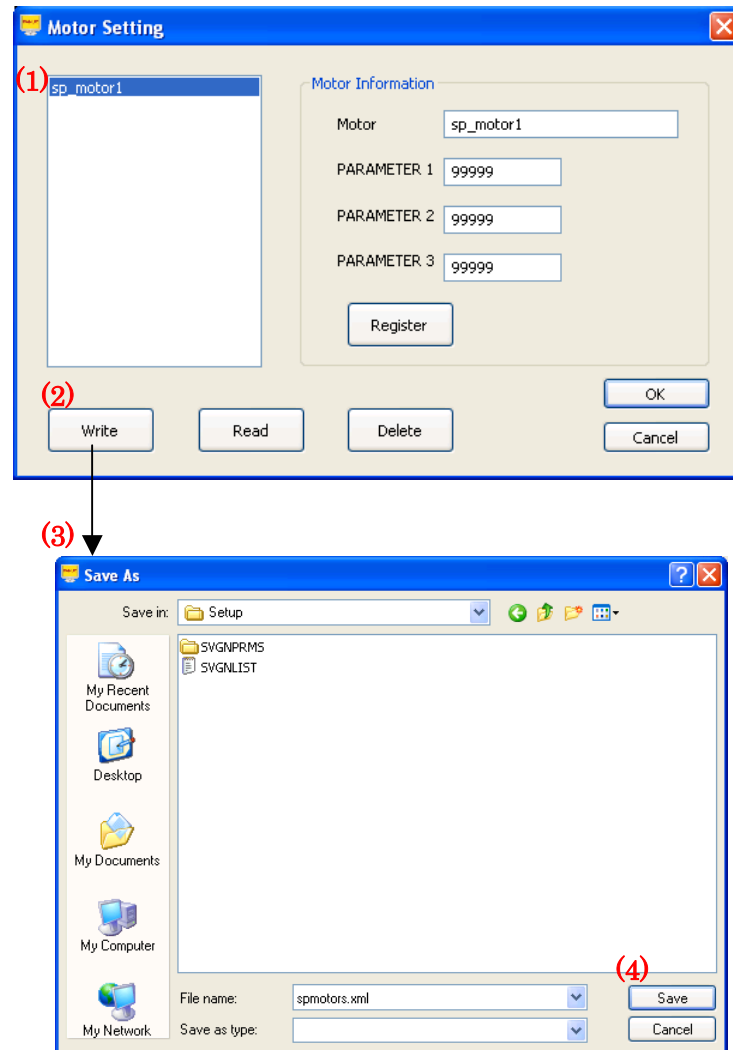


Fig. 4.4.4.3 Motor setting file output

4.4.4.4 Load motor setting

It can load motor setting that was created in beforehand, and then can edit.

1. Click Read button (1) on motor setting screen.
2. File Open dialog appears.
3. Select motor setting file, then click Open button (3).
4. Motors that are included in the selected motor setting file are added to motor list (4).

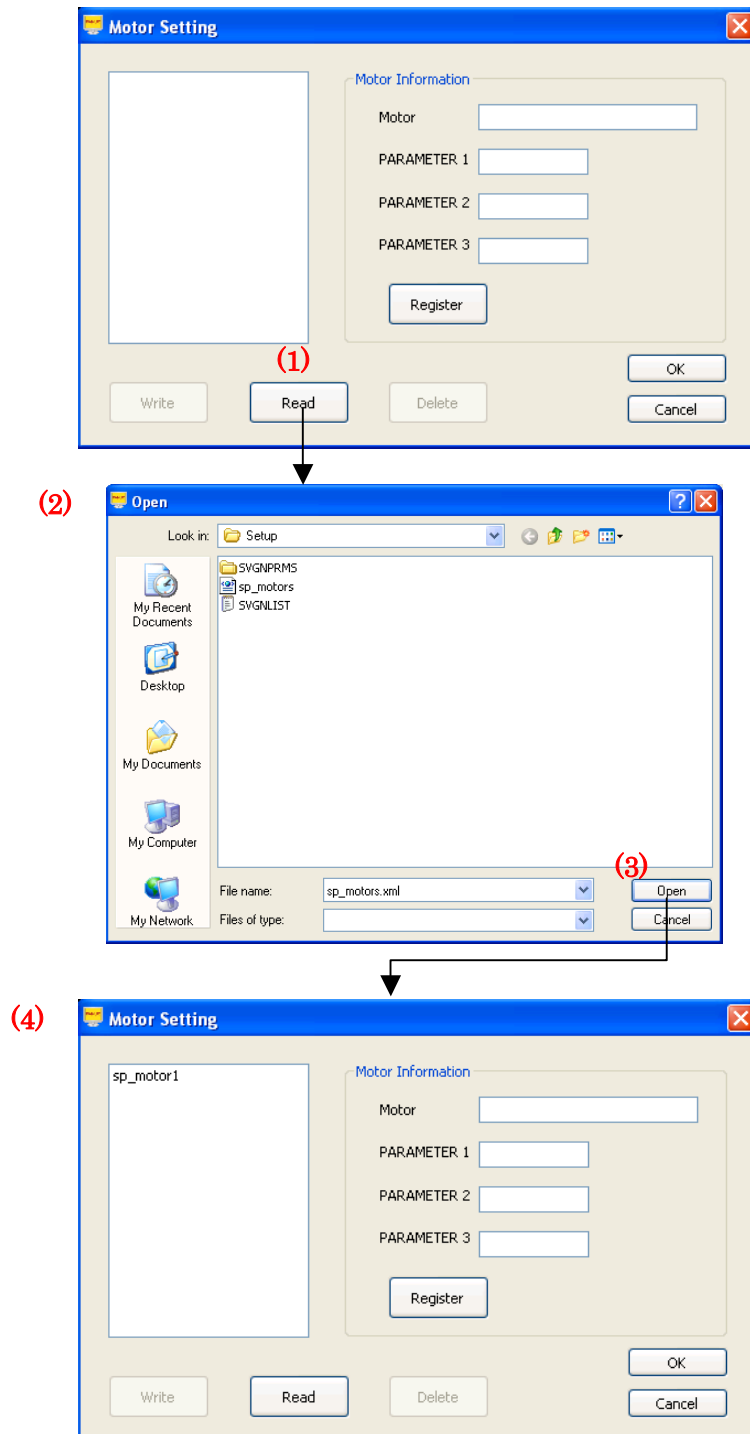


Fig. 4.4.4.4 Motor setting load

4.4.4.5 Delete motor setting

It can delete needless motor setting.

1. Select needless motor on motor list. (1)
2. Click Delete button (2).
3. Confirmation prompt appears. (3) If select yes to the prompt, and then the motor is deleted.

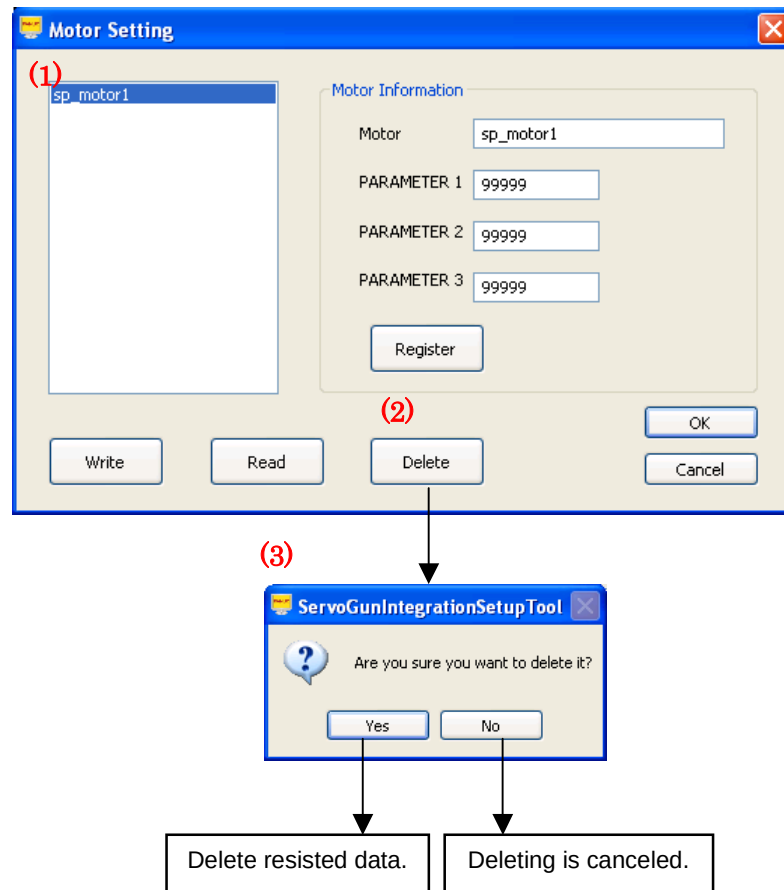


Fig. 4.4.4.5 Motor setting delete

4.4.5 Motor File Address

To select via Motor type pull down menu in Gun axis Setting, specify path of created motor setting file.

1. Select "SetUp"- "Motor File Address"(1).
2. "Motor File address" screen (2) appears.
3. Specify path of motor setting file that includes motor data to select via Motor type pull down menu.
4. Click file select button (3) to appear file select dialog (4).
5. Click OK button (5) when setting is complete.
6. Added motor can be selected via Motor type pull down menu in Gun axis Setting (6).

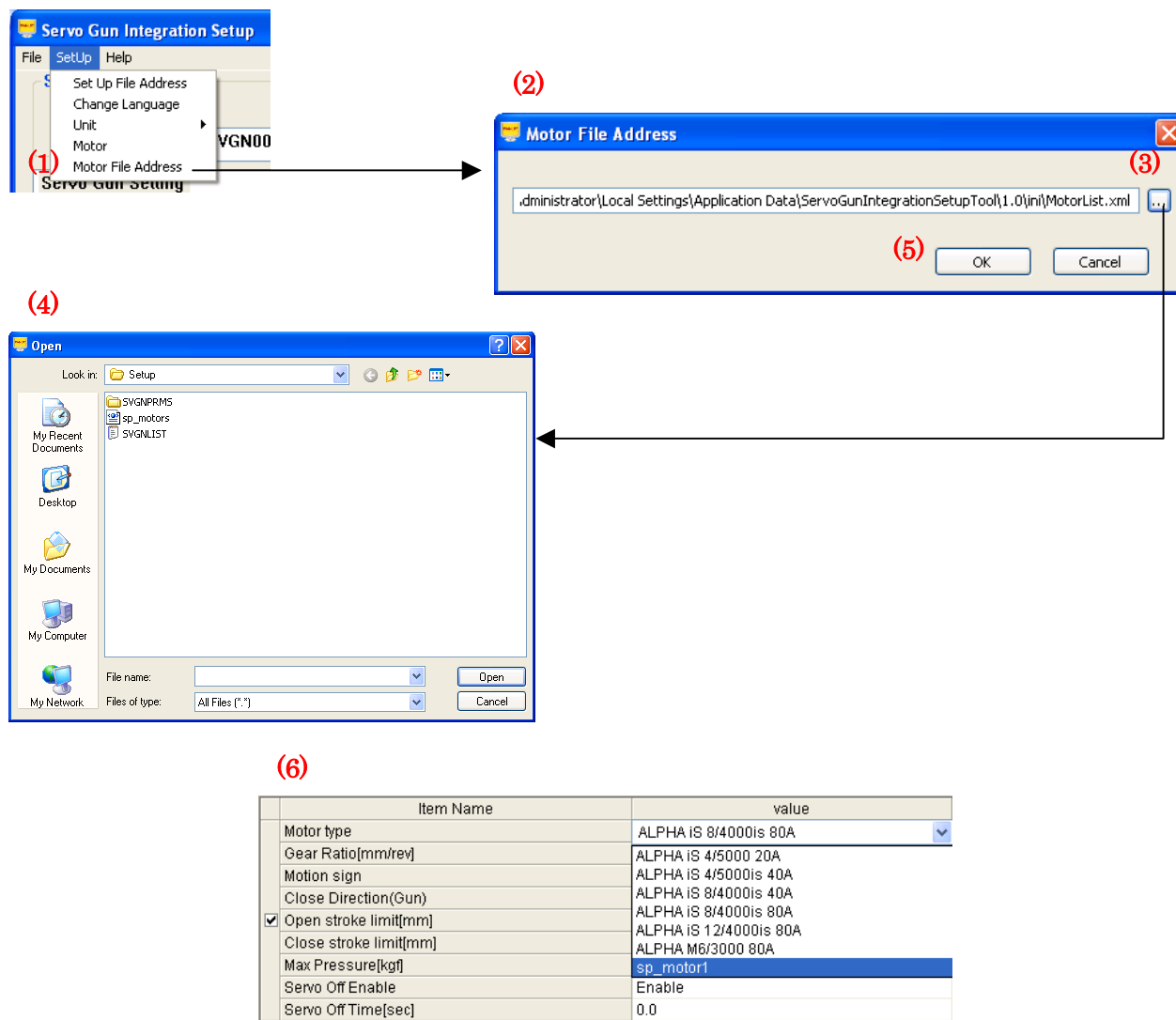


Fig. 4.4.5 Motor file address setting

4.5 HELP

It shows Software version.

1. Select "Help"-"Version"(1).
2. "Version Information" screen (2) appears.

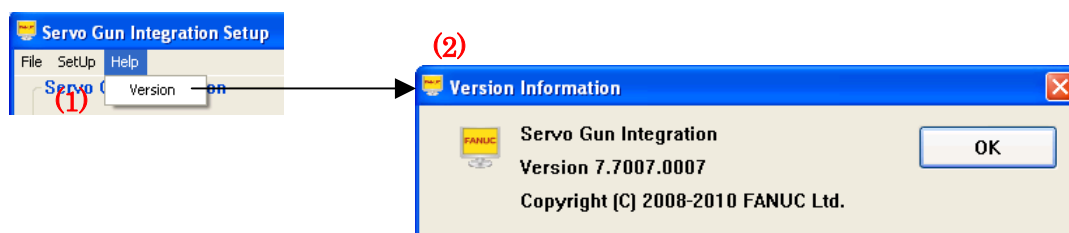


Fig. 4.5 Version

4.6 DETAIL SETTING SCREEN

4.6.1 Preparation for Setting

Enter servo gun name in edit box of servo gun information area within 20 characters.

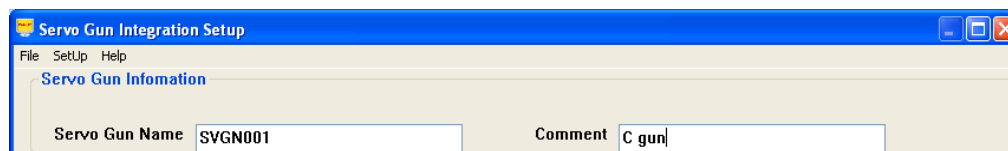


Fig. 4.6.1 Servo Gun name and comment input

Setup file cannot be created with existed same servo gun name.

If using existed same servo gun name, messages whether setup data is updated or not is prompted.

Enter servo gun comment in edit box of servo gun information area within 10 characters if you need it.

If you do not enter servo gun comment, servo gun name is applied to servo gun comment.

If servo gun name is over 11 characters and servo gun comment is nothing, 10 characters of servo gun name is applied to servo gun comment.

4.6.2 Servo Gun Function Setting Screen

Select “Servo Gun Function setting” on servo gun setting category list, then display the setting items on detail setting screen. These items set up general function of servo gun.

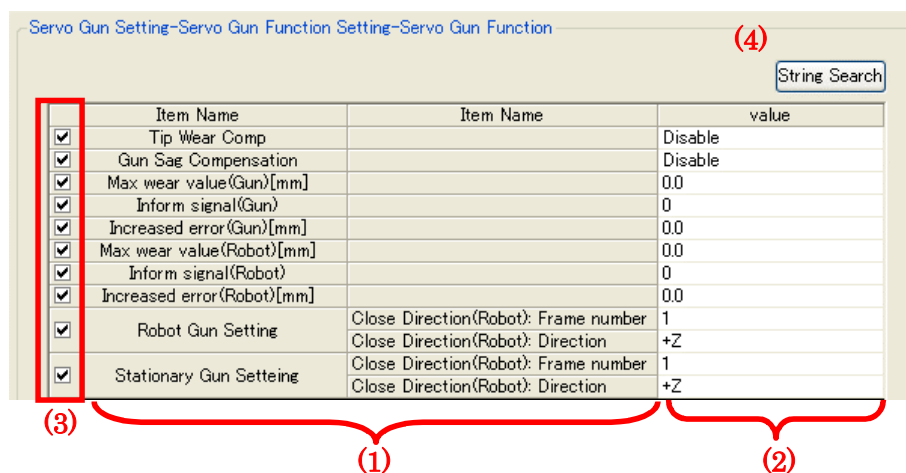


Fig. 4.6.2 Servo Gun function setting screen

- (1) It displays setting items.
- (2) It specifies setup parameter.
- (3) It decides whether data is output or not with check box. Only item that checked is output to setup file.
- (4) It searches text on screen.

NOTE

“Close direction (Robot): Frame number” and “Close direction (robot): Direction” have to be specified for both of robot gun system that robot carries gun and stationary gun system that robot handles work panel.

When a setup file is loaded, select which system is. Setting of the selected system is applied.

4.6.3 Servo Gun Schedule Setting Screen

Select “Weld Schedule” or “Distance Schedule” or “Manual Backup Schedule” via “Servo Gun Schedule Setting” on servo gun setting category list, then display the setting items on detail setting screen. These items set up schedules of servo gun.

The following example is weld schedule screen.

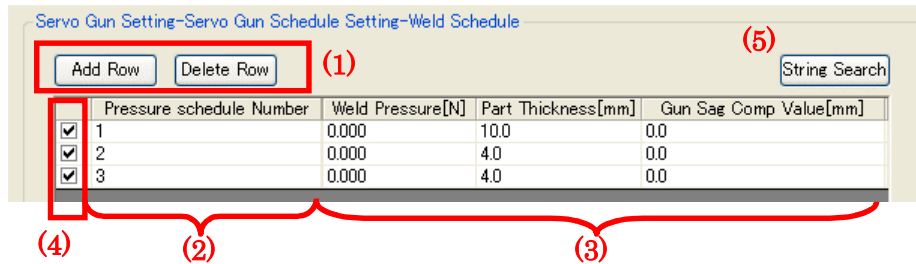


Fig. 4.6.3 Servo Gun schedule setting screen

- (1) It adds or deletes schedule setting line.
- (2) This is a line to set schedule number.
- (3) It specifies setting parameter.
- (4) It decides whether data is output or not with check box. Only item that checked is output to setup file.
- (5) It searches text on screen.

Operate weld schedule setting screen with the following.

- Firstly, enter schedule number that you want to set into schedule number cell.
Note that pressure schedule number and distance schedule number are limited within 99 and manual backup schedule number is limited within 30.
- Then specifies setting parameter into setting cell.
- If you want to add new schedule, click “Add Row”. Then new schedule line is added.
- If you want to delete needless schedule, select needless schedule number cell, and then click “Delete Row”. Then the selected schedule line is deleted.
- About operation in other schedule setting screen, the above is common.

4.6.4 Servo Gun Axis Setting Screen

Select “Gun axis Setting” on servo gun setting category list, then display the setting items on detail setting screen.

These items set up servo gun hardware settings.

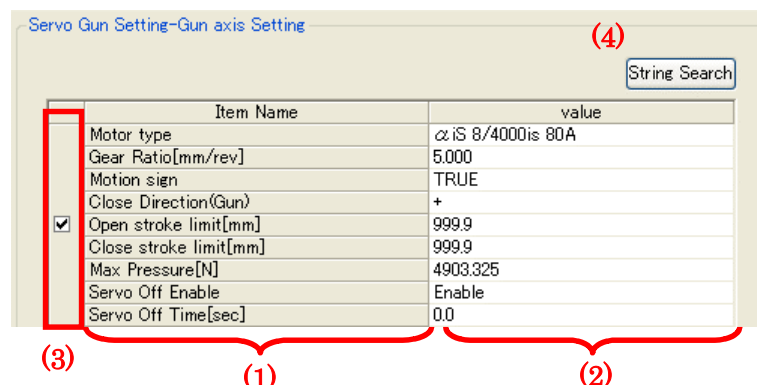


Fig. 4.6.4 (a) Servo Gun axis setting screen

- (1): It displays setting items.
- (2): It specifies setup parameter.
- (3): It decides whether data is output or not with check box. Only item that checked is output to setup file.
Servo gun axis setting cannot output individual item.
- (4): It searches text on screen.

About setting motor type, select motor name via pull down menu. If there is no special motor that you want to select, need to add special motor. Refer to 4.4.4 Motor Settings. When special motor is added, it can be selected via pull down menu.

Servo Gun Setting-Gun axis Setting

String Search

	Item Name	value
	Motor type	α IS 8/4000is 80A
	Gear Ratio[mm/rev]	α IS 4/5000 20A
	Motion sign	α IS 4/5000is 40A
	Close Direction(Gun)	α IS 8/4000is 40A
☒	Open stroke limit[mm]	α IS 8/4000is 80A
	Close stroke limit[mm]	α IS 12/4000is 80A
	Max Pressure[N]	4903.325
	Servo Off Enable	Enable
	Servo Off Time[sec]	0.0

Fig. 4.6.4 (b) Motor type selection

5 OPERATION TO LOAD SETTING DATA

In this chapter, the operation to load robot initial setting and servo gun setting input on PC to Robot controller is described.

In Robot controller, the initial setting file is loaded by 'Initial Setting Load function' software option (A05B-2500-J980) or 'Initial Setup Interface function' software option (A05B-2600-J947). Initial Setting Load function/Initial Setup Interface function, the current setting in Robot controller can be output to the file with same format as the initial setting file.

In order to execute the Initial Setting Load function in R-30iA/R-30iA Mate controller, teach pendant of Robot controller must be *iPendant*.

In case that the software option 'Servo gun setup package' (A05B-2500-J979) or 'Servo gun setup interface package' (A05B-2600-J890) is installed in Robot controller, it is possible to load the setup file created by Servo Gun Integration Setup Tool.

5.1 PREPARE IN ROBOT CONTROLLER

The setting file output by Robot Integration Setup Tool is loaded in Initial Setting Load screen in Robot controller. You can load the initial setting files output by PC into Robot controller with one of following methods.

- Memory Card Load the initial setting file from the memory card stored the initial setting file.
- FTP Load the initial setting file by the FTP communication with PC.
- USB Memory Load the initial setting file from the USB memory stored the initial setting file.

The setting file output by Servo Gun Integration Setup Tool is loaded in Robot Setting screen in controlled start menu. In Robot Setting screen, the setting file is loaded from memory card.

It is possible to load the setting file output by Servo Gun Integration Setup Tool in Initial Setting Load screen. In Initial Setting Load screen, the setting file for servo gun can be loaded by one of above three methods like the robot initial setting file. However, the servo gun axis setting in the servo gun setting can be loaded only in the Robot Setting screen in controlled start menu.

5.1.1 Load form Memory Card

In order to load the initial setting file output by Robot Integration Setup Tool, copy all subfolders and files in the output folder of Robot Integration Setup Tool on PC into a memory card. And, insert that memory card into Robot controller.

In order to load the setting file output by Servo Gun Integration Setup Tool, copy all subfolders and files in SVGNPRMS folder in the output folder of Servo Gun Integration Setup Tool on PC into a memory card. And, insert that memory card into Robot controller.

5.1.2 Load with FTP Connection

Set up Robot controller as an FTP client that is connected to the FTP server on the PC ('Robot Integration Setup Tool' runs on).

1. Set up PC as an FTP server beforehand.
2. Set the IP address of Robot controller at 'SETUP Host Communication' screen.
3. Set the IP address of the PC at the same screen.
4. Set up parameters to connect the FTP Server on the PC at the 'SETUP Clients screen'.

With the FTP connection established, loading operation with FTP can be available.

5.1.3 Load from USB Memory Card

In order to load the initial setting file output by Robot Integration Setup Tool, copy all subfolders and files in the output folder of Robot Integration Setup Tool on PC into a USB memory. And, insert that USB memory into Robot controller.

In order to load the setting file output by Servo Gun Integration Setup Tool, copy all subfolders and files in SVGNPRMS folder in the output folder of Servo Gun Integration Setup Tool on PC into a USB memory. And, insert that USB memory into Robot controller.

5.2 PROCEDURE TO LOAD INITIAL SETTING FILE

5.2.1 Display Initial Setting Load Screen

Display 'Initial Setting Load' screen by the following operations.

1. Press MENU key, and select 'Browser'.

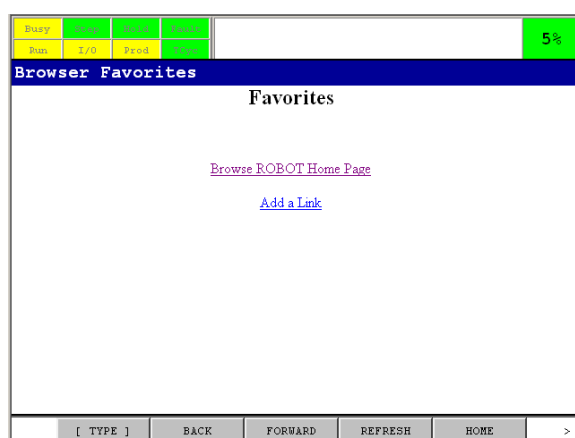


Fig. 5.2.1 (a) Browser screen

2. Press F1 [TYPE] key to display screen change menu.

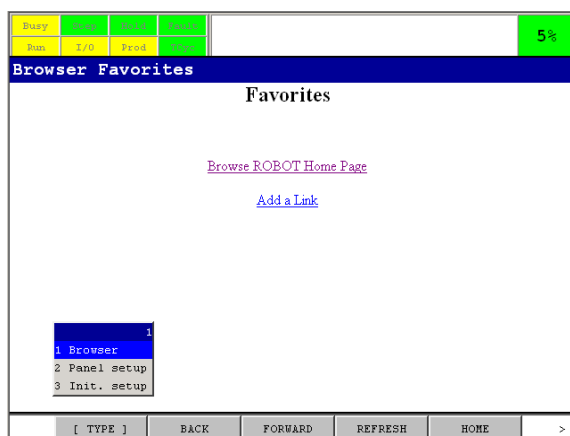


Fig. 5.2.1 (b) Browser screen menu

3. Select 'Init.setup' in the menu to display 'Initial Setting Load' screen.

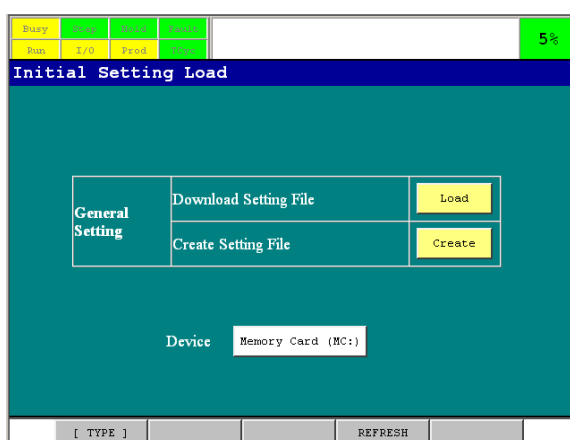


Fig. 5.2.1 (c) Initial setting load screen (1)

In case that the software option 'Servo gun setup package' (A05B-2500-J979) or 'Servo gun setup interface package' (A05B-2600-J890) is installed in Robot controller, the screen to load the initial setting file and Servo gun setup file is displayed as follows.

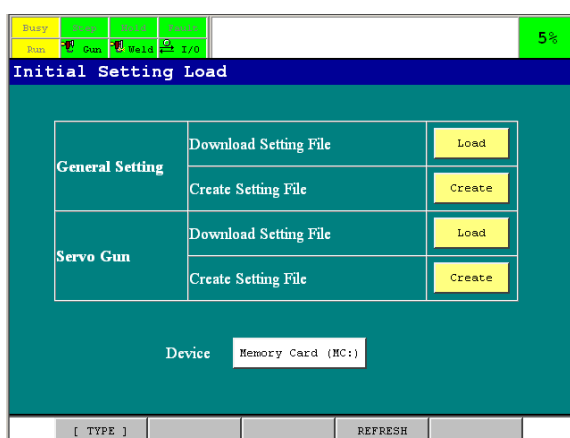


Fig. 5.2.1 (d) Initial setting load screen (2)

5.2.2 Attention about Initial Setting Load Screen

About display of Initial Setting Load screen, there are some attentions as follows.

1. In case that the software series and version of Robot controller matches one of the followings.

- Software series 7DA4 all aversion
- Software series 7DA5 version 09 or before
- Software series 7DA7 version 01

If the item 'Init. setup' does not appear in the F1[TYPE] menu, set the following system variables by manual.

(In case that Servo gun setup package/Servo gun setup interface package is not ordered.)

System variable	Data
\$TX_SCREEN[i].\$DESTINATION	../frh/ncrobot/nris01.stm
\$TX_SCREEN[i].\$SCREEN_NAME	Init. setup

(i: Find the number which is not used in the array of \$TX_SCREEN, and set data.)

(In case that Servo gun setup package/Servo gun setup interface package is ordered.)

System variable	Data
\$TX_SCREEN[i].\$DESTINATION	../frh/ncrobot/nris07.stm
\$TX_SCREEN[i].\$SCREEN_NAME	Init. setup

(i: Find the number which is not used in the array of \$TX_SCREEN, and set data.)

To confirm whether Servo gun setup package/Servo gun setup interface package is ordered, check whether the item 'Servo gun setup package J979' or 'Servo gun setup interface J890' exists in the software configuration screen. The software configuration screen can be displayed by the operation [MENU] - [STATUS] - [Version ID] - [Config].

2. In the case of other software series and version.
If the item 'Init. setup' appears twice in the menu, please clear above system variables.

5.2.3 Loading Initial Setting File

Load the initial setting files as follows.

1. Select 'Memory card' or 'USB disk' or 'FTP client' as an input device.

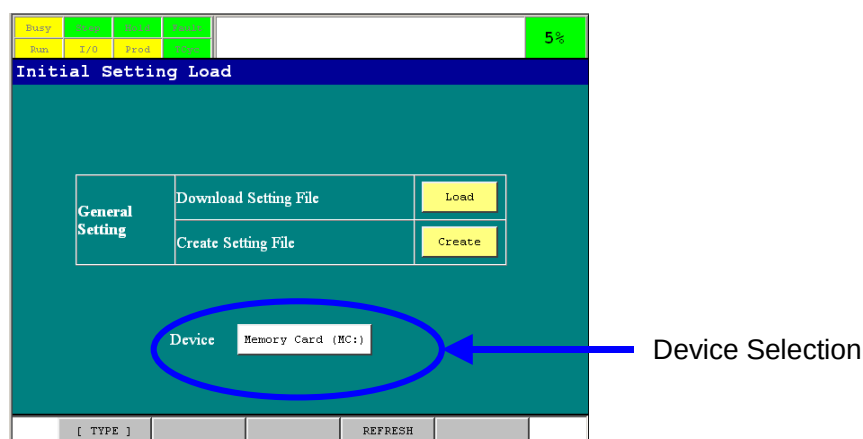


Fig. 5.2.3 (a) Device selection

Memory card , USB memory and FTP client tags are shown on the device selection list. To select 'FTP Client', FTP client settings are required before.

2. Press 'Load' button inside 'General Setting' section.

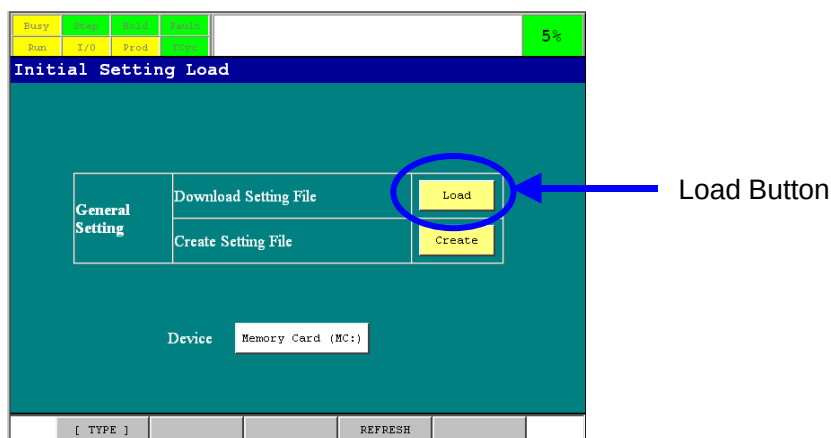


Fig. 5.2.3 (b) Load button

3. Following screen is displayed. Select the initial setting files. The initial setting files are designated by the Cell Name which the Robot belongs to, and the Robot Name.

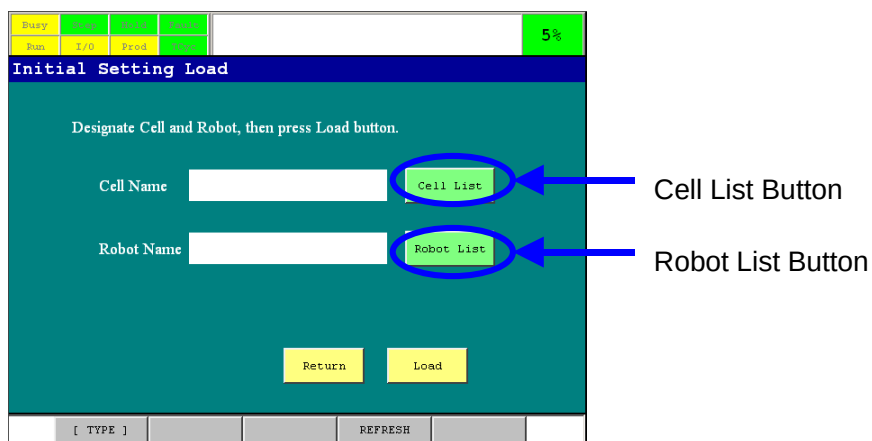


Fig. 5.2.3 (c) Cell list button, Robot list button

Input the Cell Name and Robot Name directly, or select one from cell list and robot list stored in the device.

To select the cell name and the robot name, select the cell name first, and select robot name from the robot list. With no cell is selected, the robot list is not shown. Press 'Cell List' button.

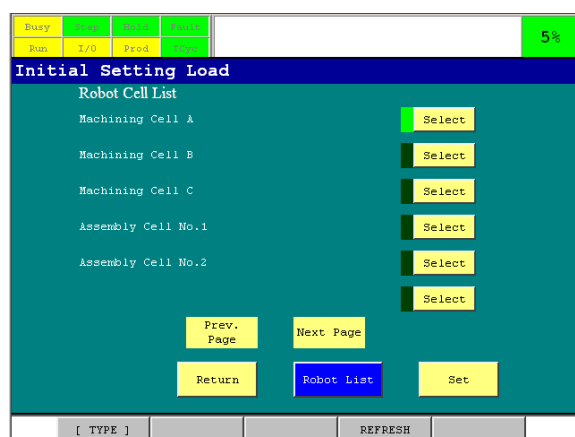


Fig. 5.2.3 (d) Cell list

Select a target cell and press 'Robot List' button. Robot name list in the cell is displayed.

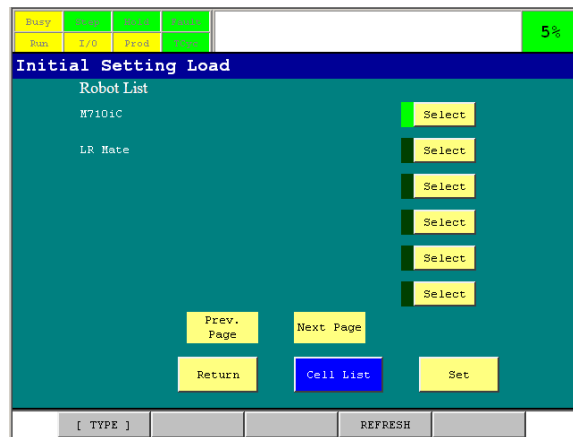


Fig. 5.2.3 (e) Robot list

Select a target robot and press 'Set' button. Both the cell name and robot name are set.

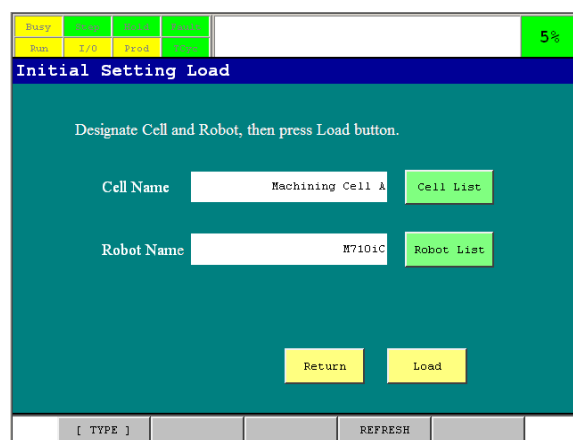


Fig. 5.2.3 (f) Cell name and Robot name input

4. Press 'Load' button. Robot controller starts to load the initial setting file.

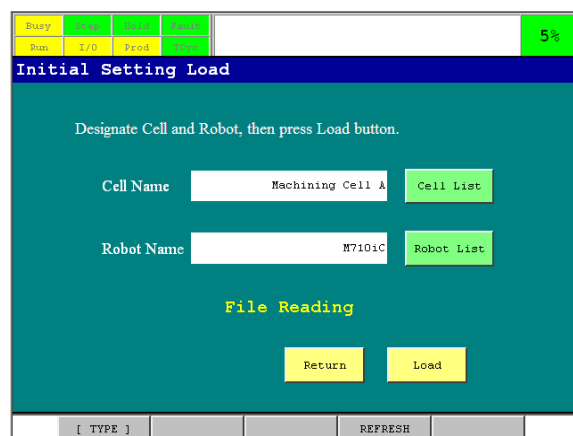


Fig. 5.2.3 (g) File reading

5. When Robot controller successfully loads the files, the following screen is displayed.

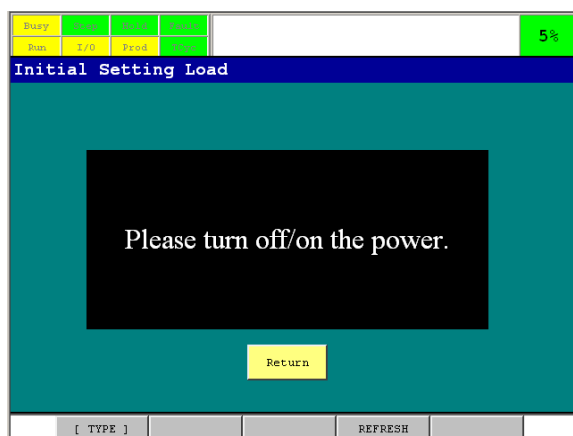


Fig. 5.2.3 (h) File read end

Reboot Robot controller definitely to apply the change.

5.2.4 Store Current Setting of Robot Controller

Store the current setting of Robot controller by the following operations.

1. Select 'Memory card' or 'USB disk' or 'FTP Client' as a target device.

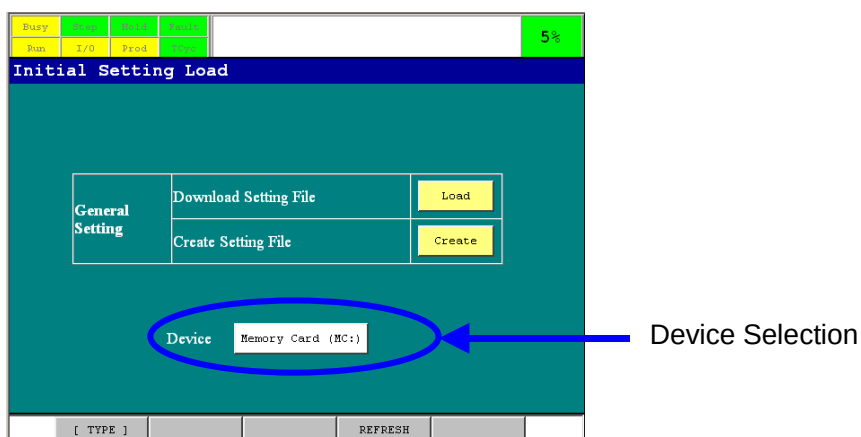


Fig. 5.2.4 (a) Device selection

Memory card , USB memory and FTP Client tags are shown on the device selection list. To select 'FTP Client', FTP client settings are required.

2. Press 'Create' button inside 'General Setting' section.

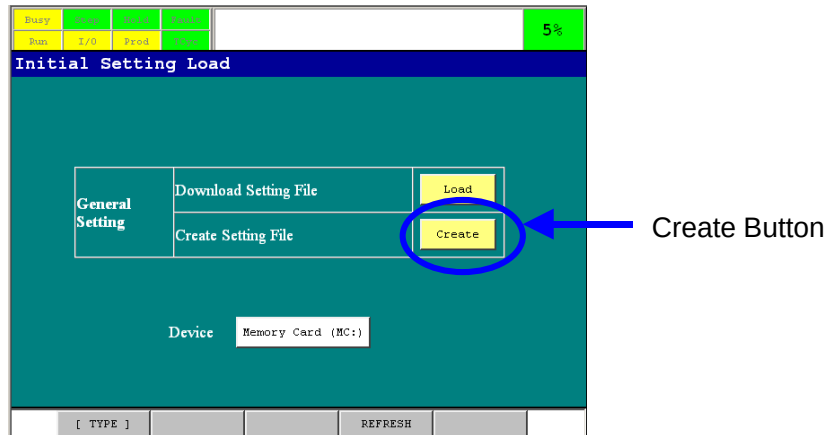


Fig. 5.2.4 (b) Create button

3. The following screen is displayed. To store the setting, select 'Yes'.

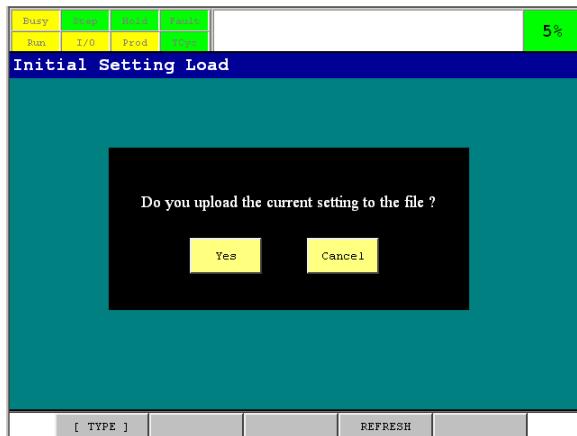


Fig. 5.2.4 (c) Initial setting file create

4. When Robot controller successfully stores the setting, Initial Setting Load screen is displayed.

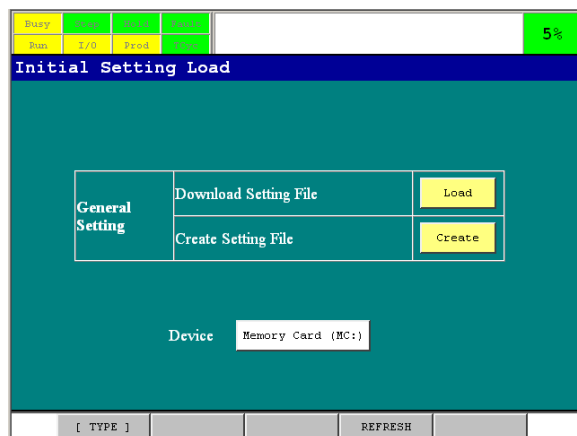


Fig. 5.2.4 (d) Initial Setting Load screen

A subfolder 'ROBOCEL00' is created on the target device. This folder contains the following files.

- CELLDATA.TXT (an administrative file)
- ROBDAT01.PRM (Initial setting files format)

You can load this 'ROBDAT01.PRM' into Robot Integration Setup Tool with the menu, 'Initialization File Reading'. In this case, the Robot Name becomes 'Robot'.

Only the setting for Function setting is described in the file created by this operation. The setting for I/O setting is not described.

5.2.5 Load Servo Gun Setting File

If “Servo gun setup package” (A05B-2500-J979) option software or “Servo gun setup interface package” (A05B-2600-J890) option software is installed, load servo gun setup files with the following operation.

5.2.5.1 Loading setup file via Robot Maintenance screen

Setup files can be loaded via robot maintenance screen in controlled start with the following procedure.

1. Start up Controlled Start.
Power up with pressing Prev key and Next key , then select Controlled start on configuration menu.
2. Display robot maintenance screen.
Press Menu key, select “9. MAINTENACE ”.
3. Insert memory card that has **SVGNPRMS folder includes setup files on root** to slot on controller.

NOTE

Do not edit setup files that are created by Servo Gun Integration Setup Tool.
If setup files are edited, setting process cannot be executed correctly.

4. Move cursor “Servo Gun Axes”(1), then press F5 GunSet (2).
Note that F5 GunSet key appears only when Servo gun setup package option is installed.

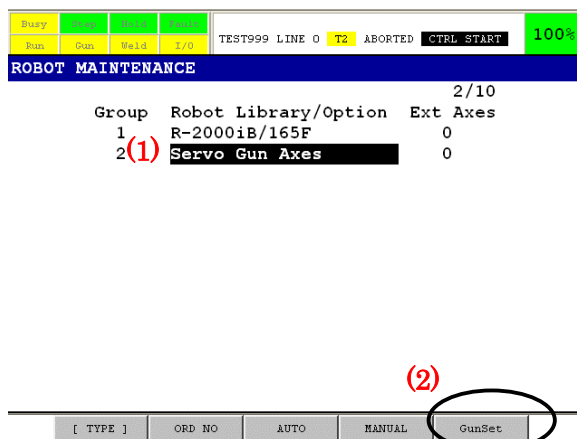


Fig. 5.2.5.1 (a) Robot Maintenance screen

5. The following screen is displayed. Enter servo gun name that you want to set up with in 20 characters. If you want to escape from this screen, enter 0 as name.

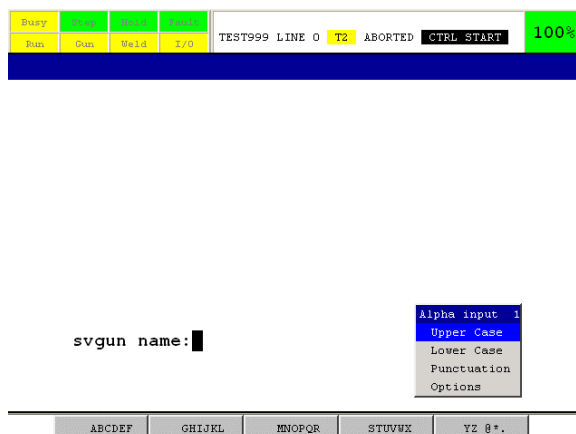


Fig. 5.2.5.1 (b) Servo Gun name input

Error message is prompted in the following cases.

- Memory card is not inserted correctly.
- Servo gun name is not correct.
- There is not setup file of entered servo gun name.
- There is no correct setup file.

6. Confirmation screen that shows hardware configuration and servo gun name is displayed.

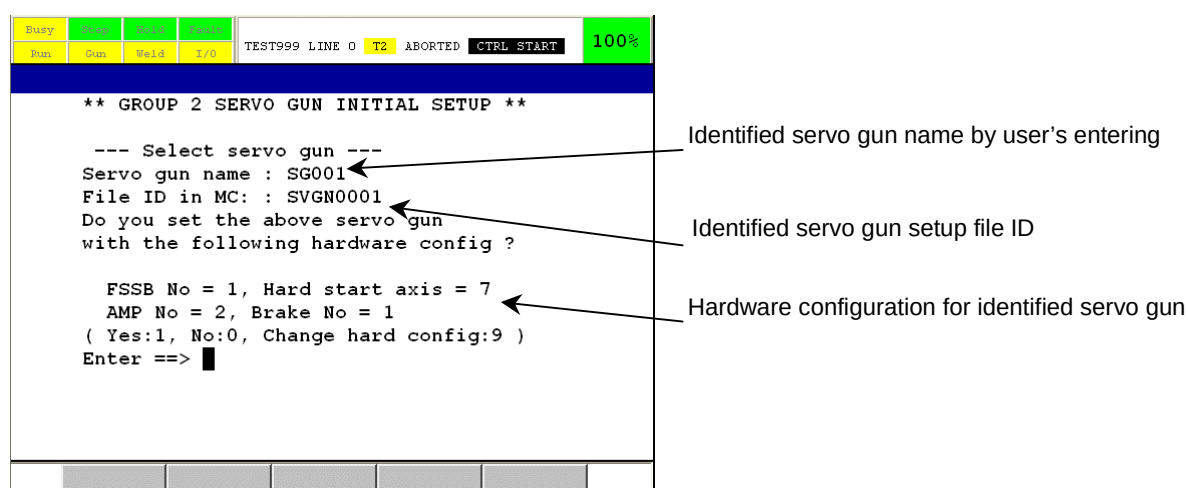


Fig. 5.2.5.1 (c) Servo Gun setting

If hardware configuration and servo gun name are correct on the screen, enter 1 to response yes. Then servo gun axis setting is loaded.

If you want to cancel setup, enter 0 to response no. Then it returns to robot maintenance screen.

If you want to change hardware configuration, enter 9. Then select item number of configuration that you want to change, and change configuration as follows.

NOTE

In this screen, amplifier number and brake number can be changed.

If you want to change FSSB line number or hardware start axis number, change the data in manual setup screen via F4 Manual key.

Note that FSSB line number or hardware start axis number can be change when there is no axis that the setup is complete.

Procedure to change hardware configuration (Ex. Changing brake number)

1. Enter 9 to select "Change hard config" on confirmation screen (1).
2. Enter 2 to select brake number setting on changing hardware screen (2).
3. Enter brake number on changing brake number screen (3).
4. Enter 0 to return to confirmation screen (4).
5. Changed configuration is applied on confirmation screen (5).

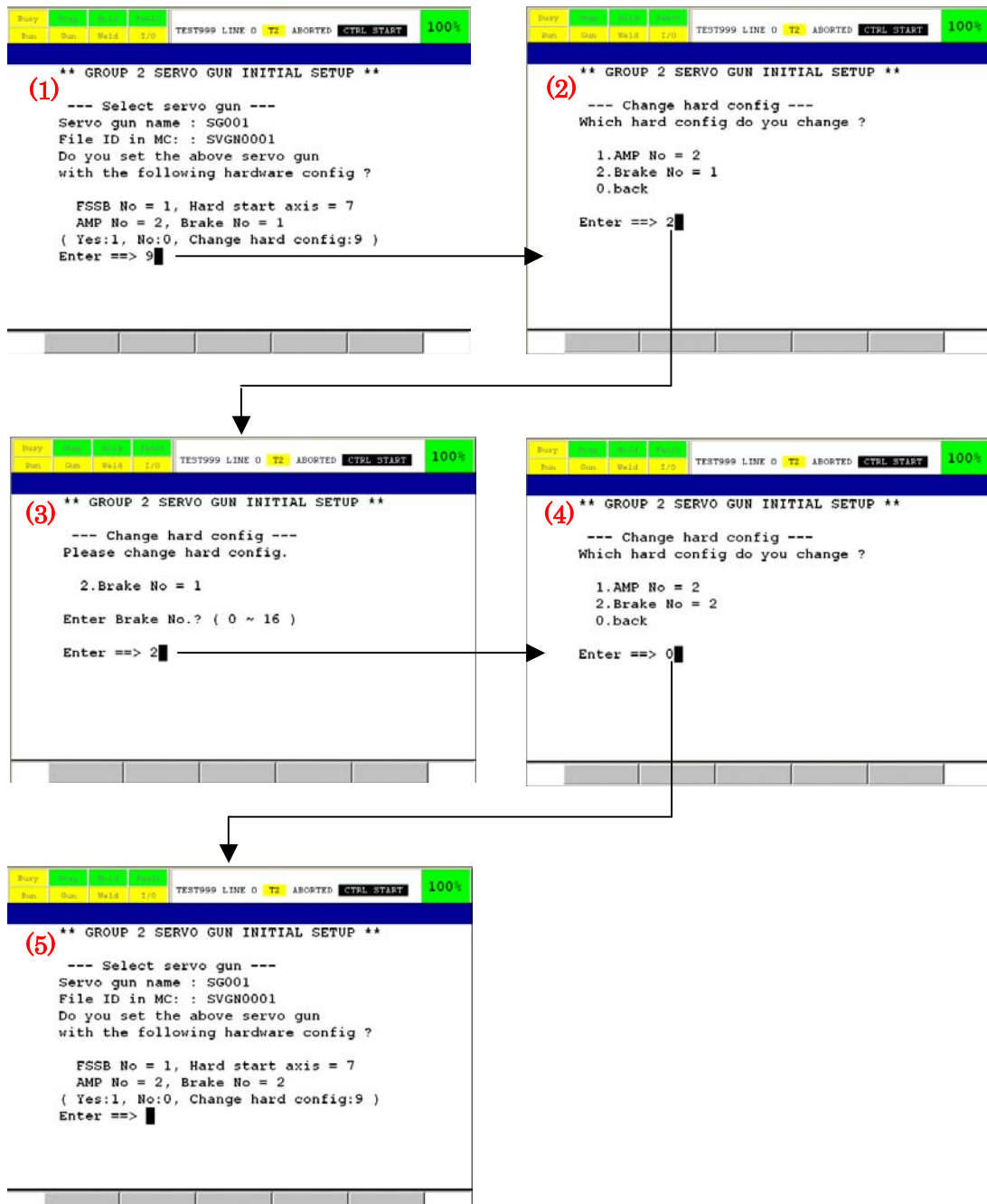


Fig. 5.2.5.1 (d) Hardware configuration change

7. When servo gun axis setting is complete, next other servo gun settings is started.

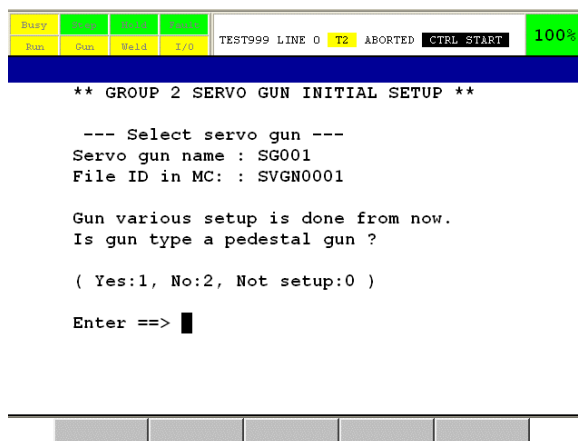


Fig. 5.2.5.1 (e) Servo Gun setting

Enter 1 for stationary gun system that robot handles work panel.

If not, enter 2 for robot gun system that robot carries gun.

If you want to setup other servo gun settings now, enter 0 to return robot maintenance screen.

8. When all settings is over, it returns to robot maintenance screen automatically.

Procedure to delete servo gun axis

Perform usual operation to delete axis if you want to delete axis setting via manual setting screen with F4 MANUAL.

5.2.5.2 Load servo gun setting file in initial setting load screen

It is possible to load the servo gun setting in Initial Setting Load screen that is displayed after cold start. However, the servo gun axis setting in servo gun setting can be loaded only in the Robot Setting screen in controlled start menu.

Table 5.2.5.2 Load the servo gun setting

Setting	Robot Setting screen	Initial Setting Load screen
Servo gun function setting	Loadable	Loadable
Servo gun schedule setting	Loadable	Loadable
Servo gun axis setting	Loadable	Not loadable

1. Select 'Memory card' or 'USB disk' or 'FTP client' as an input device.

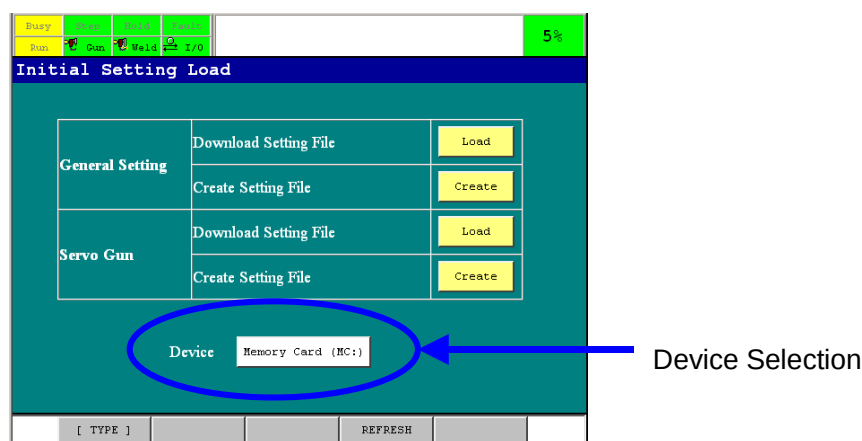


Fig. 5.2.5.2 (a) Device selection

Memory card, USB memory and FTP client tags are shown on the device selection list. To select 'FTP Client', FTP client settings are required.

- Press 'Load' button inside 'Servo Gun' section.

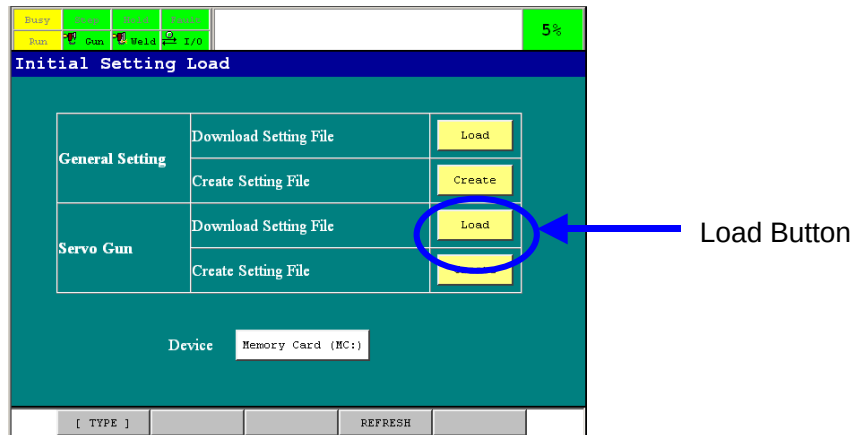


Fig. 5.2.5.2 (b) Load button

- Following screen is displayed. Select the servo gun setting file. Designate which servo gun setting file is loaded by the gun name.

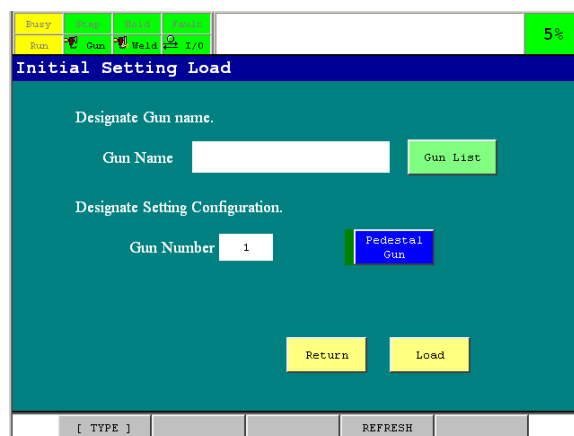


Fig. 5.2.5.2 (c) Setting file load screen

Input the Gun Name directly, or select one from the gun list stored in the target device. To see the list of the stored servo gun setting files, press 'Gun List'.

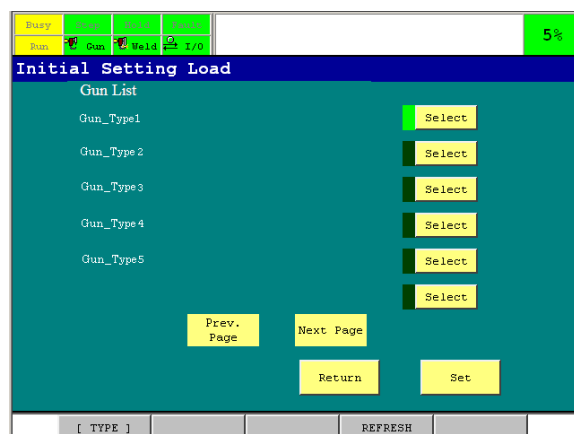


Fig. 5.2.5.2 (d) Gun list screen

Select the target gun by its name and press 'Set'. The gun name is set.

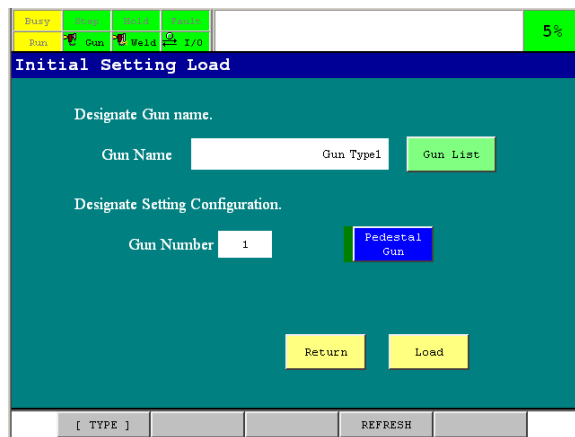


Fig. 5.2.5.2 (e) Gun name input

In addition, set the following parameters about configuration of the servo gun.

Gun Number: In order to use multiple servo guns with one controller, set the identification number of the target gun. When you use only one servo gun with Robot controller, input '1' definitely.

Pedestal gun: If the target servo gun is a pedestal gun, press 'Pedestal Gun' and make the green lamp turned on. To cancel the pedestal gun setting, press the button again and make the lamp turned off.

4. Press 'Load'. Robot controller starts to load the servo gun setting file.

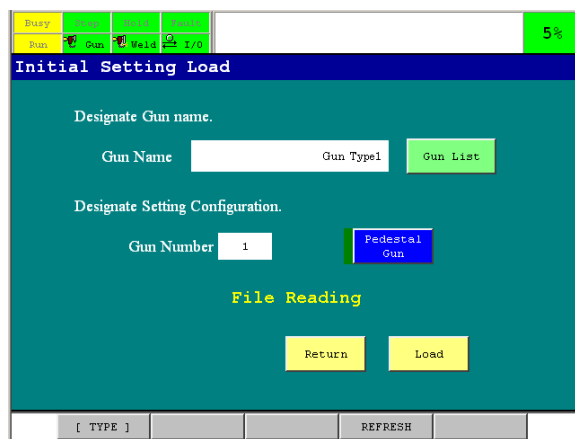


Fig. 5.2.5.2 (f) Setting file reading

5. When Robot controller successfully loads the files, Initial Setting Load screen is displayed.

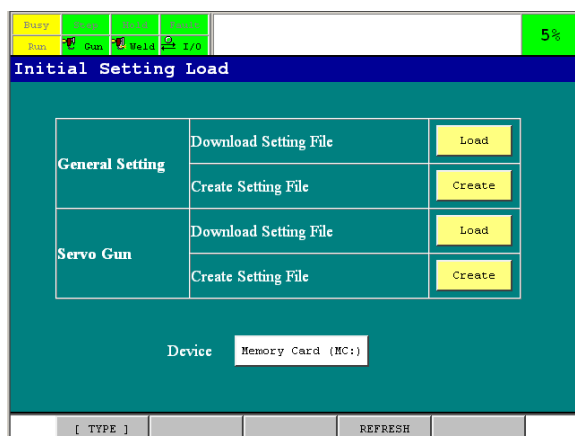


Fig. 5.2.5.2 (g) Initial Setting Load screen

- When Robot controller fails in loading the files, the error message 'Reading some system variable failed' is displayed. Refer to ¥Device(MC, etc)¥SVGNPRMS¥SGERROUT.TXT when this message is displayed. The system variables that cannot be read are recorded in SGERROUT.TXT. The system variables not recorded in SGERROUT.TXT are loaded successfully.

5.2.6 Store Current Setting of Servo Gun

Store the current setting of the servo gun by the follow operations.

- Select 'Memory card' or 'USB Disk' or 'FTP Client' as a target device.

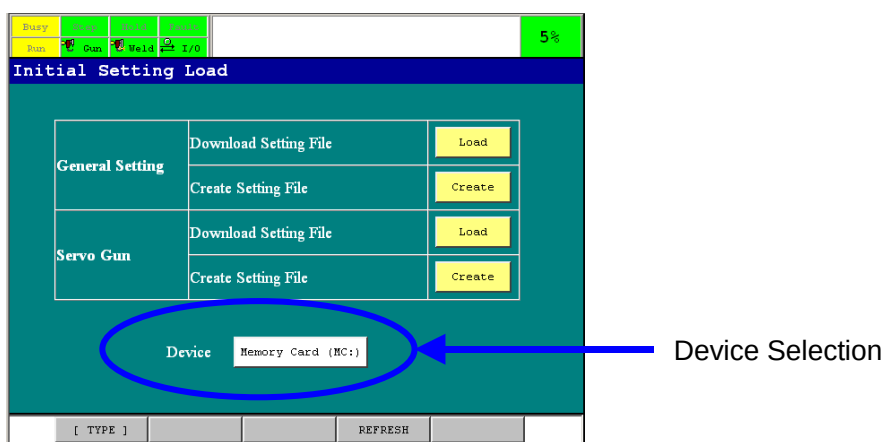


Fig. 5.2.6 (a) Device selection

Memory card, USB memory and FTP client tags are shown on the device selection list. To select 'FTP Client', FTP client settings are required before.

- Press 'Create' button inside 'Servo Gun' section.

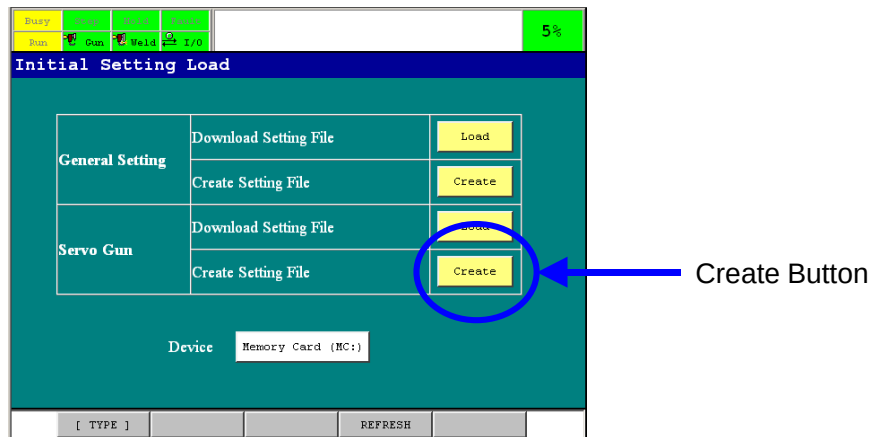


Fig. 5.2.6 (b) Create button

3. The following screen is displayed. To store the setting, designate the gun number and select 'Exec.'.

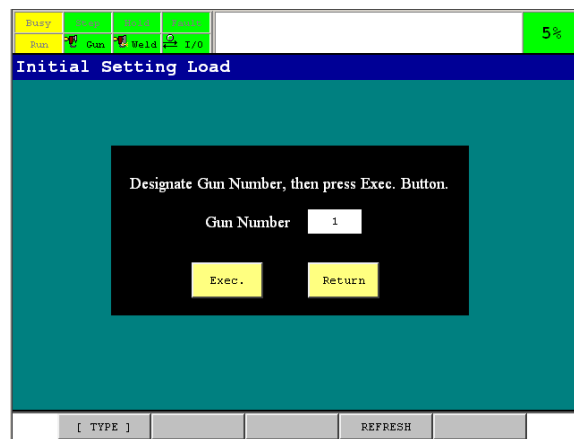


Fig. 5.2.6 (c) Setting file create

4. When Robot controller successfully stores the setting, Initial Setting Load screen is displayed.

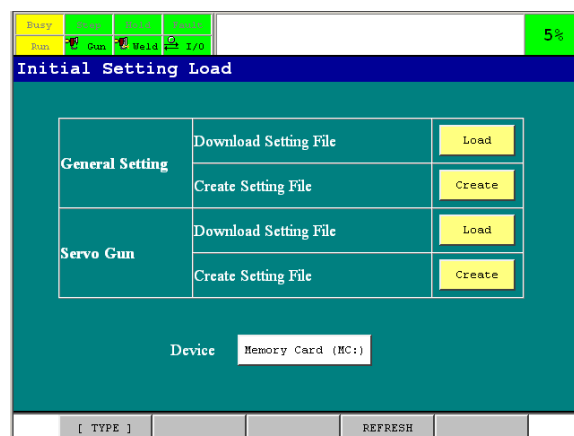


Fig. 5.2.6 (d) Initial Setting Load screen

The following file is created in the subfolder 'SVGNPRMS' on the target device.

SVGN0000.PRM (Servo gun setting file format)

You can load this 'SVGN0000.PRM' into Servo Gun Integration Setup Tool with the menu 'Setup File Input'. In this case, the servo gun name becomes 'SVGN0000'.

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REVISION RECORD

Edition	Date	Contents
03	Apr., 2014	• Revised as the manual for R-30iB/R-30iB Mate controller.
02	Apr., 2010	• Setting for Soft float is added in Initial Setting Tool • Setting for Payload is added in Initial Setting Tool • Setting of Register Value is added in Initial Setting Tool
01	Apr., 2009	

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